

Current MoE Reasoning Experiments

1 Scope

This report tracks the three active workflows: `gepaLLMAsJudge`, `probeTest`, and `correlation_pipeline`. The first two were called Experiments 0a and 0b in the previous combined manuscript. Their full protocols, intermediate results, tables, limitations, and external-validation results are retained here under their current names.

The previous combined manuscript for numbered Experiments 1–5 is preserved at `old/report/legacy_full_report.tex` together with its compiled PDF. Those archived results remain available for provenance but are not mixed into the current experimental plan.

2 Experimental Setup

Current experimental scope. The active repository contains three complementary Qwen workflows. `gepaLLMAsJudge` is an annotation-capability study that optimizes Qwen3.6-27B and Qwen3.8-27B prompts for sentence-level Schoenfeld episode classification. It does not provide evidence about Qwen router dynamics. `probeTest` is a representation-probe replication that teacher-forces the 38 released DeepSeek-R1 Schoenfeld traces through the official GPTQ 4-bit Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 checkpoint and tests whether the next sentence’s label is linearly decodable at its causal pre-sentence boundary. The separate `correlation_pipeline` generates benchmark solutions with Qwen3.5-35B-A3B, replays them through the matching checkpoint, and relates router and hidden-state summaries to correctness and reasoning-event labels.

GEPA judge design. The GEPA workflow starts from 3,125 annotated units in 38 released DeepSeek-R1 SAT responses. Complete responses are assigned to partitions before their units are flattened, preventing neighboring units from crossing splits. A local `llama.cpp` server supplies Qwen predictions. Each request contains the SAT problem and the previous, current, and next response units. The original context-aware seed instruction contains 21 audited synthetic contrastive demonstrations. The Qwen3.8 alignment instead imports the 40-example prompt selected by the earlier final fit as its external seed, so it does not dynamically sample new demonstrations. The implementation supports inverse-frequency weighted exact match and an informative same-model LLM-as-a-judge reward. Prompt-selection rules are evaluated with nested response-grouped cross-validation; the final-fit runs use a response-grouped 26/6/6 train/validation/test split.

Schoenfeld boundary-probe design. The probe workflow excludes the 38 synthetic `</think>` closing units, which are all labelled Monitor, leaving the paper-matched 3,087 units. Qwen receives each SAT instruction as the user message and the released response as a teacher-forced assistant continuation. For a target unit, the feature is the hidden state immediately before its first token, so the probe sees the problem and preceding response but cannot read the target sentence. Hidden-state index 0 is the embedding output and indices 1–40 are the successive language-model layers.

At each index, seven separate target-versus-rest logistic regressions use raw 2,048-dimensional activations, balanced class weights, L2 regularization with $C = 1$, and no standardization or PCA. The paper-matched

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protocol uses a target-stratified 80/20 sentence split with seed 42. This is a replication choice rather than a strong generalization design: all 38 response IDs occur in both partitions. The reported layer for each target maximizes accuracy on that same test split. For qualitative transfer inspection, the selected probes are applied to 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. The seven binary scores are independently trained and uncalibrated; their argmax is an inspection label, not a multiclass probability or new gold annotation. Only boundary vectors are stored; Qwen does not regenerate the released or benchmark solutions.

Correlation-pipeline design. The current pipeline separates vLLM generation from a teacher-forced Hugging Face forward pass. The September 10, 2026 untagged run covers 4,647 attempts from 1,547 source problems on MATH-500, AIME 2024, AIME 2025, OlympiadBench, AMC 2023, and Minerva Math. AIME and AMC use 32 attempts per problem; the other datasets use one. GPQA-Diamond and MMLU-Pro are absent from this snapshot. Generation serves Qwen/Qwen3.5-35B-A3B-GPTQ-Int4; replay uses unsloth/Qwen3.5-35B-A3B with runtime 4-bit quantization, so the stages use different 4-bit formats. All 4,647 attempts enter the binary analysis, including 371 unparseable answers counted as incorrect. There are 3,588 correct attempts (0.772 pooled accuracy), and 394 traces reach the 32,768-token budget. These failures account for 393 of 1,059 errors.

Replay retains compact features and selected top- k expert IDs at six decoder layers. The analysis includes whole-continuation correlations, complete-group avg@32 and within-problem contrasts, metric-pair and pooled expert-ID screens, and new full-reasoning and absolute-position views. The position reference uses the corpus mean of 12,477.026 reasoning tokens, with ten disjoint windows and overflow. Late-window features are missing for shorter traces. No sentence classifier or Schoenfeld class view was used. The pipeline is operationally complete for these six benchmarks; the analyses do not remove termination, length, or token-composition confounds, and pooled numeric expert IDs do not identify individual layer-specific expert modules.

Table model and metric conventions. Every table caption identifies the evaluated model or models and the metrics shown. For classification, accuracy is the fraction of exact label matches, balanced accuracy is the unweighted mean of class recalls, precision and recall have their standard true-positive definitions, macro-F1 is the unweighted mean of per-class F1, and AUROC measures score ranking across the positive and negative classes. Cohen’s κ is chance-corrected nominal agreement, Kendall’s τ_b is rank agreement under the fixed episode-label ordering, and the GEPA composite averages their equal-weight $[0, 1]$ rescalings. Routing entropy is Shannon entropy of the softmax router distribution; the correlation pipeline’s switch rate is the fraction of adjacent tokens whose top-1 expert changes; its top- k overlap is the adjacent-set intersection divided by k (not Jaccard overlap); and router margin is the top-1 minus top-2 router probability. Top- k mass sums the selected probabilities, while the boundary margin is $p_{(k)} - p_{(k+1)}$. Unless a caption states otherwise, N and support are observation counts, r is a Pearson-family correlation coefficient, and confidence intervals are percentile bootstrap intervals.

Artifact provenance. The active packages are located at `src/moe_exp/gepaLLMAsJudge`, `src/moe_exp/probeTest`, and `src/moe_exp/correlation_pipeline`. GEPA runs through `run_gepaLLMAsJudge.sh` and writes to `results/gepaLLMAsJudge`; boundary-probe outputs remain under `results/probeTest`. The correlation workflow is launched by its local `run_all.sh` and uses its documented output directory.

The Qwen judge artifacts formerly called Experiment 0a, including the final Qwen3.8 model-alignment run, and the Qwen3.5 boundary-probe artifacts formerly called Experiment 0b are complete. The current vLLM correlation generation, replay, and analysis are complete for six math benchmarks, including full-reasoning and position views. GPQA-Diamond was analyzed in an earlier GGUF run but is absent from this snapshot; MMLU-Pro remains unrun. Historical OLMoE and numbered-experiment outputs are archived and are not pooled with the estimates in this report.

Table 1. Preliminary gepaLLMAsJudge agreement. The composite rescales Cohen’s κ and Kendall’s τ_b from $[-1, 1]$ to $[0, 1]$ and averages them. Coverage was 100% in every row. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, Cohen’s κ , Kendall’s τ_b , their rescaled composite, and valid-output coverage.

Evaluation	Accuracy	κ	τ_b	Composite
Seed validation	0.6388	0.5581	0.4810	0.7598
Optimized validation	0.6855	0.6074	0.5847	0.7980
Optimized test	0.6607	0.5710	0.6066	0.7944
Few-shot validation	0.6806	0.6078	0.5830	0.7977
Few-shot test	0.6994	0.6178	0.6667	0.8211

3 gepaLLMAsJudge

The gepaLLMAsJudge workflow evaluates whether prompt optimization improves a local judge for the seven adapted Schoenfeld episode labels: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. The model was the Q4_K_XL quantization of Qwen3.6-27B served by llama.cpp on one RTX 3090. All splits preserve complete responses, preventing neighboring units from crossing partitions.

Preliminary single-split runs. With seed 42, the 38 responses were divided into 26 training, 6 validation, and 6 test documents (2,382/407/336 units). These first runs classified the current unit without neighboring context and optimized per-unit exact match.

The matched few-shot run used the same seed, document split, model, and GEPA budget, adding seven gold training examples to the instruction. It performs better than the optimized base condition on the held-out test set: accuracy rises from 66.07% to 69.94% (+3.87 percentage points), Cohen’s κ from 0.5710 to 0.6178, Kendall’s τ_b from 0.6066 to 0.6667, and composite agreement from 0.7944 to 0.8211. GEPA itself does not improve the few-shot validation result: the seed and selected optimized prompts both score 68.06% accuracy. The current comparison therefore favors few-shot prompting for held-out generalization, whereas the measurable GEPA optimization gain is confined to the base-prompt condition.

Context-aware nested cross-validation. The revised protocol excludes the same six responses from model selection and runs five outer folds over the remaining 32 responses. Within each fold, five responses form the inner validation set and all others outside the outer fold form the GEPA training set. Each request includes the problem statement, previous unit, current unit, and next unit. The seed instruction includes 21 audited synthetic contrastive demonstrations. GEPA receives inverse-frequency weighted exact match, whose split mean is balanced accuracy; the safety selector compares either balanced accuracy or macro-F1 and rejects any candidate whose class recall falls by more than the configured threshold.

Informative same-model reward. The current implementation can replace the static GEPA reward with an LLM-as-a-judge call to the same Qwen3.6-27B model. For each candidate label, the judge returns a short free-text explanation plus four diagnostic scores from 1 to 5: gold agreement, functional fit, contextual coherence, and boundary precision. Their normalized weighted mean is used only as GEPA’s optimization signal, with respective weights 0.40, 0.25, 0.15, and 0.20. Gold-label accuracy, balanced accuracy, macro-F1, and per-class recall remain the independent selection and reporting criteria. The judge runs deterministically with model reasoning disabled. Every judge call is persisted as JSONL with its component scores, normalized score, and free-text explanation, permitting a direct audit of feedback quality. Tables 2 and 3 remain static-reward cross-validation baselines; the completed LLM-judge final fit is reported separately below because its reused validation set is a prompt-development set rather than an outer-fold estimate.

Table 2. Aggregate selected-prompt outer-fold results over 2,789 units. “Balanced light” is the earlier 0.10 recall-gate run; the macro-F1 runs use a 0.05 gate. The six excluded responses were not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, and the rescaled κ/τ_b composite.

Selection / budget	Accuracy	Balanced acc.	Macro-F1	Composite
Balanced / light	0.6701	0.6753	0.6480	0.8031
Macro-F1 / light	0.6734	0.6796	0.6505	0.8063
Macro-F1 / heavy	0.6669	0.6737	0.6437	0.8030

Table 3. Per-class outer-fold performance for the best current configuration, macro-F1 selection with the light GEPA budget. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: class support, precision, recall, and F1.

Class	Support	Precision	Recall	F1
Read	318	0.798	0.770	0.784
Analyze	685	0.757	0.632	0.689
Plan	205	0.524	0.590	0.555
Implement	650	0.880	0.722	0.793
Explore	193	0.679	0.492	0.571
Verify	538	0.669	0.606	0.636
Monitor	200	0.364	0.945	0.526

Macro-F1 selection with the light budget is the strongest cross-validated configuration, but its advantage over balanced-accuracy selection is small: +0.32 accuracy points, +0.43 balanced-accuracy points, and +0.24 macro-F1 points. At the response level it wins on 15 documents, loses on 13, and ties on 4, so the observed gain is not stable across documents. The light run selects a GEPA-optimized prompt in three of five folds; the recall gate retains the seed prompt in the other two because Plan or Monitor recall falls too far.

Increasing the budget does not help. The heavy run selects an optimized prompt in only two folds and is 0.65 accuracy points and 0.68 macro-F1 points below the light run. Paired predictions show 38 units correct only under light versus 20 correct only under heavy. This difference is concentrated in fold 5, where light discovers and selects a strong candidate whereas heavy retains the seed after its candidate violates the Analyze recall gate. Independent heavy search is therefore not a superset of light search and can discard a better prompt found by a smaller run.

The revised objective improves overall balance only marginally and does not resolve Monitor overprediction. The model emits Monitor 519 times for 200 gold Monitor units; the leading errors are Verify→Monitor (127), Plan→Monitor (65), and Explore→Monitor (42). The annotation audit flags 180/3,125 units (5.76%), including 145 multiline units and 49 structural/control markers such as `</think>` and “Final Answer.” These artifacts plausibly inflate Monitor recall while depressing precision and should be reviewed before further optimization.

All runs retain 100% final-output coverage and complete without OOM failures, but the light and heavy jobs log 31 and 28 truncated LM responses during optimization. Moreover, the outer folds currently evaluate only the safety-selected prompt, not both seed and optimized prompts, so they do not directly estimate GEPA’s paired outer-fold gain. Finally, the six responses excluded from current CV were inspected in the preliminary experiments; they are locked from the current code path but no longer constitute a pristine research holdout. The judge should therefore not replace validated annotation until structural artifacts are adjudicated, decoding is constrained, paired outer evaluation is added, and a genuinely unseen external test set is designated.

Table 4. Context-aware LLM-judge final-fit results on the 407-unit prompt-development validation set. Coverage is 100% in both rows. The locked test is not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , Kendall’s τ_b , absolute change, and valid-output coverage.

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
21-example seed	0.6683	0.6765	0.6132	0.5988	0.5904
GEPA optimized	0.7543	0.7769	0.7270	0.6968	0.6832
Absolute change	+0.0860	+0.1003	+0.1138	+0.0980	+0.0929

Table 5. Per-class recall and optimized F1 for the completed LLM-judge final fit. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: class support, seed and optimized recall, and optimized per-class F1.

Class	Support	Seed recall	Opt. recall	Opt. F1
Read	45	0.778	0.756	0.791
Analyze	122	0.598	0.680	0.744
Plan	32	0.531	0.813	0.722
Implement	103	0.786	0.825	0.825
Explore	17	0.471	0.706	0.686
Verify	70	0.571	0.714	0.763
Monitor	18	1.000	0.944	0.557

Completed LLM-judge final fit. The August 11, 2026 run uses seed 42, the 21-example context-aware few-shot prompt, and the same-model informative reward on the original response-grouped 26/6/6 split (2,382/407/336 units). Seed and optimized prompts both return a valid label for all 407 validation units. Table 4 reports exact gold-label metrics rather than the LLM judge’s diagnostic score.

The optimized prompt fixes 49 seed errors and regresses on 14 previously correct units, a net gain of 35. All six validation responses improve, with response-level accuracy gains from 2.0 to 13.8 percentage points. Its primary effect is to curb the seed prompt’s overuse of Monitor: Monitor predictions fall from 74 to 43, including nine corrections from Monitor to true Plan and 15 from Monitor to true Verify. Recall rises most for Plan (+28.13 points), Explore (+23.53), Verify (+14.29), and Analyze (+8.20). Monitor recall falls slightly from 1.000 to 0.944, but precision rises from 0.243 to 0.395. The selected prompt still has weak Monitor precision and Analyze recall, and the remaining errors concentrate at the Analyze/Implement/Verify boundaries and Verify→Monitor. Implement is the only class whose F1 decreases, from 0.839 to 0.825, because its additional predictions include more false positives.

The internal same-model reward rises from 0.7264 for the seed to 0.8196 for the selected candidate. The optimization trace contains 145 candidates, 144 full validation evaluations, and 60,767 persisted judge calls, of which 148 are judge-output errors. A text audit finds 37 validation units for which the judge explicitly disputes the corpus gold label in at least one evaluation, showing that the reward itself detects annotation-sensitive boundaries rather than consistently treating gold as semantically authoritative. The structural audit flags 29 validation units; their accuracy remains 72.4% under both prompts, so the gain comes entirely from otherwise unflagged units.

GEPA expands the instruction from 21 to 40 demonstrations and from 7,648 to 32,717 characters. The extra rules and examples target observed corpus edge cases, raising inference cost and the risk of learning annotation

Table 6. Final Qwen3.8 model-alignment results on the 407-unit development validation set. Both prompts have 100% valid-output coverage; the 336-unit locked test is not evaluated. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL, used for both classification and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , and Kendall’s τ_b .

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
Imported 40-example seed	0.7174	0.7400	0.6938	0.6523	0.6122
GEPA optimized	0.7125	0.7288	0.6743	0.6475	0.6009
Absolute change	−0.0049	−0.0112	−0.0196	−0.0049	−0.0113

Table 7. Per-class recall in the final Qwen3.8 alignment. The selection gate rejects any candidate whose recall drops by more than 0.10 for any class. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL. *Metrics*: class support and seed versus GEPA-optimized recall.

Class	Support	Seed recall	Opt. recall
Read	45	0.778	0.756
Analyze	122	0.549	0.533
Plan	32	0.750	0.563
Implement	103	0.874	0.874
Explore	17	0.529	0.647
Verify	70	0.700	0.786
Monitor	18	1.000	0.944

conventions rather than a portable episode taxonomy. More importantly, GEPA evaluates 145 candidates against the same six-response validation set used for final selection. The nominal 407 units are also correlated within responses, and only one optimization seed is available. The result files explicitly record `locked_test_evaluated: false` and `test: null`. The observed gain is therefore strong evidence of successful prompt optimization on the development distribution, but not a held-out generalization estimate. The prompt should be frozen and compressed or ablated before a one-time test on genuinely unseen responses.

Final Qwen3.8 model-alignment pass. The August 27–28, 2026 alignment run uses Qwen3.8-27B UD-Q4_K_XL, seed 42, the heavy GEPA budget, and medium reasoning effort for the same-model judge. It keeps the same response-grouped 2,382/407/336 train/validation/locked-test split. Rather than drawing new few-shot examples, it imports the previous 40-example Qwen3.6 optimized prompt as its external seed. Gold-label agreement supplies 40% of the internal judge reward; functional fit, contextual coherence, and boundary precision supply the remaining 60%. Final selection uses exact balanced accuracy with a maximum allowed per-class recall drop of 0.10.

The optimized candidate changes 49 labels: it fixes 17 seed errors, breaks 19 seed-correct cases, and changes 13 errors to other errors. Its largest failure is Plan recall, which falls by 0.188 and therefore breaches the recall gate; the selected prompt is byte-identical to the imported seed. An unclustered exact McNemar test on the 17 versus 19 discordant correctness outcomes gives $p = 0.868$, offering no evidence of an aggregate improvement, although this unit-level calculation understates response clustering.

This negative exact-label result contrasts with the internal GEPA objective, whose best candidate reward rises from 0.7684 to 0.7762. Because only 40% of that reward is gold agreement, the discrepancy is consistent with optimizing the judge’s broader functional criteria without improving corpus-label classification. The optimized instruction is shorter (2,591 versus 5,201 words) but internally inconsistent: an early rule treats “Let me compute

Table 8. Accuracy-selected Schoenfeld boundary probes on the sentence-level test split. Prevalence is over all 3,087 units. Δ Maj. subtracts the always-negative accuracy. Balanced accuracy is derived from the saved test confusion matrix. Layer selection and evaluation use the same test split and are therefore selection-optimistic. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen hidden-state encoder, with one-versus-rest logistic probes. *Metrics*: target prevalence, selected hidden-state layer, accuracy, accuracy minus the majority baseline, balanced accuracy, F1, and AUROC.

Target	Prevalence	Layer	Accuracy	Δ Maj.	Balanced acc.	F1	AUROC
Read	12.21%	34	0.9239	+0.0461	0.8476	0.7044	0.9338
Analyze	25.62%	36	0.8107	+0.0669	0.7669	0.6465	0.8348
Plan	7.29%	39	0.9175	-0.0096	0.7405	0.4848	0.8551
Implement	21.87%	38	0.8964	+0.1151	0.8617	0.7714	0.9305
Explore	6.74%	40	0.9223	-0.0103	0.7486	0.4894	0.8884
Verify	20.38%	27	0.8754	+0.0792	0.8509	0.7260	0.9122
Monitor	5.90%	35	0.9304	-0.0106	0.7546	0.4819	0.9031

that” as Monitor while a later rule treats “Let me calculate” as Plan. Plan-to-Monitor errors rise from 6 to 10, consistent with that boundary confusion. The best internal candidate appears by iteration 3 and remains the best through iteration 19, so the remaining heavy search adds roughly 14.5 hours without a better candidate. The appropriate conclusion is that the final alignment completed correctly and the safety gate worked: GEPA found no safe improvement over the existing seed. The complete artifact is stored at `results/gepaLLMAsJudge/qwen3.8-27b-medium-final-s42-v3`.

4 probeTest

The probeTest workflow tests a different annotation mechanism from the generative judge in gepaLLMAsJudge. It asks whether Qwen’s representation of the prefix immediately before a sentence contains linearly accessible information about the upcoming Schoenfeld episode. It therefore measures prospective prefix-state decodability, not semantic classification after reading the sentence and not the latent dynamics of a Qwen-generated trace.

Within-corpus results. Table 8 reports the accuracy-selected layer for each one-versus-rest target. AUROC is 0.835–0.934 and balanced accuracy is 0.740–0.862, demonstrating substantial within-split ranking signal. Implement is strongest at the natural threshold (F1 0.771), followed by Verify (0.726) and Read (0.704). Analyze is intermediate. Plan, Explore, and Monitor have F1 below 0.50. Their nominal accuracies are also 0.96–1.06 percentage points below an always-negative majority classifier, so accuracy alone gives a misleading impression for these rare targets.

The signal emerges rapidly and is not maximal at the model output. Mean one-versus-rest AUROC across targets rises from 0.624 at the embedding output to 0.873 at layer 10 and 0.892 at layer 20, peaks at 0.899 at layer 30, and falls to 0.867 at layer 40. Mean F1 peaks at layer 28. Accuracy-based target selection sometimes obscures this profile: for Plan, the selected layer 39 has AUROC 0.855 whereas layer 21 reaches 0.922. AUROC or balanced accuracy, chosen inside a validation fold, better matches the scientific question of decodability.

The fits are numerically stable—none of the 287 regressions reports a convergence warning—but statistical overfitting remains visible. At the selected layers, train–test accuracy gaps are 16.7 points for Analyze, 10.1 for Implement, approximately 7.8 for Plan and Explore, and 4.5–5.5 for the other targets. Explore reaches 100% training accuracy. With 2,048 features, 2,469 training units, strong within-response dependence, and only 38

Table 9. Argmax inspection-label counts on external reasoning units. These are raw prediction frequencies, not accuracies. Table 10 compares them with the subsequent provisional pragmatic review. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit counts and seven-way argmax prediction counts.

Dataset	Units	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
GSM8K	96	0	2	3	53	0	0	38
MATH	88	23	7	1	45	0	1	11
PRM800K	139	34	18	0	71	0	1	15
ProcessBench	258	69	18	12	117	0	17	25
Total	581	126	45	16	286	0	19	89

distinct responses, the effective independent sample size is much closer to the number of responses than to the number of sentences. Test-based layer selection adds further optimism.

External benchmark inspection. All 80 requested examples complete without omitted units, yielding 581 sentence boundaries. Table 9 shows that the seven-way inspection behavior collapses mainly to Implement, Read, and Monitor. Implement alone receives 49.2% of argmax assignments, whereas Explore is never selected. The transfer distribution also differs sharply from training: Analyze falls from 25.6% of gold units to 7.7% of predictions and Verify from 20.4% to 3.3%.

The independent binary scores do not form a clean multiclass decision rule. Seventy-four units (12.7%) have no score above 0.5 but are still assigned an argmax label, while 180 (31.0%) activate more than one probe and 99 (17.0%) have a top-two margin below 0.1. A conservative display rule requiring a top score of at least 0.8 and a margin of at least 0.2 retains only 318/581 units (54.7%), mostly Implement, Read, and Monitor. This threshold is a useful abstention heuristic, not a calibrated guarantee of correctness.

Provisional pragmatic adjudication. To make the qualitative inspection measurable, every displayed unit was subsequently reviewed using the problem, the current unit, its neighbors, and the adapted Schoenfeld precedence rule. The annotation is a hybrid rule-assisted Codex pragmatic review, not human consensus or authoritative benchmark gold. It records one primary label for conventional single-label statistics and, where two functions are genuinely defensible, an accepted set containing both labels. This occurs for 187/581 units (32.2%), mostly at boundaries where a unit both announces and executes an operation or where a structural marker also carries a direct result. Of the annotations, 394 have high, 150 medium, and 37 low confidence; the low-confidence cases remain priority items for human review.

The probe agrees with the primary pragmatic choice on 292/581 units (50.3%) and with at least one accepted label on 353/581 (60.8%). Thus recognizing genuine taxonomy overlap recovers 61 additional compatible decisions, but it does not make the reference authoritative. ProcessBench has the strongest compatible agreement (68.6%), followed by GSM8K (62.5%); MATH has the lowest strict agreement (40.9%). Table 10 reports the complete dataset-level results.

The class-level comparison in Table 11 sharpens the interpretation. Implement is the only operationally strong external label (precision 0.804, recall 0.667, F1 0.729). Read is moderately useful (F1 0.524). Analyze and Plan are severely under-produced, while Verify and Monitor have low recall. Explore has 11 accepted reference memberships but is never predicted, hence zero recall. Because multi-label units contribute to two reference classes, accepted supports overlap and do not sum to 581.

Table 10. Agreement with the provisional pragmatic review. Primary accuracy requires the preferred label; compatible accuracy accepts either label on a multi-label unit. Macro-F1 and macro recall treat each accepted label as a positive reference membership. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit count, primary accuracy, compatible accuracy, multi-label count, accepted-membership macro-F1, and macro recall.

Dataset	Units	Primary acc.	Compatible acc.	Multi-label	Macro-F1	Macro recall
GSM8K	96	49.0%	62.5%	20	0.136	0.129
MATH	88	40.9%	53.4%	36	0.211	0.228
PRM800K	139	46.0%	49.6%	44	0.241	0.302
ProcessBench	258	56.2%	68.6%	87	0.396	0.350
Total	581	50.3%	60.8%	187	0.330	0.306

Table 11. Per-label comparison with accepted pragmatic memberships. Support counts every unit for which the label is accepted, including secondary labels on ambiguous units. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: accepted-label support, argmax prediction count, precision, recall, and F1.

Label	Accepted support	Probe predictions	Precision	Recall	F1
Read	84	126	0.437	0.655	0.524
Analyze	84	45	0.356	0.190	0.248
Plan	90	16	0.375	0.067	0.113
Implement	345	286	0.804	0.667	0.729
Explore	11	0	0.000	0.000	0.000
Verify	51	19	0.632	0.235	0.343
Monitor	103	89	0.382	0.330	0.354

Pragmatically, Implement is the most coherent transferred label: it marks long calculation, substitution, and algebra runs. Read often marks givens or restatements, particularly near the start of MATH and PRM800K solutions. Monitor behaves primarily as a solution-boundary or opening-state detector: all 20 GSM8K first units and 18/20 ProcessBench first units receive Monitor, whereas 19/20 MATH and 17/20 PRM800K first units receive Read. Because the target sentence is unseen at this point, this source dependence must arise from the problem, prefix position, and formatting rather than the displayed unit’s semantics. Analyze and Plan appear plausibly on some explanatory or phase-change boundaries but have low coverage. Explore is unusable for external tagging in the present model.

Verify has a narrower but potentially useful role as a review or reconsideration flag. All 17 ProcessBench Verify predictions occur at or after the benchmark’s annotated first error, including six within one trace that repeatedly reinterprets the question before ending with the wrong sign. This is not an error detector: errors generally occur early, Verify may merely mark late position or reflective language, and no PRM800K post-error unit is labelled Verify. It can prioritize passages for inspection but cannot judge correctness.

The qualitative stage also exposes an input problem independent of the classifier. Thirteen units are split immediately after abbreviations such as “Mr.”, producing fragments including “Therefore, Mr.” and inconsistent scores. External use should first adopt abbreviation- and LaTeX-aware segmentation. With the current outputs, the defensible practical representation is a coarse three-phase view—setup (Read and opening Monitor), execution

Table 12. Prespecified measures for the expanded correlation analysis.

Analysis	Unit	Measure	Reason
Continuous feature versus correctness or a binary reasoning-event indicator	Generation trace	Point-biserial r_{pb}	The predictor is continuous and the outcome is binary; the signed coefficient is exactly Pearson's r after coding the outcome as 0/1.
Mean feature versus avg@ N correctness	Source problem	Spearman's ρ	Avg@ N is bounded, often tied, and not assumed to have a linear interval relationship with a feature. Ranking tests monotone association and one row per problem avoids treating repeated attempts as independent.
Continuous feature–feature and continuous expert-rate associations	Trace or source problem	Spearman's ρ	These exploratory scans ask whether quantities vary monotonically without assuming comparable scales or a linear relationship.
Correct-versus-incorrect feature difference within a repeated problem	Mixed-outcome source problem	Mean paired contrast	This directly controls for the problem by forming one correct-minus-incorrect contrast per problem; uncertainty is obtained by resampling problems.

(Analyze, Plan, and Implement), and review (Verify)—plus an explicit unknown/ambiguous state. The seven raw argmax labels should not be used as automatic cognitive annotations.

5 Qwen Correlation Pipeline

5.1 Statistical measures and literature basis

The corrected analysis selects the association measure from the scale of the outcome and the unit of analysis, as summarized in Table 12. The numerical results below use the September 10, 2026 vLLM untagged analysis snapshot and supersede the earlier tables. Generation/replay coverage and full-reasoning/position analysis are complete for the six executed math benchmarks, but the token-budget audit still flags frequent truncation. Consequently, these are fixed-budget observational results, not uncensored reasoning-length estimates or evidence that a routing intervention improves correctness.

For a continuous feature X and binary target $Y \in \{0, 1\}$, the first coefficient is

$$r_{pb} = \text{Corr}(X, Y). \quad (1)$$

Thus positive values mean that the feature tends to be higher for correct (or event-present) traces, and negative values mean that it tends to be lower. Liu et al. directly use point-biserial correlation for continuous language-model quantities against binary answer correctness, making it the closest precedent for this case [3].

For repeated-attempt datasets, correctness and each predictor are first aggregated once per source problem. Only groups containing exactly the declared avg@ N unique sample identifiers, no duplicate generations detected by the saved fingerprints, and a complete set of binary outcomes enter the primary Spearman analysis. Spearman correlation is appropriate here because its target is monotone rank association rather than a specifically linear change in avg@ N . Hong et al. correlate a scalar projection of residual-stream states with graded reasoning scores using Spearman correlation [2]; Xu et al. use Spearman correlation between language-model-derived conceptual quantities and human ratings [4]; and Hewitt and Manning use it to compare geometry derived from contextual representations with gold syntactic structure [1]. These studies are analogous precedents for rank correlation with representation-derived quantities; they do not by themselves establish Spearman as a unique

Table 13. Coverage and fixed-budget scores for the September 10 run. Accuracy uses all attempts, including unparseable answers as incorrect: pass@1 for single attempts and avg@32 for AIME and AMC. The pooled row is descriptive, not a benchmark macro-average. Lexical columns give trace percentages. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: problem/trace counts, all-attempt accuracy, limit hits, and lexical-event prevalence.

Dataset	Problems	Traces	Correct	Accuracy	Limit hits	Backtrack%	Contradict%	Self-correct%
MATH-500	500	500	344	0.688	20 (4.0%)	25.4	14.6	16.6
AIME 2024	30	960	820	0.854	106 (11.0%)	72.2	49.6	58.2
AIME 2025	30	960	789	0.822	144 (15.0%)	73.6	46.6	63.6
OlympiadBench	675	675	323	0.479	94 (13.9%)	51.1	39.0	41.6
AMC 2023	40	1,280	1,240	0.969	12 (0.9%)	41.9	33.8	30.1
Minerva Math	272	272	72	0.265	18 (6.6%)	38.6	18.4	29.8
Pooled	1,547	4,647	3,588	0.772	394 (8.5%)	54.1	37.5	43.0

choice for avg@32. Zhang et al. additionally report Pearson and Spearman correlations together when relating model uncertainty, including entropy-based measures, to graded human hallucination judgments [5].

Uncertainty is reported with 95% percentile bootstrap intervals that resample whole source problems, keeping attempts from the same problem together. Xu et al. provide relevant precedent for bootstrap intervals and use 10,000 resamples in one category–feature analysis [4]; the resample count in this pipeline remains explicit and configurable rather than being justified by that number alone. Benjamini–Hochberg q -values are computed separately within each declared dataset/scope, target, aggregation, or cross-feature family. Naive trace-level p -values are retained only as diagnostics when a dataset contains repeated attempts, and layerwise and expert-identity screens remain exploratory.

5.2 Completed vLLM replay and fixed-budget audit

The September 10, 2026 analysis of reasoning-vllm-untagged-v1 contains 4,647 attempts from 1,547 source problems across six math benchmarks. It supersedes the September 8 numerical snapshot in this report. GPQA-Diamond and MMLU-Pro are absent from this run and from every total below; GPQA results from the earlier run are not pooled into this snapshot. All 4,647 generation texts match their saved replay continuations and analysis SHA-256 fingerprints. Every repeated group is complete: 30 AIME 2024, 30 AIME 2025, and 40 AMC 2023 problems each have 32 distinct attempts, with no duplicate sample IDs or exact duplicate generations. The other three datasets use one attempt per problem.

Generation uses vLLM serving Qwen/Qwen3.5-35B-A3B-GPTQ-Int4, at temperature 0.6 and top- p 0.95 with a 32,768-token completion budget. Saved messages and continuations are replayed through unsloth/Qwen3.5-35B-A3B with runtime 4-bit quantization. The generation and replay checkpoints share the model family but use different quantization formats and engines. Replay computes compact features at decoder indices 27, 34, 35, 36, 38, and 39, selected using the separate probeTest run. Hidden features describe block inputs, not the exact post-attention router inputs. Router confidence is $1 - H(p)/\log 256$, where p is the softmax router distribution over all experts. The main analysis tests 83 scalar features; entropy is omitted from separate testing because it is an affine transform of confidence. Aggregate-feature uncertainty uses 500 problem-bootstrap replicates and seed 42. The additional views cover full reasoning and absolute token positions; no sentence classifier or Schoenfeld class views were used in this untagged run.

Correctness denominator and comparability. All-attempt accuracy is $3,588/4,647 = 0.772$. The scoring contract counts 371 unparseable answers with known gold answers as incorrect. Generation manifests instead condition on scored answers: MATH-500 reports $344/480 = 0.717$, whereas its all-attempt score is $344/500 = 0.688$. Table 13 uses

Table 14. Marginal correctness correlations over the entire saved continuation. Router/hidden quantities are token- and selected-layer means. Bold marks a saved 95% problem-cluster interval excluding zero; intervals are pointwise and are not length/termination-adjusted effects or multiplicity-adjusted discoveries. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: point-biserial correctness correlations.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.322	−0.711	−0.695	−0.374	−0.152	−0.290
Router confidence	0.259	0.523	0.442	0.216	0.055	0.079
Top-1/top-2 margin	0.188	0.536	0.424	0.183	0.204	0.031
Top-8 probability mass	0.223	0.203	0.244	0.135	−0.023	0.054
Top-8/top-9 margin	0.152	0.010	0.117	0.065	−0.111	0.010
Hidden-state norm	0.249	0.024	−0.060	0.074	0.463	0.030
Hidden-state step distance	0.214	0.385	0.352	0.090	0.479	0.076
Hidden/router geometry	−0.189	−0.198	−0.250	−0.107	−0.152	−0.156
Top-1 switch rate	−0.028	−0.287	0.169	0.047	−0.063	0.033
Adjacent top-8 overlap	0.049	0.018	−0.234	−0.032	−0.134	−0.056

the latter denominator throughout. The independent audit verifies the stored outcomes and their propagation; it does not independently rejudge mathematical answers. Differences from the earlier GGUF run cannot be attributed to the serving engine alone: the generated samples, quantization, and benchmark composition differ, and there is no controlled engine comparison here.

Completion and truncation. The analysis and both requested view families are complete, but the budget audit remains frequent_limit_hits. There are 394 limited attempts (8.5%), including 370 without final assistant content. Limited attempts account for 393 of 1,059 errors (37.1%); the remaining 666 errors occur among non-limit traces. Accuracy is $1/394 = 0.003$ among limited traces and $3,587/4,253 = 0.843$ among non-limit traces. Token-limit failures therefore remain a substantial confound, although they no longer constitute most errors in the pooled sample. Their contribution differs sharply by benchmark: 106/140 AIME 2024 errors and 144/171 AIME 2025 errors are limited, compared with 20/156 MATH-500 errors and 12/40 AMC errors.

5.3 Trace-level correctness associations

Appendix B provides the complete aggregate reasoning-only correctness table, all aggregate position metrics, and every metric pair in the saved trace-level scans for the continuation, full reasoning, and position windows. The tables below retain the selected results for discussion.

Token count is negatively associated with correctness in all six datasets ($r_{pb} = -0.152$ to -0.711), with all six pointwise cluster intervals excluding zero. Confidence and top-1/top-2 margin are positive with intervals excluding zero on MATH-500, both AIME sets, and OlympiadBench. Their AMC and Minerva intervals include zero. Thus the previous seven-dataset claim of uniformly supported positive confidence does not describe this run. Top-8 mass follows the same four-dataset pattern, whereas the top-8/top-9 boundary margin has an interval excluding zero only on MATH-500.

Mean hidden-state step distance is positive in every dataset, with intervals excluding zero in five; Minerva remains uncertain. Hidden norm is positive with intervals excluding zero on MATH-500, OlympiadBench, and AMC, although Olympiad’s lower endpoint is only 0.0014. Hidden/router geometry is negative with intervals excluding zero in five datasets, with AMC uncertain. Switch rates and adjacent top-8 overlap remain heterogeneous: only AIME 2024’s switch-rate interval excludes zero, and no overlap interval does. These are marginal associations, not independently adjusted predictors or evidence that changing a hidden-state or routing quantity changes correctness.

Table 15. Descriptive sensitivity after excluding token-limit hits. Conditioning on termination changes the sample and breaks complete avg@32 groups. No confidence intervals or significance claims are attached to these post-hoc coefficients. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: retained trace/error counts and post-hoc point-biserial coefficients.

Dataset	Non-limit traces	Incorrect	Token count	Confidence	Margin
MATH-500	480	136	-0.150	0.009	-0.131
AIME 2024	854	34	-0.351	0.233	0.248
AIME 2025	816	27	-0.294	0.207	0.190
OlympiadBench	581	258	-0.150	-0.012	-0.068
AMC 2023	1,268	28	0.107	-0.352	-0.249
Minerva Math	254	183	-0.321	-0.015	-0.093

Table 16. Mean feature versus per-problem avg@32. The saved 95% intervals resample whole problems; BH q is adjusted within each dataset’s 83-feature mean-feature family. All 100 repeated groups are complete. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: problem-level Spearman correlation, bootstrap intervals, and BH adjustment.

Dataset	Mean feature	Problems	ρ	95% CI	q
AIME 2024	Token count	30	-0.791	[-0.916, -0.587]	1.62×10^{-5}
	Confidence	30	0.606	[0.314, 0.809]	0.009
	Margin	30	0.560	[0.226, 0.768]	0.013
AIME 2025	Token count	30	-0.892	[-0.958, -0.776]	3.09×10^{-9}
	Confidence	30	0.647	[0.390, 0.827]	0.002
	Margin	30	0.533	[0.228, 0.752]	0.015
AMC 2023	Token count	40	0.041	[-0.263, 0.328]	0.940
	Confidence	40	-0.255	[-0.599, 0.206]	0.372
	Margin	40	-0.067	[-0.429, 0.282]	0.866

5.4 Sensitivity to token-limit hits

Excluding limited traces leaves moderate positive confidence correlations on AIME 2024 (0.233) and AIME 2025 (0.207), but approximately zero on MATH-500 (0.009), OlympiadBench (-0.012), and Minerva (-0.015). AMC changes from +0.055 in the full sample to -0.352 among non-limit traces; its margin also changes from +0.204 to -0.250, while length changes from -0.152 to +0.108. These sign changes are descriptive diagnostics, not significance-tested differences between coefficients. Conditioning on termination changes the population and can induce selection bias. The check is not substituted for all-attempt benchmark scores or complete-group avg@32 analyses, and it does not establish termination-independent prediction.

5.5 Problem-level and within-problem analyses

At the problem level, both AIME datasets show negative length and positive confidence and margin associations under both the pointwise-interval and within-family $q < 0.05$ criteria. Confidence gives $\rho = 0.606$ on AIME 2024 ($q = 0.009$) and 0.647 on AIME 2025 ($q = 0.002$). AMC does not reproduce this pattern: confidence is -0.255, margin -0.067, and length +0.041, with all three intervals crossing zero. Its near-ceiling accuracy limits the rank analysis: 34 of 40 problems have no incorrect attempt. Problem-level difficulty associations and attempt-to-attempt variation answer different questions and need not have matching signs.

Table 17. Mean correct-minus-incorrect difference within mixed-outcome problems. Intervals resample those problems; confidence and margin differences are scaled by 10^3 . No length or termination adjustment is applied. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: paired feature differences and problem-bootstrap intervals.

Dataset	Feature	Mixed problems	Difference	95% CI
AIME 2024	Token count	12	−9, 589	[−13, 335, −5, 654]
	$10^3 \times$ confidence	12	2.883	[0.749, 5.316]
	$10^3 \times$ margin	12	0.475	[0.097, 0.928]
AIME 2025	Token count	14	−8, 251	[−11, 443, −5, 435]
	$10^3 \times$ confidence	14	2.313	[0.675, 5.053]
	$10^3 \times$ margin	14	0.536	[0.195, 1.064]
AMC 2023	Token count	6	−16, 563	[−25, 460, −8, 016]
	$10^3 \times$ confidence	6	9.799	[4.159, 15.323]
	$10^3 \times$ margin	6	2.103	[0.940, 3.198]

Table 18. Selected metric-pair associations over entire continuations. These are descriptive Spearman coefficients from the 13-feature scan (78 pairs per scope). Dependence among predictors limits interpretations based on several correlated summaries. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: trace-level feature–feature Spearman correlations.

Metric pair	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Length / confidence	−0.707	−0.668	−0.824	−0.868	−0.849	−0.303
Confidence / margin	0.629	0.688	0.805	0.801	0.641	0.633
Top-8 mass / mean gap	1.000	1.000	1.000	1.000	1.000	1.000

All nine displayed within-problem intervals exclude zero: correct attempts are shorter and have higher confidence and margin. These comparisons use only 12, 14, and 6 mixed-outcome problems on AIME 2024, AIME 2025, and AMC, respectively. Correct attempts are 8,251–16,563 tokens shorter on average. Matching question identity therefore leaves major differences in length, termination, and token content. AMC illustrates the distinction particularly clearly: its positive paired confidence difference coexists with a negative non-limit trace coefficient and an uncertain negative problem-level rank coefficient. None of these analyses alone identifies a causal routing benefit.

5.6 Metric-pair and expert-rate screens

Confidence and token count have negative Spearman associations in every dataset ($\rho = -0.303$ to -0.868). These summaries cannot be treated as independent evidence. Top-8 mass and the top- k /non-top- k mean probability gap have effectively unit rank correlation. For selected mass S , $k = 8$, and $E = 256$, this dependence is algebraic:

$$G = \frac{S}{k} - \frac{1-S}{E-k} = \frac{E}{k(E-k)}S - \frac{1}{E-k}. \quad (2)$$

The relation is preserved by token/layer averaging, so both quantities should not be included together in an unregularized model or counted as independent corroboration. The main metric-pair scan covers 13 aggregate features and 78 pairs per scope, with separate trace and complete-repeated-problem tables.

The expert-rate screen covers all 256 numeric expert IDs and retains 128 of 65,234 eligible ordered top-1/top-2 pairs. Selection uses routing support alone, requiring at least four traces. There are 640 features: 256 top-1 rates, 256 any-top-8 rates, and 128 ordered-pair rates.

Table 19. Post-hoc largest absolute problem-level coefficient per dataset in the 640-feature expert-rate screen. BH adjusts over 640 features per dataset. Pairs describe top-1/top-2 IDs in the same routing decision, not temporal transitions. Rates pool six layers; numeric IDs do not identify individual layer-specific modules. No bootstrap intervals or held-out selection validation are available. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: problem-level expert-rate Spearman maxima and within-dataset BH adjustment.

Dataset	Pooled-ID rate	Problems	ρ	q
AIME 2024	Top-1 204 / top-2 242	30	0.766	5.10×10^{-4}
AIME 2025	Top-1 200 / top-2 58	30	0.812	3.38×10^{-5}
AMC 2023	ID 242 in top-8	40	-0.551	0.086

The strongest AIME problem-level candidates survive their dataset’s BH adjustment, but AMC’s largest absolute coefficient does not ($q = 0.086$). The 200/58 pair has $\rho = 0.723$ on AIME 2024 and 0.812 on AIME 2025, but -0.347 on AMC ($q = 0.207$). Rates pool decisions across the six selected layers, where the same numeric ID refers to different parameterized experts. The screen therefore does not identify a portable individual expert or a validated intervention target. Layer-qualified (ℓ, e) rates, length/termination controls, independent selection and evaluation, and actual interventions are needed before assigning functional roles. Trace-level expert p - and q -values do not repair repeated-attempt dependence.

5.7 Full-reasoning and position views

The new views isolate saved reasoning content from the final answer and then partition reasoning tokens into fixed absolute-position windows. Their shared reference is the mean of all 4,647 traces: 12,477.026 reasoning tokens. Window boundaries are $\lceil i \cdot 12,477.026 / 10 \rceil$ for $i = 0, \dots, 10$; the first covers zero-based offsets $[0, 1,248)$, and overflow starts at 12,478. These windows are disjoint, not cumulative prefixes or per-trace relative deciles. Every original attempt and its correctness are retained; an unreached window has missing routing features rather than zero values. Position-token count is excluded from the correlation predictors. No Schoenfeld episode labels or class-specific findings are available.

Full-reasoning confidence coefficients remain close to the entire-continuation coefficients, but the fixed-window pattern differs substantially. In the first window, confidence is negative on AIME 2024 (-0.153), AIME 2025 (-0.209), and AMC (-0.361), despite positive full-reasoning point estimates. AIME coefficients turn positive in later windows; AMC reaches 0.724 in bin 9 and 0.830 in overflow. These observations show that a whole-trace mean hides strong position and sample-composition differences. They do not show that confidence becomes causally beneficial later in reasoning.

Coverage falls from 4,647 traces in bin 0 to 2,049 in bin 9 and 1,825 in overflow. AMC, for example, falls from 1,280 to 317 and 238; MATH-500 falls from 500 to 65 and 54. Late estimates condition on reaching those positions, and the full-budget termination confound remains. Cross-view comparisons are exploratory; neither the per-view BH families nor the saved pointwise bootstrap intervals adjust for selecting a sign change across windows. A common-cohort analysis, explicit length/termination controls, and problem-respecting validation are required before interpreting this as a within-trace temporal effect.

5.8 Lexical events, storage, and implications

The lexical detectors flag 2,513 backtracking, 1,741 contradiction, and 2,000 self-correction traces. Their pooled point-biserial associations with token count are 0.566, 0.425, and 0.509, respectively, with positive point estimates in every dataset. Backtracking has negative confidence coefficients in all six datasets, while self-correction is negative in five and slightly positive on Minerva (0.048). These are surface-phrase detectors applied to whole

Table 20. Confidence versus whole-attempt correctness in full reasoning and fixed absolute-position windows. Cells give r_{pb} (finite traces). Ten disjoint windows divide the corpus mean of 12,477.026 reasoning tokens; overflow starts at token offset 12,478. These are not per-trace relative deciles. Late-window samples contain only traces reaching those positions; cross-window sign changes are exploratory and not adjusted temporal effects. *Model(s)*: Qwen3.5-35B-A3B, GPTQ-Int4 generation and runtime-4-bit replay. *Metrics*: position-specific confidence–correctness point-biserial correlations and finite trace counts.

View	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Full reasoning	0.255 (500)	0.518 (960)	0.435 (960)	0.208 (675)	0.044 (1,280)	0.112 (272)
Bin 0	−0.030 (500)	−0.153 (960)	−0.209 (960)	0.043 (675)	−0.361 (1,280)	−0.020 (272)
Bin 1	−0.050 (429)	−0.148 (960)	−0.093 (960)	0.070 (668)	−0.421 (1,269)	0.070 (269)
Bin 2	−0.157 (347)	−0.033 (960)	0.067 (960)	0.023 (634)	−0.423 (1,172)	0.108 (254)
Bin 3	−0.118 (268)	0.072 (956)	0.113 (942)	0.024 (564)	−0.391 (1,038)	−0.004 (234)
Bin 4	0.020 (204)	0.137 (927)	0.150 (900)	−0.016 (494)	−0.190 (876)	−0.066 (202)
Bin 5	0.122 (171)	0.183 (859)	0.149 (847)	0.017 (427)	0.081 (767)	0.011 (162)
Bin 6	0.246 (139)	0.184 (816)	0.122 (803)	−0.003 (384)	0.363 (684)	0.075 (133)
Bin 7	0.385 (106)	0.206 (748)	0.118 (752)	0.093 (342)	0.454 (559)	0.077 (102)
Bin 8	0.467 (83)	0.218 (679)	0.157 (724)	0.121 (314)	0.589 (436)	0.093 (83)
Bin 9	0.392 (65)	0.265 (605)	0.185 (701)	0.152 (291)	0.724 (317)	0.137 (70)
Overflow	0.539 (54)	0.474 (540)	0.376 (664)	0.195 (270)	0.830 (238)	0.237 (59)

attempts, not independently validated reasoning episodes; the view analyses reuse those whole-attempt flags rather than detecting local events.

Replay retains 61,964,775 continuation tokens. The projected selected-layer raw payload is 1.904 TB (1.731 TiB), while persisted compact tensor artifacts total 5.958 GB and per-dataset files total 8.047 GB. No raw router/hidden tensors or expert weights are saved. Compact tensor bytes are approximately 0.313% of the projected raw payload; the expanded reasoning-view summaries are also retained in the dataset files.

For *Guiding MoE Reasoning via Routing Intervention*, this run supplies AIME confidence/margin candidates, exploratory pooled expert-ID associations, and a position-dependent pattern requiring controlled follow-up. Priorities are to prespecify how to handle remaining token-limit failures, validate routing features beyond length and benchmark on held-out problems, and compare position windows on common cohorts. GPQA-Diamond and MMLU-Pro require separate runs before broader coverage claims. Neither full-trace correlations nor expert-screen maxima establish that increasing confidence or forcing an expert improves reasoning.

6 Analysis

6.1 Conclusions supported so far

- (1) The static-reward, context-aware GEPA study reaches 0.650 cross-validated macro-F1 with the light budget. The advantage over balanced-accuracy selection is small and the heavy search is worse. In the separate completed same-model LLM-judge final fit, development-set balanced accuracy rises from 0.677 to 0.777 and macro-F1 from 0.613 to 0.727, primarily through reduced Monitor overprediction. This establishes successful prompt optimization on the development distribution, not held-out generalization. The final Qwen3.8-27B alignment does not reproduce that gain: balanced accuracy falls from 0.740 for the imported seed to 0.729 for the GEPA candidate, and the prespecified recall gate correctly retains the seed.
- (2) The boundary probes show substantial sentence-split decodability of upcoming Schoenfeld episodes, with selected-layer AUROC of 0.835–0.934 and the strongest thresholded performance for Implement, Verify, and Read. This establishes accessible prefix-state signal within familiar responses, not generalization to unseen reasoning traces. External application is best interpreted as coarse phase inspection: a provisional

pragmatic review gives 50.3% strict and 60.8% compatible agreement, with 187/581 multi-label units. Implement is strongest (F1 0.729), while the remaining labels are incomplete external tags. ProcessBench Verify predictions identify review-heavy post-error passages but do not identify correctness.

- (3) The September 10 vLLM correlation analysis completes 4,647 attempts from 1,547 problems on six math benchmarks. All-attempt accuracy is $3,588/4,647 = 0.772$, with 371 unparseable answers counted as incorrect. Confidence has positive correctness associations with pointwise intervals excluding zero on MATH-500, both AIME sets, and OlympiadBench; AMC and Minerva remain uncertain. The 394 token-limit hits account for 393 of 1,059 errors. Among non-limit traces, confidence remains moderately positive on AIME (0.233 and 0.207), is near zero on MATH-500, OlympiadBench, and Minerva, and is negative on AMC (−0.352). Complete-problem confidence associations survive BH adjustment on both AIME datasets but not AMC. Correct attempts within mixed-outcome problems have higher confidence yet are 8,251–16,563 tokens shorter. New fixed-position views show early negative and later positive confidence coefficients on AIME and AMC, while coverage shrinks. Expert-rate maxima remain exploratory pooled-ID candidates.

6.2 What has not been established

Neither GEPA final fit provides an unbiased test estimate. The earlier run selects among 145 candidates on the same six-response validation set, has only one optimization seed, and leaves the locked 336-unit test unevaluated. The result does not establish that the expanded 40-demonstration prompt transfers to new responses or that the same-model reward is an independent evaluator. The Qwen3.8 alignment reuses that selected prompt as its seed, optimizes and selects on the same validation partition, and also leaves the locked test untouched; retaining the seed is evidence that this search found no safe development-set improvement, not evidence of generalization.

The boundary-probe run does not establish response-grouped generalization, calibrated seven-way episode classification, or the cognitive states of Qwen’s own generations. It uses Qwen to encode teacher-forced DeepSeek-R1 text, shares all 38 response IDs between train and test, and selects layers on the reported test set. Its external argmax outputs have no authoritative human or benchmark episode ground truth. Agreement with the provisional hybrid review is diagnostic and should not be called validated transfer accuracy.

The correlation results do not establish a dataset-invariant routing predictor or a causal mechanism. No length-conditioned or dataset-adjusted model has been fit, and the 32,768-token budget censors 8.5% of traces. Repeated-attempt comparisons control question identity but not length, termination, or token content. The current run excludes GPQA-Diamond and MMLU-Pro. Raw pooled coefficients mix datasets with different accuracy, length, and routing baselines. Non-limit sensitivity checks select on termination; late-position views select on reaching the window. Neither is an unbiased correction for these confounds. The untagged views contain no Schoenfeld class evidence. Layerwise and expert maxima are selected on the analyzed data, and numeric expert IDs are pooled across distinct layer-local modules. Family-specific BH adjustment does not supply held-out validation or correct selection across position views. Across all three workflows, correlation and linear decodability should not be read as causal evidence.

7 Discussion

The completed GEPA final fit demonstrates that same-model informative feedback can guide a large prompt toward the development labels: it fixes 49 seed errors while introducing 14 regressions and improves every validation response. Its optimization trace also shows the central risk. The prompt expands from 21 to 40 demonstrations and is selected after 145 candidate evaluations on those same six responses. The next defensible test is therefore to freeze and ablate or compress the prompt, quantify response-level uncertainty, and evaluate it once on genuinely unseen responses.

The Qwen3.8 alignment is an important negative result. Although its internal same-model reward increases, exact corpus-label metrics decline and Plan recall breaches the prespecified 0.10 safety threshold. Its shorter optimized instruction also contains conflicting guidance for phrases such as “Let me compute that,” plausibly contributing to more Plan-to-Monitor errors. The safety gate therefore performs its intended role: optimization reward alone is insufficient evidence for deployment, and the byte-identical selected seed should now be frozen rather than optimized further on this validation set.

The episode probes add a complementary representation-level result: the prefix before a sentence often encodes the phase likely to follow, and the signal is strongest in middle-to-late layers rather than at the final model output. The current split cannot determine whether this reflects a portable episode representation, trace-specific transition regularities, position, or formatting. A response-grouped nested evaluation should select both layer and regularization on inner folds, evaluate once on unseen responses, repeat over seeds, and report response-bootstrap intervals. Standardized or norm-matched features and simple position, previous-unit, and formatting baselines are needed to separate hidden-state information from layer scale and trace structure.

For external use, the probe outputs should be treated as abstaining phase suggestions rather than annotations. Abbreviation- and LaTeX-aware segmentation should precede evaluation; independent one-versus-rest scores should either be calibrated on a common multiclass validation set or retain multiple labels. The pragmatic review confirms substantial ambiguity and only 60.8% compatible agreement. Independent human annotators should resolve the 37 low-confidence units and assess inter-annotator agreement before any transfer-accuracy claim. Cross-model recognition, as studied here, should also be separated from Qwen’s own generation by running the protocol on Qwen-generated traces.

The new replay narrows the correlation interpretation. Negative length associations recur across all six datasets, but supported positive confidence associations occur in four, and the non-limit check leaves moderate positive coefficients only on AIME. AMC has a negative non-limit confidence coefficient, a positive within-question difference, and an uncertain negative problem-level coefficient. These results concern different populations and aggregation levels. Fixed-position windows further reveal early negative and later positive coefficients while progressively excluding shorter traces. A whole-trace confidence mean therefore cannot be interpreted as a uniform signal throughout reasoning.

The next analysis should prespecify treatment of budget failures, jointly examine length, termination, token composition, and dataset, compare windows on common cohorts, and validate selected predictors on held-out problems. Simply discarding limited traces or comparing late windows to early windows with different membership does not resolve selection bias. Layer-qualified expert rates are needed before the pooled-ID screen can motivate a particular expert intervention. GPQA-Diamond and MMLU-Pro require separate execution under the current generation/replay configuration to extend coverage.

8 Conclusion

The three active workflows support deliberately limited claims. GEPA improves the earlier same-model-judge development result substantially, but the final Qwen3.8 alignment finds no safe improvement over that imported prompt and neither run has been tested on an untouched response set. The boundary probes reveal strong within-split linear access to upcoming episode labels, especially for Implement, Verify, and Read, but their sentence split leaks every response across train and test and their external outputs collapse toward a coarse setup, execution, and review taxonomy. The six-benchmark correlation study supports AIME confidence/margin associations at both trace and problem levels, but the results vary with benchmark, termination, aggregation, and token position. Token-limit failures remain substantial, and later position windows select longer traces. Expanded metric-pair, expert-rate, and reasoning-view screens provide hypotheses for controlled evaluation without establishing an independent or causal routing mechanism.

The immediate evidential requirements are a frozen-prompt GEPA evaluation on unseen responses, response-grouped nested boundary probes with independently validated external labels, a prespecified response to remaining generation censoring, extension of the current run to GPQA-Diamond and MMLU-Pro, and length-/termination-aware and common-cohort position analyses with problem-respecting validation before routing interventions are evaluated.

9 Immediate Work

The agreed order of work is:

- (1) freeze and archive the seed prompt retained by the completed final GEPA alignment, and reserve the locked test for a prespecified evaluation;
- (2) prespecify a response to the remaining token-limit failures and extend the current vLLM run to GPQA-Diamond and MMLU-Pro;
- (3) fit a prespecified length-/termination-aware, dataset-aware correctness model with problem-respecting validation and exploratory layerwise screening;
- (4) test repeated-attempt and position-window confidence contrasts under length, termination, token-composition, and common-cohort controls, and qualify pooled expert-ID candidates by layer; and
- (5) expand to additional MoE families only after the pilot pipeline and disk budget are stable.

The detailed acceptance criteria and transcript ambiguities are maintained in NEXT_STEPS.md at the repository root.

A Reproducibility Checklist

Table 21. Living inventory for the three active workflows. *Model(s)*: Qwen3.6-27B and Qwen3.8-27B Q4_K_XL, Qwen/Qwen3.5-35B-A3B-GPTQ-Int4, and Qwen3.5-35B-A3B correlation-run variants. *Metrics*: workflow status, sample counts, headline metrics, and next actions; this is an inventory rather than a common-metric evaluation.

Item	Status	Note / next action
gepaLLMAsJudge	Cross-validation and two final-fit validations complete; test pending	Static-reward grouped cross-validation over 2,789 units gives 67.34% accuracy and 65.05% macro-F1 for light-budget GEPA; heavy is worse. The same-model LLM-judge final fit improves validation accuracy from 66.83% to 75.43% and macro-F1 from 0.613 to 0.727, but reuses six responses across 145 candidate evaluations. In the final Qwen3.8 alignment, GEPA instead reduces balanced accuracy from 0.740 to 0.729 and Plan recall from 0.750 to 0.563, so the safety gate correctly retains the imported 40-example seed. Neither run evaluates the 336-unit locked test. Freeze the selected seed and evaluate it only under a prespecified held-out protocol.
probeTest	Complete; grouped and human validation pending	Qwen3.5 GPTQ probes cover 3,087 units, seven targets, and 41 hidden-state indices; selected-layer AUROC is 0.835–0.934. A provisional hybrid review of 581 external units gives 50.3% primary and 60.8% compatible agreement, with 187 multi-label and 37 low-confidence units; Explore is never predicted. Grouped cross-validation, robust segmentation, multiclass calibration, and independent human adjudication remain pending.
correlation_pipeline	Six math benchmarks and requested views complete	The September 10 vLLM untagged run covers 4,647 attempts from 1,547 problems, with all-attempt accuracy 0.772. There are 394 token-limit hits, including 393 of 1,059 errors. Confidence is positively associated with correctness with intervals excluding zero on four datasets; AMC and Minerva are uncertain. Non-limit confidence is moderately positive on AIME and negative on AMC. Problem-level AIME confidence survives BH adjustment. Full-reasoning and ten fixed-position windows plus overflow are complete, with strong sample attrition in later windows; no class views were run. Generation uses GPTQ-Int4 in vLLM and replay uses runtime 4-bit quantization. Validate length, termination, and position effects on held-out problems, qualify expert IDs by layer, and extend this run to GPQA-Diamond and MMLU-Pro.

A.1 Correlation snapshot and independent checks

The current numerical source is `results/correlation_pipeline/reasoning-vllm-untagged-v1/analysis/unsloth--Qwen3.5-35B-A3B/correlations.json`, written September 10, 2026. Companion CSVs contain trace, problem, and expert features. The `views-v1` subdirectory contains full reasoning, ten position bins, overflow, and their coverage and correlation outputs. The `forward-stage summary.json` records extraction and storage metadata. The primary analysis JSON SHA-256 is `a1b575098cc48774b6628ffea388c1868f3c82ee19ab98e505214124348d60cc`. The saved configuration uses 500 bootstrap samples, seed 42, minimum expert support of four traces, and a cap of 128 ordered pairs. The position reference uses all 4,647 traces and has corpus SHA-256 `78c133703974adfd5ff804941569c7d95c53128d69b2fdc29bb36b0695f1e133`. Results are reported at saved precision before table rounding.

`report/audit_correlation_run.py` independently matches all generation and replay texts to their analysis fingerprints, verifies layer selection, feature values, stored outcomes, and the common completion budget, and recomputes binary and mean-feature problem-level Spearman coefficients from the CSV. It also recomputes the trace metric-pair screen and the three reported expert-rate maxima, and verifies outcome alignment across views. It checks binary coefficients in all 12 reasoning/position views against their finite trace observations and computes the explicitly post-hoc non-limit sensitivity table and storage totals. The primary binary and problem-level maximum absolute discrepancies are below 1.6×10^{-13} and 1.2×10^{-16} , respectively; every view's binary discrepancy is below 1.4×10^{-13} . This verifies propagation of saved correctness, not an independent mathematical rejudging of answers.

The audit is read-only and defaults to this run. From the repository root in the project Python environment, execute `python report/audit_correlation_run.py`; an alternative snapshot can be selected with `-run-dir`. Bootstrap intervals and BH values in the primary tables are taken from the saved analysis, not refit after inspecting results. The report update does not regenerate traces, change scores, or overwrite experiment artifacts. The earlier GGUF snapshot remains under `results/correlation_pipeline/analysis` and is not mixed into these results.

B Complete aggregate correlation tables

These tables expand the selected results in the main text using the same September 10, 2026 reasoning-v1lm-untagged-v1 snapshot. All tables use Qwen3.5-35B-A3B: GPTQ-Int4 generation through vLLM and runtime-4-bit replay through the unsloth checkpoint, with selected decoder indices 27, 34, 35, 36, 38, and 39. Generation uses temperature 0.6, top- p 0.95, and a 32,768-token completion budget. MATH-500, OlympiadBench, and Minerva use one attempt per problem; AIME 2024/2025 and AMC 2023 use 32. Finite trace counts appear either alongside coefficients or in a separate row when shared by all metrics in a panel. The unit is an attempt, including repeated attempts from the same problem. Coefficients are descriptive: no significance markings or uncertainty intervals are implied. Saved uncertainty estimates, where available, remain in the source JSON files; source hashes are recorded in `correlation_tables_sources.json`.

The full-reasoning view excludes final-answer tokens. Position windows are disjoint absolute spans of one tenth of the corpus mean (12,477.026 tokens), with an additional overflow window. Missing windows are excluded pairwise; late windows therefore describe changing cohorts. Token counts are omitted as position predictors. All coefficients remain subject to the length, termination, and repeated-attempt limitations discussed in the main text. These tables cover aggregate means, not the exploratory layerwise screen. No class tagging is used.

B.1 Full reasoning versus correctness

Table 22. Full reasoning: aggregate features versus whole-attempt correctness. Point-biserial r (finite traces); model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.322 (500)	−0.728 (960)	−0.715 (960)	−0.375 (675)	−0.173 (1,280)	−0.307 (272)
Router confidence	0.255 (500)	0.518 (960)	0.435 (960)	0.208 (675)	0.044 (1,280)	0.112 (272)
Top-1/top-2 margin	0.199 (500)	0.535 (960)	0.420 (960)	0.182 (675)	0.202 (1,280)	0.069 (272)
Top-8 probability mass	0.221 (500)	0.254 (960)	0.293 (960)	0.142 (675)	−0.038 (1,280)	0.095 (272)
Top-8/top-9 margin	0.148 (500)	0.072 (960)	0.191 (960)	0.078 (675)	−0.126 (1,280)	0.050 (272)
Top-8/non-top-8 mean gap	0.221 (500)	0.254 (960)	0.293 (960)	0.142 (675)	−0.038 (1,280)	0.095 (272)
Hidden-state norm	0.256 (500)	0.064 (960)	−0.010 (960)	0.087 (675)	0.466 (1,280)	0.068 (272)
Hidden-state step distance	0.227 (500)	0.405 (960)	0.389 (960)	0.117 (675)	0.489 (1,280)	0.136 (272)
Hidden/router geometry	−0.168 (500)	−0.030 (960)	−0.010 (960)	−0.074 (675)	−0.120 (1,280)	−0.237 (272)
Top-1 switch rate	−0.026 (500)	−0.299 (960)	0.136 (960)	0.039 (675)	−0.066 (1,280)	0.016 (272)
Adjacent top-8 overlap	0.039 (500)	0.081 (960)	−0.166 (960)	−0.020 (675)	−0.139 (1,280)	−0.047 (272)

B.2 All aggregate metrics by position

Table 23. Router confidence: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	−0.030 (500)	−0.153 (960)	−0.209 (960)	0.043 (675)	−0.361 (1,280)	−0.020 (272)
Bin 1	−0.050 (429)	−0.148 (960)	−0.093 (960)	0.070 (668)	−0.421 (1,269)	0.070 (269)
Bin 2	−0.157 (347)	−0.033 (960)	0.067 (960)	0.023 (634)	−0.423 (1,172)	0.108 (254)
Bin 3	−0.118 (268)	0.072 (956)	0.113 (942)	0.024 (564)	−0.391 (1,038)	−0.004 (234)
Bin 4	0.020 (204)	0.137 (927)	0.150 (900)	−0.016 (494)	−0.190 (876)	−0.066 (202)
Bin 5	0.122 (171)	0.183 (859)	0.149 (847)	0.017 (427)	0.081 (767)	0.011 (162)
Bin 6	0.246 (139)	0.184 (816)	0.122 (803)	−0.003 (384)	0.363 (684)	0.075 (133)
Bin 7	0.385 (106)	0.206 (748)	0.118 (752)	0.093 (342)	0.454 (559)	0.077 (102)
Bin 8	0.467 (83)	0.218 (679)	0.157 (724)	0.121 (314)	0.589 (436)	0.093 (83)
Bin 9	0.392 (65)	0.265 (605)	0.185 (701)	0.152 (291)	0.724 (317)	0.137 (70)
Overflow	0.539 (54)	0.474 (540)	0.376 (664)	0.195 (270)	0.830 (238)	0.237 (59)

Table 24. Top-1/top-2 margin: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	-0.116 (500)	0.182 (960)	-0.042 (960)	0.006 (675)	-0.274 (1,280)	-0.062 (272)
Bin 1	-0.081 (429)	0.156 (960)	-0.043 (960)	0.086 (668)	-0.211 (1,269)	-0.084 (269)
Bin 2	-0.157 (347)	0.217 (960)	0.020 (960)	0.014 (634)	-0.162 (1,172)	-0.060 (254)
Bin 3	-0.075 (268)	0.241 (956)	0.065 (942)	0.036 (564)	-0.127 (1,038)	-0.150 (234)
Bin 4	0.110 (204)	0.213 (927)	0.110 (900)	0.001 (494)	0.117 (876)	-0.142 (202)
Bin 5	0.196 (171)	0.212 (859)	0.124 (847)	0.056 (427)	0.311 (767)	-0.148 (162)
Bin 6	0.303 (139)	0.244 (816)	0.083 (803)	-0.012 (384)	0.550 (684)	-0.135 (133)
Bin 7	0.433 (106)	0.237 (748)	0.103 (752)	0.135 (342)	0.548 (559)	-0.054 (102)
Bin 8	0.467 (83)	0.243 (679)	0.101 (724)	0.077 (314)	0.645 (436)	-0.031 (83)
Bin 9	0.379 (65)	0.301 (605)	0.134 (701)	0.122 (291)	0.740 (317)	0.049 (70)
Overflow	0.500 (54)	0.505 (540)	0.339 (664)	0.180 (270)	0.801 (238)	0.182 (59)

Table 25. Top-8 probability mass: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	-0.023 (500)	-0.366 (960)	-0.193 (960)	-0.125 (675)	-0.123 (1,280)	-0.036 (272)
Bin 1	-0.039 (429)	-0.361 (960)	-0.201 (960)	-0.035 (668)	-0.416 (1,269)	0.055 (269)
Bin 2	-0.097 (347)	-0.285 (960)	-0.130 (960)	-0.008 (634)	-0.429 (1,172)	0.140 (254)
Bin 3	-0.053 (268)	-0.196 (956)	-0.048 (942)	-0.001 (564)	-0.359 (1,038)	0.076 (234)
Bin 4	0.118 (204)	-0.080 (927)	-0.019 (900)	-0.015 (494)	-0.182 (876)	0.018 (202)
Bin 5	0.202 (171)	-0.024 (859)	0.038 (847)	-0.016 (427)	0.107 (767)	0.015 (162)
Bin 6	0.304 (139)	-0.031 (816)	0.020 (803)	-0.021 (384)	0.360 (684)	0.144 (133)
Bin 7	0.400 (106)	-0.024 (748)	0.076 (752)	0.022 (342)	0.447 (559)	0.118 (102)
Bin 8	0.506 (83)	0.070 (679)	0.112 (724)	0.069 (314)	0.555 (436)	0.125 (83)
Bin 9	0.460 (65)	0.106 (605)	0.165 (701)	0.090 (291)	0.640 (317)	0.142 (70)
Overflow	0.583 (54)	0.367 (540)	0.311 (664)	0.155 (270)	0.727 (238)	0.294 (59)

Table 26. Top-8/top-9 margin: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	-0.024 (500)	-0.402 (960)	-0.181 (960)	-0.135 (675)	-0.018 (1,280)	-0.034 (272)
Bin 1	-0.070 (429)	-0.397 (960)	-0.184 (960)	-0.051 (668)	-0.397 (1,269)	0.071 (269)
Bin 2	-0.093 (347)	-0.312 (960)	-0.134 (960)	-0.027 (634)	-0.453 (1,172)	0.144 (254)
Bin 3	-0.071 (268)	-0.236 (956)	-0.052 (942)	-0.031 (564)	-0.407 (1,038)	0.064 (234)
Bin 4	0.083 (204)	-0.124 (927)	-0.032 (900)	-0.026 (494)	-0.245 (876)	0.004 (202)
Bin 5	0.177 (171)	-0.066 (859)	0.024 (847)	-0.038 (427)	0.043 (767)	-0.007 (162)
Bin 6	0.286 (139)	-0.072 (816)	0.023 (803)	-0.025 (384)	0.283 (684)	0.149 (133)
Bin 7	0.379 (106)	-0.072 (748)	0.066 (752)	0.015 (342)	0.365 (559)	0.093 (102)
Bin 8	0.475 (83)	0.020 (679)	0.112 (724)	0.057 (314)	0.459 (436)	0.117 (83)
Bin 9	0.495 (65)	0.043 (605)	0.148 (701)	0.096 (291)	0.533 (317)	0.141 (70)
Overflow	0.597 (54)	0.284 (540)	0.261 (664)	0.139 (270)	0.610 (238)	0.329 (59)

Table 27. Top-8/non-top-8 mean gap: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	-0.023 (500)	-0.366 (960)	-0.193 (960)	-0.125 (675)	-0.123 (1,280)	-0.036 (272)
Bin 1	-0.039 (429)	-0.361 (960)	-0.201 (960)	-0.035 (668)	-0.416 (1,269)	0.055 (269)
Bin 2	-0.097 (347)	-0.285 (960)	-0.130 (960)	-0.008 (634)	-0.429 (1,172)	0.140 (254)

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Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 3	−0.053 (268)	−0.196 (956)	−0.048 (942)	−0.001 (564)	−0.359 (1,038)	0.076 (234)
Bin 4	0.118 (204)	−0.080 (927)	−0.019 (900)	−0.015 (494)	−0.182 (876)	0.018 (202)
Bin 5	0.202 (171)	−0.024 (859)	0.038 (847)	−0.016 (427)	0.107 (767)	0.015 (162)
Bin 6	0.304 (139)	−0.031 (816)	0.020 (803)	−0.021 (384)	0.360 (684)	0.144 (133)
Bin 7	0.400 (106)	−0.024 (748)	0.076 (752)	0.022 (342)	0.447 (559)	0.118 (102)
Bin 8	0.506 (83)	0.070 (679)	0.112 (724)	0.069 (314)	0.555 (436)	0.125 (83)
Bin 9	0.460 (65)	0.106 (605)	0.165 (701)	0.090 (291)	0.640 (317)	0.142 (70)
Overflow	0.583 (54)	0.367 (540)	0.311 (664)	0.155 (270)	0.727 (238)	0.294 (59)

Table 28. Hidden-state norm: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	0.035 (500)	−0.320 (960)	−0.087 (960)	−0.080 (675)	0.175 (1,280)	−0.048 (272)
Bin 1	0.041 (429)	−0.292 (960)	−0.190 (960)	−0.128 (668)	0.089 (1,269)	−0.117 (269)
Bin 2	0.109 (347)	−0.276 (960)	−0.220 (960)	−0.062 (634)	0.115 (1,172)	−0.114 (254)
Bin 3	0.196 (268)	−0.296 (956)	−0.193 (942)	−0.057 (564)	0.220 (1,038)	−0.094 (234)
Bin 4	0.291 (204)	−0.211 (927)	−0.192 (900)	−0.028 (494)	0.248 (876)	−0.008 (202)
Bin 5	0.299 (171)	−0.186 (859)	−0.167 (847)	−0.078 (427)	0.447 (767)	−0.076 (162)
Bin 6	0.401 (139)	−0.204 (816)	−0.154 (803)	−0.019 (384)	0.589 (684)	0.047 (133)
Bin 7	0.429 (106)	−0.167 (748)	−0.162 (752)	−0.029 (342)	0.643 (559)	0.077 (102)
Bin 8	0.462 (83)	−0.096 (679)	−0.114 (724)	−0.022 (314)	0.725 (436)	0.087 (83)
Bin 9	0.508 (65)	−0.087 (605)	−0.100 (701)	0.028 (291)	0.778 (317)	0.080 (70)
Overflow	0.572 (54)	0.214 (540)	0.026 (664)	0.107 (270)	0.837 (238)	0.185 (59)

Table 29. Hidden-state step distance: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	−0.092 (500)	0.039 (960)	−0.008 (960)	−0.206 (675)	0.102 (1,280)	−0.143 (272)
Bin 1	−0.087 (429)	−0.082 (960)	−0.023 (960)	−0.166 (668)	0.179 (1,269)	−0.209 (269)
Bin 2	0.002 (347)	−0.137 (960)	−0.032 (960)	−0.137 (634)	0.210 (1,172)	−0.108 (254)
Bin 3	0.133 (268)	−0.100 (956)	−0.018 (942)	−0.114 (564)	0.335 (1,038)	−0.145 (234)
Bin 4	0.194 (203)	−0.082 (927)	0.051 (900)	−0.076 (494)	0.429 (876)	−0.060 (202)
Bin 5	0.271 (171)	−0.121 (858)	0.073 (847)	−0.030 (426)	0.518 (767)	−0.075 (162)
Bin 6	0.335 (139)	−0.094 (816)	0.031 (803)	−0.050 (383)	0.649 (684)	−0.066 (133)
Bin 7	0.403 (106)	−0.084 (748)	0.029 (751)	0.048 (342)	0.744 (559)	−0.004 (102)
Bin 8	0.427 (83)	−0.095 (679)	0.057 (724)	−0.027 (314)	0.781 (436)	0.066 (83)
Bin 9	0.440 (65)	−0.009 (605)	0.036 (701)	0.030 (291)	0.809 (317)	0.099 (70)
Overflow	0.514 (54)	0.274 (540)	0.253 (664)	0.103 (270)	0.861 (238)	0.176 (59)

Table 30. Hidden/router geometry: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	0.048 (500)	0.136 (960)	0.016 (960)	0.026 (675)	−0.017 (1,280)	0.020 (272)
Bin 1	0.091 (429)	0.190 (960)	0.158 (960)	0.122 (668)	0.086 (1,269)	−0.050 (269)
Bin 2	0.080 (347)	0.204 (960)	0.174 (960)	0.109 (634)	0.137 (1,172)	−0.228 (254)
Bin 3	0.075 (268)	0.218 (956)	0.158 (942)	0.048 (564)	0.009 (1,038)	−0.174 (234)
Bin 4	−0.165 (203)	0.164 (927)	0.099 (900)	0.081 (494)	−0.087 (876)	−0.146 (202)
Bin 5	−0.105 (171)	0.186 (858)	0.078 (847)	0.045 (426)	−0.181 (767)	−0.002 (162)
Bin 6	−0.239 (139)	0.242 (816)	0.154 (803)	0.035 (383)	−0.275 (684)	−0.074 (133)
Bin 7	−0.281 (106)	0.183 (748)	0.133 (751)	0.046 (342)	−0.394 (559)	−0.110 (102)
Bin 8	−0.297 (83)	0.132 (679)	0.029 (724)	0.104 (314)	−0.340 (436)	−0.071 (83)

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Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 9	−0.416 (65)	0.121 (605)	0.052 (701)	0.028 (291)	−0.464 (317)	−0.127 (70)
Overflow	−0.569 (54)	−0.027 (540)	0.133 (664)	0.084 (270)	−0.531 (238)	−0.363 (59)

Table 31. Top-1 switch rate: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	0.068 (500)	−0.071 (960)	0.246 (960)	0.005 (675)	0.132 (1,280)	0.056 (272)
Bin 1	0.062 (429)	−0.188 (960)	0.164 (960)	−0.032 (668)	0.147 (1,269)	0.072 (269)
Bin 2	0.130 (347)	−0.238 (960)	0.069 (960)	0.004 (634)	0.136 (1,172)	−0.002 (254)
Bin 3	0.125 (268)	−0.235 (956)	0.109 (942)	−0.003 (564)	0.153 (1,038)	−0.000 (234)
Bin 4	0.106 (203)	−0.212 (927)	0.063 (900)	−0.004 (494)	0.102 (876)	0.016 (202)
Bin 5	−0.004 (171)	−0.234 (858)	0.054 (847)	−0.051 (426)	−0.048 (767)	−0.093 (162)
Bin 6	−0.015 (139)	−0.256 (816)	0.134 (803)	−0.040 (383)	−0.112 (684)	−0.059 (133)
Bin 7	−0.149 (106)	−0.228 (748)	0.122 (751)	0.063 (342)	−0.194 (559)	0.104 (102)
Bin 8	−0.171 (83)	−0.216 (679)	0.160 (724)	−0.000 (314)	−0.296 (436)	0.087 (83)
Bin 9	−0.231 (65)	−0.263 (605)	0.113 (701)	−0.003 (291)	−0.399 (317)	−0.009 (70)
Overflow	−0.392 (54)	−0.303 (540)	0.109 (664)	0.026 (270)	−0.552 (238)	−0.045 (59)

Table 32. Adjacent top-8 overlap: position-specific point-biserial r versus whole-attempt correctness (finite traces). Model and protocol as specified above.

Window	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0	−0.004 (500)	−0.293 (960)	−0.210 (960)	−0.085 (675)	−0.081 (1,280)	−0.079 (272)
Bin 1	−0.019 (429)	−0.188 (960)	−0.211 (960)	−0.040 (668)	−0.276 (1,269)	−0.058 (269)
Bin 2	−0.057 (347)	−0.098 (960)	−0.114 (960)	−0.020 (634)	−0.327 (1,172)	0.057 (254)
Bin 3	−0.110 (268)	−0.031 (956)	−0.119 (942)	−0.015 (564)	−0.321 (1,038)	0.055 (234)
Bin 4	−0.044 (203)	−0.005 (927)	−0.120 (900)	−0.035 (494)	−0.251 (876)	0.030 (202)
Bin 5	−0.017 (171)	0.052 (858)	−0.083 (847)	−0.019 (426)	−0.105 (767)	0.096 (162)
Bin 6	−0.054 (139)	0.072 (816)	−0.120 (803)	0.024 (383)	−0.010 (684)	0.121 (133)
Bin 7	0.015 (106)	0.049 (748)	−0.086 (751)	−0.091 (342)	0.040 (559)	−0.031 (102)
Bin 8	0.099 (83)	0.096 (679)	−0.096 (724)	0.001 (314)	0.098 (436)	0.006 (83)
Bin 9	0.273 (65)	0.121 (605)	−0.012 (701)	0.034 (291)	0.182 (317)	0.041 (70)
Overflow	0.540 (54)	0.252 (540)	−0.024 (664)	−0.002 (270)	0.402 (238)	0.150 (59)

B.3 Correlations organized by target metric

Each table fixes one target metric, lists the other metrics by name in the rows, and compares the six benchmarks in columns, matching the layout of the correctness table. Cells give descriptive Spearman ρ ; a separate sample count row is used when all coefficients in a panel have the same finite count. Self-correlations are omitted. A correlation appears under both target orientations for ease of reading; these are the same association, not independent results. The algebraically redundant mass/gap pair is retained to reproduce the saved scan. Character and step counts occur only in the entire-continuation scan; token count is excluded from position scans.

B.3.1 Entire continuation.

Table 33. Entire continuation: metrics versus token count. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Router confidence	−0.707	−0.668	−0.824	−0.868	−0.849	−0.303
Top-1/top-2 margin	−0.379	−0.479	−0.710	−0.716	−0.496	−0.253

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8 probability mass	-0.253	0.006	-0.325	-0.425	-0.343	-0.079
Top-8/top-9 margin	0.025	0.238	-0.061	-0.158	-0.034	-0.077
Top-8/non-top-8 mean gap	-0.253	0.006	-0.325	-0.425	-0.343	-0.079
Hidden-state norm	0.369	0.469	0.374	0.383	0.360	0.026
Hidden-state step distance	0.172	-0.354	-0.271	-0.161	0.370	-0.253
Hidden/router geometry	0.260	-0.020	0.206	0.284	0.191	0.318
Top-1 switch rate	0.234	0.316	0.003	0.066	0.301	-0.120
Adjacent top-8 overlap	-0.165	0.023	0.237	0.062	-0.244	0.196
Character count	0.986	0.970	0.976	0.982	0.978	0.973
Step count	0.266	-0.273	-0.036	-0.226	0.167	0.027

Table 34. Entire continuation: metrics versus router confidence. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.707	-0.668	-0.824	-0.868	-0.849	-0.303
Top-1/top-2 margin	0.629	0.688	0.805	0.801	0.641	0.633
Top-8 probability mass	0.559	0.300	0.493	0.655	0.581	0.718
Top-8/top-9 margin	0.318	0.034	0.205	0.410	0.312	0.601
Top-8/non-top-8 mean gap	0.559	0.300	0.493	0.655	0.581	0.718
Hidden-state norm	-0.355	-0.237	-0.342	-0.282	-0.227	0.034
Hidden-state step distance	-0.174	0.230	0.207	0.150	-0.303	-0.007
Hidden/router geometry	-0.232	-0.183	-0.222	-0.348	-0.303	-0.361
Top-1 switch rate	-0.415	-0.492	-0.300	-0.180	-0.412	-0.381
Adjacent top-8 overlap	0.377	0.152	-0.039	0.115	0.377	0.470
Character count	-0.666	-0.677	-0.831	-0.835	-0.800	-0.263
Step count	-0.081	0.029	-0.086	0.186	-0.228	0.019

Table 35. Entire continuation: metrics versus top-1/top-2 margin. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.379	-0.479	-0.710	-0.716	-0.496	-0.253
Router confidence	0.629	0.688	0.805	0.801	0.641	0.633
Top-8 probability mass	0.271	0.134	0.249	0.418	0.378	0.225
Top-8/top-9 margin	0.145	-0.046	-0.007	0.209	0.235	0.022
Top-8/non-top-8 mean gap	0.271	0.134	0.249	0.418	0.378	0.225
Hidden-state norm	-0.227	-0.196	-0.298	-0.294	0.012	-0.258
Hidden-state step distance	0.133	0.304	0.386	0.222	-0.198	0.145
Hidden/router geometry	-0.148	-0.083	-0.152	-0.258	-0.130	-0.170
Top-1 switch rate	-0.514	-0.631	-0.382	-0.342	-0.593	-0.592
Adjacent top-8 overlap	0.057	0.018	-0.210	-0.039	0.206	0.334
Character count	-0.395	-0.566	-0.768	-0.728	-0.513	-0.336
Step count	-0.036	0.022	-0.158	0.133	-0.222	-0.090

Table 36. Entire continuation: metrics versus top-8 probability mass. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.253	0.006	-0.325	-0.425	-0.343	-0.079
Router confidence	0.559	0.300	0.493	0.655	0.581	0.718
Top-1/top-2 margin	0.271	0.134	0.249	0.418	0.378	0.225
Top-8/top-9 margin	0.916	0.915	0.900	0.915	0.900	0.935

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	0.299	0.575	0.403	0.299	0.352	0.524
Hidden-state step distance	-0.023	0.132	0.104	0.173	0.025	-0.048
Hidden/router geometry	-0.461	-0.530	-0.403	-0.498	-0.474	-0.481
Top-1 switch rate	-0.144	0.122	0.032	0.006	-0.154	-0.205
Adjacent top-8 overlap	0.746	0.400	0.431	0.519	0.673	0.603
Character count	-0.146	0.125	-0.243	-0.313	-0.212	0.055
Step count	0.108	-0.166	-0.136	0.139	-0.102	-0.008

Table 37. Entire continuation: metrics versus top-8/top-9 margin. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.025	0.238	-0.061	-0.158	-0.034	-0.077
Router confidence	0.318	0.034	0.205	0.410	0.312	0.601
Top-1/top-2 margin	0.145	-0.046	-0.007	0.209	0.235	0.022
Top-8 probability mass	0.916	0.915	0.900	0.915	0.900	0.935
Top-8/non-top-8 mean gap	0.916	0.915	0.900	0.915	0.900	0.935
Hidden-state norm	0.510	0.698	0.577	0.509	0.541	0.614
Hidden-state step distance	0.114	0.135	0.158	0.199	0.156	-0.000
Hidden/router geometry	-0.389	-0.492	-0.336	-0.409	-0.433	-0.464
Top-1 switch rate	-0.012	0.294	0.184	0.086	-0.049	0.019
Adjacent top-8 overlap	0.625	0.300	0.380	0.503	0.583	0.425
Character count	0.123	0.352	0.024	-0.045	0.088	0.074
Step count	0.140	-0.197	-0.165	0.107	-0.042	0.070

Table 38. Entire continuation: metrics versus top-8/non-top-8 mean gap. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.253	0.006	-0.325	-0.425	-0.343	-0.079
Router confidence	0.559	0.300	0.493	0.655	0.581	0.718
Top-1/top-2 margin	0.271	0.134	0.249	0.418	0.378	0.225
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.916	0.915	0.900	0.915	0.900	0.935
Hidden-state norm	0.299	0.575	0.403	0.299	0.352	0.524
Hidden-state step distance	-0.023	0.132	0.104	0.173	0.025	-0.048
Hidden/router geometry	-0.461	-0.530	-0.403	-0.498	-0.474	-0.481
Top-1 switch rate	-0.144	0.122	0.032	0.006	-0.154	-0.205
Adjacent top-8 overlap	0.746	0.400	0.431	0.519	0.673	0.603
Character count	-0.146	0.125	-0.243	-0.313	-0.212	0.055
Step count	0.108	-0.166	-0.136	0.139	-0.102	-0.008

Table 39. Entire continuation: metrics versus hidden-state norm. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.369	0.469	0.374	0.383	0.360	0.026
Router confidence	-0.355	-0.237	-0.342	-0.282	-0.227	0.034
Top-1/top-2 margin	-0.227	-0.196	-0.298	-0.294	0.012	-0.258
Top-8 probability mass	0.299	0.575	0.403	0.299	0.352	0.524
Top-8/top-9 margin	0.510	0.698	0.577	0.509	0.541	0.614
Top-8/non-top-8 mean gap	0.299	0.575	0.403	0.299	0.352	0.524

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden-state step distance	0.248	-0.115	-0.115	0.070	0.282	0.202
Hidden/router geometry	-0.233	-0.263	-0.125	-0.146	-0.162	-0.275
Top-1 switch rate	0.205	0.239	0.201	0.164	0.019	0.213
Adjacent top-8 overlap	0.199	0.315	0.360	0.342	0.297	0.167
Character count	0.399	0.521	0.418	0.453	0.399	0.155
Step count	0.054	-0.115	-0.192	-0.076	-0.026	-0.029

Table 40. Entire continuation: metrics versus hidden-state step distance. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.172	-0.354	-0.271	-0.161	0.370	-0.253
Router confidence	-0.174	0.230	0.207	0.150	-0.303	-0.007
Top-1/top-2 margin	0.133	0.304	0.386	0.222	-0.198	0.145
Top-8 probability mass	-0.023	0.132	0.104	0.173	0.025	-0.048
Top-8/top-9 margin	0.114	0.135	0.158	0.199	0.156	-0.000
Top-8/non-top-8 mean gap	-0.023	0.132	0.104	0.173	0.025	-0.048
Hidden-state norm	0.248	-0.115	-0.115	0.070	0.282	0.202
Hidden/router geometry	-0.106	-0.170	-0.090	-0.268	-0.142	-0.075
Top-1 switch rate	0.130	0.163	0.195	0.285	0.365	0.201
Adjacent top-8 overlap	-0.357	-0.451	-0.500	-0.357	-0.327	-0.368
Character count	0.165	-0.334	-0.295	-0.146	0.385	-0.260
Step count	-0.069	0.129	-0.135	0.062	-0.031	-0.035

Table 41. Entire continuation: metrics versus hidden/router geometry. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.260	-0.020	0.206	0.284	0.191	0.318
Router confidence	-0.232	-0.183	-0.222	-0.348	-0.303	-0.361
Top-1/top-2 margin	-0.148	-0.083	-0.152	-0.258	-0.130	-0.170
Top-8 probability mass	-0.461	-0.530	-0.403	-0.498	-0.474	-0.481
Top-8/top-9 margin	-0.389	-0.492	-0.336	-0.409	-0.433	-0.464
Top-8/non-top-8 mean gap	-0.461	-0.530	-0.403	-0.498	-0.474	-0.481
Hidden-state norm	-0.233	-0.263	-0.125	-0.146	-0.162	-0.275
Hidden-state step distance	-0.106	-0.170	-0.090	-0.268	-0.142	-0.075
Top-1 switch rate	0.026	-0.081	0.008	-0.045	0.050	0.053
Adjacent top-8 overlap	-0.330	-0.173	-0.185	-0.219	-0.301	-0.237
Character count	0.192	-0.098	0.143	0.200	0.106	0.242
Step count	0.091	0.100	-0.006	-0.111	0.072	0.004

Table 42. Entire continuation: metrics versus top-1 switch rate. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.234	0.316	0.003	0.066	0.301	-0.120
Router confidence	-0.415	-0.492	-0.300	-0.180	-0.412	-0.381
Top-1/top-2 margin	-0.514	-0.631	-0.382	-0.342	-0.593	-0.592
Top-8 probability mass	-0.144	0.122	0.032	0.006	-0.154	-0.205
Top-8/top-9 margin	-0.012	0.294	0.184	0.086	-0.049	0.019
Top-8/non-top-8 mean gap	-0.144	0.122	0.032	0.006	-0.154	-0.205
Hidden-state norm	0.205	0.239	0.201	0.164	0.019	0.213
Hidden-state step distance	0.130	0.163	0.195	0.285	0.365	0.201

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden/router geometry	0.026	-0.081	0.008	-0.045	0.050	0.053
Adjacent top-8 overlap	-0.392	-0.462	-0.430	-0.467	-0.493	-0.774
Character count	0.269	0.438	0.108	0.133	0.344	-0.032
Step count	0.057	-0.028	0.119	0.060	0.145	0.138

Table 43. Entire continuation: metrics versus adjacent top-8 overlap. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.165	0.023	0.237	0.062	-0.244	0.196
Router confidence	0.377	0.152	-0.039	0.115	0.377	0.470
Top-1/top-2 margin	0.057	0.018	-0.210	-0.039	0.206	0.334
Top-8 probability mass	0.746	0.400	0.431	0.519	0.673	0.603
Top-8/top-9 margin	0.625	0.300	0.380	0.503	0.583	0.425
Top-8/non-top-8 mean gap	0.746	0.400	0.431	0.519	0.673	0.603
Hidden-state norm	0.199	0.315	0.360	0.342	0.297	0.167
Hidden-state step distance	-0.357	-0.451	-0.500	-0.357	-0.327	-0.368
Hidden/router geometry	-0.330	-0.173	-0.185	-0.219	-0.301	-0.237
Top-1 switch rate	-0.392	-0.462	-0.430	-0.467	-0.493	-0.774
Character count	-0.083	0.049	0.258	0.122	-0.154	0.208
Step count	0.178	0.012	0.038	0.029	-0.010	-0.080

Table 44. Entire continuation: metrics versus character count. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.986	0.970	0.976	0.982	0.978	0.973
Router confidence	-0.666	-0.677	-0.831	-0.835	-0.800	-0.263
Top-1/top-2 margin	-0.395	-0.566	-0.768	-0.728	-0.513	-0.336
Top-8 probability mass	-0.146	0.125	-0.243	-0.313	-0.212	0.055
Top-8/top-9 margin	0.123	0.352	0.024	-0.045	0.088	0.074
Top-8/non-top-8 mean gap	-0.146	0.125	-0.243	-0.313	-0.212	0.055
Hidden-state norm	0.399	0.521	0.418	0.453	0.399	0.155
Hidden-state step distance	0.165	-0.334	-0.295	-0.146	0.385	-0.260
Hidden/router geometry	0.192	-0.098	0.143	0.200	0.106	0.242
Top-1 switch rate	0.269	0.438	0.108	0.133	0.344	-0.032
Adjacent top-8 overlap	-0.083	0.049	0.258	0.122	-0.154	0.208
Step count	0.297	-0.249	-0.022	-0.205	0.199	0.045

Table 45. Entire continuation: metrics versus step count. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.266	-0.273	-0.036	-0.226	0.167	0.027
Router confidence	-0.081	0.029	-0.086	0.186	-0.228	0.019
Top-1/top-2 margin	-0.036	0.022	-0.158	0.133	-0.222	-0.090
Top-8 probability mass	0.108	-0.166	-0.136	0.139	-0.102	-0.008
Top-8/top-9 margin	0.140	-0.197	-0.165	0.107	-0.042	0.070
Top-8/non-top-8 mean gap	0.108	-0.166	-0.136	0.139	-0.102	-0.008
Hidden-state norm	0.054	-0.115	-0.192	-0.076	-0.026	-0.029
Hidden-state step distance	-0.069	0.129	-0.135	0.062	-0.031	-0.035
Hidden/router geometry	0.091	0.100	-0.006	-0.111	0.072	0.004
Top-1 switch rate	0.057	-0.028	0.119	0.060	0.145	0.138

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Adjacent top-8 overlap	0.178	0.012	0.038	0.029	-0.010	-0.080
Character count	0.297	-0.249	-0.022	-0.205	0.199	0.045

B.3.2 Full reasoning.

Table 46. Full reasoning: metrics versus token count. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Router confidence	-0.671	-0.644	-0.810	-0.854	-0.830	-0.290
Top-1/top-2 margin	-0.453	-0.491	-0.724	-0.740	-0.538	-0.279
Top-8 probability mass	-0.258	-0.072	-0.413	-0.488	-0.388	-0.108
Top-8/top-9 margin	0.008	0.167	-0.172	-0.245	-0.076	-0.109
Top-8/non-top-8 mean gap	-0.258	-0.072	-0.413	-0.488	-0.388	-0.108
Hidden-state norm	0.354	0.423	0.298	0.336	0.312	0.026
Hidden-state step distance	-0.061	-0.427	-0.394	-0.328	0.221	-0.393
Hidden/router geometry	0.214	-0.169	-0.089	0.102	0.063	0.349
Top-1 switch rate	0.196	0.333	0.046	0.074	0.303	-0.121
Adjacent top-8 overlap	-0.090	-0.053	0.153	0.014	-0.270	0.197

Table 47. Full reasoning: metrics versus router confidence. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.671	-0.644	-0.810	-0.854	-0.830	-0.290
Top-1/top-2 margin	0.660	0.688	0.813	0.806	0.662	0.609
Top-8 probability mass	0.622	0.392	0.585	0.727	0.646	0.755
Top-8/top-9 margin	0.403	0.129	0.325	0.513	0.384	0.643
Top-8/non-top-8 mean gap	0.622	0.392	0.585	0.727	0.646	0.755
Hidden-state norm	-0.339	-0.170	-0.258	-0.225	-0.167	0.028
Hidden-state step distance	-0.041	0.274	0.312	0.273	-0.194	-0.017
Hidden/router geometry	-0.257	-0.053	-0.018	-0.191	-0.179	-0.318
Top-1 switch rate	-0.375	-0.495	-0.334	-0.184	-0.408	-0.349
Adjacent top-8 overlap	0.370	0.227	0.050	0.184	0.421	0.464

Table 48. Full reasoning: metrics versus top-1/top-2 margin. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.453	-0.491	-0.724	-0.740	-0.538	-0.279
Router confidence	0.660	0.688	0.813	0.806	0.662	0.609
Top-8 probability mass	0.280	0.165	0.310	0.451	0.354	0.214
Top-8/top-9 margin	0.152	-0.010	0.073	0.256	0.203	0.010
Top-8/non-top-8 mean gap	0.280	0.165	0.310	0.451	0.354	0.214
Hidden-state norm	-0.262	-0.181	-0.262	-0.282	-0.022	-0.302
Hidden-state step distance	0.163	0.342	0.463	0.312	-0.205	0.150
Hidden/router geometry	-0.167	-0.013	0.028	-0.084	-0.068	-0.071
Top-1 switch rate	-0.468	-0.634	-0.404	-0.331	-0.588	-0.573
Adjacent top-8 overlap	0.039	0.060	-0.151	-0.001	0.213	0.325

Table 49. Full reasoning: metrics versus top-8 probability mass. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.258	-0.072	-0.413	-0.488	-0.388	-0.108
Router confidence	0.622	0.392	0.585	0.727	0.646	0.755
Top-1/top-2 margin	0.280	0.165	0.310	0.451	0.354	0.214
Top-8/top-9 margin	0.921	0.911	0.908	0.922	0.900	0.930
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	0.234	0.543	0.384	0.257	0.320	0.470
Hidden-state step distance	-0.032	0.124	0.155	0.217	-0.013	-0.102
Hidden/router geometry	-0.459	-0.462	-0.292	-0.451	-0.417	-0.485
Top-1 switch rate	-0.134	0.085	-0.001	0.001	-0.142	-0.185
Adjacent top-8 overlap	0.720	0.406	0.436	0.510	0.658	0.568

Table 50. Full reasoning: metrics versus top-8/top-9 margin. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.008	0.167	-0.172	-0.245	-0.076	-0.109
Router confidence	0.403	0.129	0.325	0.513	0.384	0.643
Top-1/top-2 margin	0.152	-0.010	0.073	0.256	0.203	0.010
Top-8 probability mass	0.921	0.911	0.908	0.922	0.900	0.930
Top-8/non-top-8 mean gap	0.921	0.911	0.908	0.922	0.900	0.930
Hidden-state norm	0.432	0.666	0.546	0.447	0.503	0.567
Hidden-state step distance	0.033	0.113	0.179	0.204	0.072	-0.043
Hidden/router geometry	-0.440	-0.524	-0.359	-0.464	-0.457	-0.489
Top-1 switch rate	-0.023	0.268	0.163	0.083	-0.038	0.049
Adjacent top-8 overlap	0.625	0.278	0.368	0.491	0.570	0.381

Table 51. Full reasoning: metrics versus top-8/non-top-8 mean gap. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.258	-0.072	-0.413	-0.488	-0.388	-0.108
Router confidence	0.622	0.392	0.585	0.727	0.646	0.755
Top-1/top-2 margin	0.280	0.165	0.310	0.451	0.354	0.214
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.921	0.911	0.908	0.922	0.900	0.930
Hidden-state norm	0.234	0.543	0.384	0.257	0.320	0.470
Hidden-state step distance	-0.032	0.124	0.155	0.217	-0.013	-0.102
Hidden/router geometry	-0.459	-0.462	-0.292	-0.451	-0.417	-0.485
Top-1 switch rate	-0.134	0.085	-0.001	0.001	-0.142	-0.185
Adjacent top-8 overlap	0.720	0.406	0.436	0.510	0.658	0.568

Table 52. Full reasoning: metrics versus hidden-state norm. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.354	0.423	0.298	0.336	0.312	0.026
Router confidence	-0.339	-0.170	-0.258	-0.225	-0.167	0.028
Top-1/top-2 margin	-0.262	-0.181	-0.262	-0.282	-0.022	-0.302
Top-8 probability mass	0.234	0.543	0.384	0.257	0.320	0.470
Top-8/top-9 margin	0.432	0.666	0.546	0.447	0.503	0.567

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8/non-top-8 mean gap	0.234	0.543	0.384	0.257	0.320	0.470
Hidden-state step distance	0.102	-0.167	-0.169	-0.018	0.146	0.165
Hidden/router geometry	-0.235	-0.312	-0.213	-0.267	-0.184	-0.336
Top-1 switch rate	0.184	0.227	0.216	0.169	0.036	0.244
Adjacent top-8 overlap	0.198	0.290	0.322	0.300	0.278	0.122

Table 53. Full reasoning: metrics versus hidden-state step distance. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.061	-0.427	-0.394	-0.328	0.221	-0.393
Router confidence	-0.041	0.274	0.312	0.273	-0.194	-0.017
Top-1/top-2 margin	0.163	0.342	0.463	0.312	-0.205	0.150
Top-8 probability mass	-0.032	0.124	0.155	0.217	-0.013	-0.102
Top-8/top-9 margin	0.033	0.113	0.179	0.204	0.072	-0.043
Top-8/non-top-8 mean gap	-0.032	0.124	0.155	0.217	-0.013	-0.102
Hidden-state norm	0.102	-0.167	-0.169	-0.018	0.146	0.165
Hidden/router geometry	-0.202	-0.240	-0.142	-0.243	-0.206	-0.119
Top-1 switch rate	0.122	0.128	0.176	0.279	0.391	0.250
Adjacent top-8 overlap	-0.388	-0.431	-0.500	-0.378	-0.409	-0.427

Table 54. Full reasoning: metrics versus hidden/router geometry. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.214	-0.169	-0.089	0.102	0.063	0.349
Router confidence	-0.257	-0.053	-0.018	-0.191	-0.179	-0.318
Top-1/top-2 margin	-0.167	-0.013	0.028	-0.084	-0.068	-0.071
Top-8 probability mass	-0.459	-0.462	-0.292	-0.451	-0.417	-0.485
Top-8/top-9 margin	-0.440	-0.524	-0.359	-0.464	-0.457	-0.489
Top-8/non-top-8 mean gap	-0.459	-0.462	-0.292	-0.451	-0.417	-0.485
Hidden-state norm	-0.235	-0.312	-0.213	-0.267	-0.184	-0.336
Hidden-state step distance	-0.202	-0.240	-0.142	-0.243	-0.206	-0.119
Top-1 switch rate	-0.015	-0.202	0.031	-0.093	-0.001	-0.035
Adjacent top-8 overlap	-0.264	-0.017	-0.175	-0.245	-0.206	-0.128

Table 55. Full reasoning: metrics versus top-1 switch rate. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	0.196	0.333	0.046	0.074	0.303	-0.121
Router confidence	-0.375	-0.495	-0.334	-0.184	-0.408	-0.349
Top-1/top-2 margin	-0.468	-0.634	-0.404	-0.331	-0.588	-0.573
Top-8 probability mass	-0.134	0.085	-0.001	0.001	-0.142	-0.185
Top-8/top-9 margin	-0.023	0.268	0.163	0.083	-0.038	0.049
Top-8/non-top-8 mean gap	-0.134	0.085	-0.001	0.001	-0.142	-0.185
Hidden-state norm	0.184	0.227	0.216	0.169	0.036	0.244
Hidden-state step distance	0.122	0.128	0.176	0.279	0.391	0.250
Hidden/router geometry	-0.015	-0.202	0.031	-0.093	-0.001	-0.035
Adjacent top-8 overlap	-0.415	-0.507	-0.446	-0.487	-0.505	-0.787

Table 56. Full reasoning: metrics versus adjacent top-8 overlap. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	500	960	960	675	1,280	272
Token count	-0.090	-0.053	0.153	0.014	-0.270	0.197
Router confidence	0.370	0.227	0.050	0.184	0.421	0.464
Top-1/top-2 margin	0.039	0.060	-0.151	-0.001	0.213	0.325
Top-8 probability mass	0.720	0.406	0.436	0.510	0.658	0.568
Top-8/top-9 margin	0.625	0.278	0.368	0.491	0.570	0.381
Top-8/non-top-8 mean gap	0.720	0.406	0.436	0.510	0.658	0.568
Hidden-state norm	0.198	0.290	0.322	0.300	0.278	0.122
Hidden-state step distance	-0.388	-0.431	-0.500	-0.378	-0.409	-0.427
Hidden/router geometry	-0.264	-0.017	-0.175	-0.245	-0.206	-0.128
Top-1 switch rate	-0.415	-0.507	-0.446	-0.487	-0.505	-0.787

B.3.3 Position windows, organized by target. Each table keeps the same target throughout. Separate panels give the non-cumulative windows, with metrics in rows and benchmarks in columns.

Table 57. Position windows: metrics versus router confidence. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Top-1/top-2 margin	0.369	0.341	0.212	0.415	0.390	0.659
Top-8 probability mass	0.488	0.551	0.343	0.254	0.154	0.658
Top-8/top-9 margin	0.310	0.456	0.252	0.111	-0.083	0.425
Top-8/non-top-8 mean gap	0.488	0.551	0.343	0.254	0.154	0.658
Hidden-state norm	-0.315	0.099	0.041	-0.189	-0.371	-0.109
Hidden-state step distance	-0.420	-0.327	-0.335	-0.312	-0.439	-0.162
Hidden/router geometry	0.013	-0.097	0.094	0.085	0.224	-0.201
Top-1 switch rate	-0.397	-0.114	-0.235	-0.230	-0.396	-0.575
Adjacent top-8 overlap	0.430	0.380	0.290	0.122	0.138	0.646
Bin 1						
Finite traces	429	960	960	668	1,269	269
Top-1/top-2 margin	0.513	0.292	0.509	0.530	0.534	0.505
Top-8 probability mass	0.700	0.521	0.305	0.589	0.625	0.755
Top-8/top-9 margin	0.603	0.412	0.171	0.453	0.476	0.631
Top-8/non-top-8 mean gap	0.700	0.521	0.305	0.589	0.625	0.755
Hidden-state norm	-0.418	0.190	-0.137	-0.212	-0.215	-0.074
Hidden-state step distance	-0.479	-0.284	-0.285	-0.386	-0.506	-0.242
Hidden/router geometry	0.110	-0.175	0.073	0.033	0.076	-0.064
Top-1 switch rate	-0.193	-0.244	-0.426	-0.195	-0.449	-0.258
Adjacent top-8 overlap	0.500	0.431	0.218	0.315	0.486	0.511
Bin 2						
Finite traces	347	960	960	634	1,172	254
Top-1/top-2 margin	0.616	0.374	0.551	0.537	0.523	0.527
Top-8 probability mass	0.773	0.502	0.473	0.705	0.771	0.735
Top-8/top-9 margin	0.684	0.369	0.321	0.609	0.682	0.622
Top-8/non-top-8 mean gap	0.773	0.502	0.473	0.705	0.771	0.735
Hidden-state norm	-0.516	0.079	-0.113	-0.333	-0.110	-0.182
Hidden-state step distance	-0.459	-0.250	-0.313	-0.334	-0.426	-0.233
Hidden/router geometry	0.015	-0.098	0.049	0.022	-0.101	-0.196
Top-1 switch rate	-0.389	-0.207	-0.415	-0.200	-0.351	-0.187
Adjacent top-8 overlap	0.605	0.340	0.358	0.401	0.598	0.429
Bin 3						
Finite traces	268	956	942	564	1,038	234
Top-1/top-2 margin	0.651	0.408	0.484	0.608	0.508	0.514
Top-8 probability mass	0.839	0.494	0.588	0.728	0.797	0.743

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8/top-9 margin	0.789	0.362	0.467	0.633	0.729	0.663
Top-8/non-top-8 mean gap	0.839	0.494	0.588	0.728	0.797	0.743
Hidden-state norm	−0.393	−0.089	−0.114	−0.294	−0.014	−0.112
Hidden-state step distance	−0.366	−0.142	−0.249	−0.244	−0.362	−0.047
Hidden/router geometry	−0.242	−0.042	0.033	−0.011	−0.143	−0.212
Top-1 switch rate	−0.277	−0.258	−0.334	−0.203	−0.285	−0.010
Adjacent top-8 overlap	0.626	0.338	0.404	0.466	0.620	0.334
Bin 4						
Top-1/top-2 margin	0.600 (204)	0.448 (927)	0.453 (900)	0.542 (494)	0.541 (876)	0.487 (202)
Top-8 probability mass	0.761 (204)	0.497 (927)	0.563 (900)	0.713 (494)	0.792 (876)	0.772 (202)
Top-8/top-9 margin	0.728 (204)	0.373 (927)	0.457 (900)	0.623 (494)	0.714 (876)	0.703 (202)
Top-8/non-top-8 mean gap	0.761 (204)	0.497 (927)	0.563 (900)	0.713 (494)	0.792 (876)	0.772 (202)
Hidden-state norm	−0.185 (204)	−0.163 (927)	−0.113 (900)	−0.153 (494)	0.117 (876)	−0.182 (202)
Hidden-state step distance	−0.111 (203)	−0.044 (927)	−0.204 (900)	−0.109 (494)	−0.184 (876)	−0.052 (202)
Hidden/router geometry	−0.104 (203)	−0.005 (927)	−0.019 (900)	−0.121 (494)	−0.173 (876)	−0.085 (202)
Top-1 switch rate	−0.217 (203)	−0.206 (927)	−0.290 (900)	−0.150 (494)	−0.268 (876)	−0.025 (202)
Adjacent top-8 overlap	0.466 (203)	0.256 (927)	0.380 (900)	0.469 (494)	0.552 (876)	0.292 (202)
Bin 5						
Top-1/top-2 margin	0.673 (171)	0.452 (859)	0.539 (847)	0.500 (427)	0.528 (767)	0.523 (162)
Top-8 probability mass	0.814 (171)	0.533 (859)	0.555 (847)	0.719 (427)	0.791 (767)	0.747 (162)
Top-8/top-9 margin	0.774 (171)	0.426 (859)	0.444 (847)	0.618 (427)	0.737 (767)	0.643 (162)
Top-8/non-top-8 mean gap	0.814 (171)	0.533 (859)	0.555 (847)	0.719 (427)	0.791 (767)	0.747 (162)
Hidden-state norm	0.032 (171)	−0.103 (859)	−0.144 (847)	−0.112 (427)	0.210 (767)	−0.213 (162)
Hidden-state step distance	0.079 (171)	−0.061 (858)	−0.029 (847)	−0.026 (426)	−0.134 (767)	0.125 (162)
Hidden/router geometry	−0.285 (171)	−0.076 (858)	−0.002 (847)	−0.135 (426)	−0.214 (767)	−0.088 (162)
Top-1 switch rate	−0.166 (171)	−0.197 (858)	−0.277 (847)	−0.082 (426)	−0.184 (767)	−0.001 (162)
Adjacent top-8 overlap	0.399 (171)	0.278 (858)	0.319 (847)	0.379 (426)	0.532 (767)	0.207 (162)
Bin 6						
Top-1/top-2 margin	0.726 (139)	0.420 (816)	0.473 (803)	0.478 (384)	0.529 (684)	0.502 (133)
Top-8 probability mass	0.778 (139)	0.600 (816)	0.537 (803)	0.747 (384)	0.767 (684)	0.778 (133)
Top-8/top-9 margin	0.734 (139)	0.496 (816)	0.438 (803)	0.665 (384)	0.688 (684)	0.714 (133)
Top-8/non-top-8 mean gap	0.778 (139)	0.600 (816)	0.537 (803)	0.747 (384)	0.767 (684)	0.778 (133)
Hidden-state norm	0.133 (139)	−0.091 (816)	−0.135 (803)	−0.050 (384)	0.190 (684)	−0.092 (133)
Hidden-state step distance	0.425 (139)	−0.082 (816)	0.053 (803)	0.029 (383)	0.019 (684)	0.008 (133)
Hidden/router geometry	−0.217 (139)	−0.097 (816)	0.012 (803)	−0.060 (383)	−0.181 (684)	−0.311 (133)
Top-1 switch rate	−0.257 (139)	−0.156 (816)	−0.214 (803)	−0.031 (383)	−0.163 (684)	−0.042 (133)
Adjacent top-8 overlap	0.244 (139)	0.327 (816)	0.257 (803)	0.396 (383)	0.481 (684)	0.379 (133)
Bin 7						
Top-1/top-2 margin	0.686 (106)	0.402 (748)	0.510 (752)	0.516 (342)	0.608 (559)	0.533 (102)
Top-8 probability mass	0.795 (106)	0.596 (748)	0.559 (752)	0.722 (342)	0.797 (559)	0.748 (102)
Top-8/top-9 margin	0.767 (106)	0.481 (748)	0.477 (752)	0.651 (342)	0.729 (559)	0.662 (102)
Top-8/non-top-8 mean gap	0.795 (106)	0.596 (748)	0.559 (752)	0.722 (342)	0.797 (559)	0.748 (102)
Hidden-state norm	0.365 (106)	−0.078 (748)	−0.118 (752)	0.001 (342)	0.216 (559)	−0.042 (102)
Hidden-state step distance	0.498 (106)	−0.052 (748)	0.114 (751)	0.086 (342)	0.109 (559)	−0.000 (102)
Hidden/router geometry	−0.502 (106)	−0.071 (748)	−0.069 (751)	−0.181 (342)	−0.270 (559)	−0.262 (102)
Top-1 switch rate	−0.455 (106)	−0.129 (748)	−0.228 (751)	−0.062 (342)	−0.169 (559)	−0.172 (102)
Adjacent top-8 overlap	0.254 (106)	0.285 (748)	0.271 (751)	0.323 (342)	0.455 (559)	0.283 (102)
Bin 8						
Finite traces	83	679	724	314	436	83
Top-1/top-2 margin	0.832	0.423	0.506	0.500	0.639	0.748
Top-8 probability mass	0.864	0.662	0.612	0.726	0.792	0.789
Top-8/top-9 margin	0.789	0.557	0.520	0.651	0.717	0.715
Top-8/non-top-8 mean gap	0.864	0.662	0.612	0.726	0.792	0.789
Hidden-state norm	0.367	−0.029	−0.107	0.115	0.318	−0.015
Hidden-state step distance	0.547	0.041	0.185	0.118	0.258	0.081
Hidden/router geometry	−0.446	−0.074	−0.163	−0.234	−0.297	−0.200
Top-1 switch rate	−0.393	−0.094	−0.242	−0.083	−0.170	−0.231
Adjacent top-8 overlap	0.314	0.293	0.313	0.384	0.403	0.398
Bin 9						
Finite traces	65	605	701	291	317	70

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-1/top-2 margin	0.876	0.446	0.519	0.479	0.667	0.679
Top-8 probability mass	0.835	0.701	0.670	0.762	0.788	0.891
Top-8/top-9 margin	0.791	0.594	0.579	0.675	0.725	0.855
Top-8/non-top-8 mean gap	0.835	0.701	0.670	0.762	0.788	0.891
Hidden-state norm	0.466	-0.023	-0.055	0.165	0.424	0.172
Hidden-state step distance	0.764	-0.003	0.205	0.167	0.313	0.253
Hidden/router geometry	-0.650	-0.115	-0.178	-0.248	-0.311	-0.333
Top-1 switch rate	-0.589	-0.126	-0.206	-0.059	-0.178	-0.232
Adjacent top-8 overlap	0.287	0.344	0.335	0.455	0.399	0.538
Overflow						
Finite traces	54	540	664	270	238	59
Top-1/top-2 margin	0.879	0.620	0.675	0.703	0.691	0.871
Top-8 probability mass	0.940	0.737	0.727	0.799	0.822	0.884
Top-8/top-9 margin	0.903	0.602	0.622	0.720	0.763	0.841
Top-8/non-top-8 mean gap	0.940	0.737	0.727	0.799	0.822	0.884
Hidden-state norm	0.625	0.118	-0.147	0.284	0.507	0.412
Hidden-state step distance	0.823	0.215	0.394	0.354	0.470	0.596
Hidden/router geometry	-0.650	-0.131	-0.047	-0.276	-0.322	-0.565
Top-1 switch rate	-0.746	-0.279	-0.174	-0.085	-0.124	-0.632
Adjacent top-8 overlap	0.674	0.364	0.237	0.348	0.424	0.707

Table 58. Position windows: metrics versus top-1/top-2 margin. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	0.369	0.341	0.212	0.415	0.390	0.659
Top-8 probability mass	-0.016	-0.146	-0.420	-0.208	-0.015	0.227
Top-8/top-9 margin	-0.070	-0.226	-0.505	-0.309	-0.091	-0.081
Top-8/non-top-8 mean gap	-0.016	-0.146	-0.420	-0.208	-0.015	0.227
Hidden-state norm	-0.226	-0.303	-0.459	-0.369	-0.099	-0.449
Hidden-state step distance	0.045	0.063	0.128	-0.105	-0.069	-0.019
Hidden/router geometry	0.037	0.108	0.225	0.222	0.087	-0.064
Top-1 switch rate	-0.337	-0.320	-0.434	-0.350	-0.282	-0.672
Adjacent top-8 overlap	-0.163	-0.248	-0.421	-0.276	-0.150	0.397
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	0.513	0.292	0.509	0.530	0.534	0.505
Top-8 probability mass	0.170	-0.092	-0.233	0.040	0.249	0.111
Top-8/top-9 margin	0.114	-0.180	-0.341	-0.080	0.146	-0.119
Top-8/non-top-8 mean gap	0.170	-0.092	-0.233	0.040	0.249	0.111
Hidden-state norm	-0.324	-0.180	-0.437	-0.384	-0.129	-0.492
Hidden-state step distance	-0.050	0.098	0.025	-0.156	-0.235	0.000
Hidden/router geometry	0.074	0.080	0.224	0.130	0.058	0.150
Top-1 switch rate	-0.357	-0.510	-0.524	-0.345	-0.477	-0.619
Adjacent top-8 overlap	0.035	-0.048	-0.290	-0.110	0.144	0.342
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	0.616	0.374	0.551	0.537	0.523	0.527
Top-8 probability mass	0.303	-0.114	-0.072	0.132	0.243	0.075
Top-8/top-9 margin	0.235	-0.213	-0.211	0.044	0.173	-0.145
Top-8/non-top-8 mean gap	0.303	-0.114	-0.072	0.132	0.243	0.075
Hidden-state norm	-0.431	-0.242	-0.355	-0.431	-0.142	-0.554
Hidden-state step distance	-0.055	0.178	0.034	-0.119	-0.153	0.096
Hidden/router geometry	0.056	0.013	0.109	0.101	-0.047	0.033
Top-1 switch rate	-0.437	-0.512	-0.470	-0.390	-0.462	-0.541

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Adjacent top-8 overlap	0.152	−0.108	−0.125	−0.004	0.145	0.244
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	0.651	0.408	0.484	0.608	0.508	0.514
Top-8 probability mass	0.370	−0.104	−0.022	0.233	0.222	0.086
Top-8/top-9 margin	0.332	−0.204	−0.138	0.140	0.155	−0.100
Top-8/non-top-8 mean gap	0.370	−0.104	−0.022	0.233	0.222	0.086
Hidden-state norm	−0.399	−0.303	−0.354	−0.383	−0.109	−0.517
Hidden-state step distance	−0.098	0.188	0.077	−0.101	−0.087	0.156
Hidden/router geometry	−0.070	0.063	0.104	0.075	−0.024	0.069
Top-1 switch rate	−0.417	−0.493	−0.419	−0.358	−0.416	−0.478
Adjacent top-8 overlap	0.240	−0.073	−0.082	0.087	0.106	0.219
Bin 4						
Router confidence	0.600 (204)	0.448 (927)	0.453 (900)	0.542 (494)	0.541 (876)	0.487 (202)
Top-8 probability mass	0.265 (204)	−0.024 (927)	−0.035 (900)	0.171 (494)	0.257 (876)	0.098 (202)
Top-8/top-9 margin	0.223 (204)	−0.133 (927)	−0.131 (900)	0.071 (494)	0.169 (876)	−0.078 (202)
Top-8/non-top-8 mean gap	0.265 (204)	−0.024 (927)	−0.035 (900)	0.171 (494)	0.257 (876)	0.098 (202)
Hidden-state norm	−0.246 (204)	−0.313 (927)	−0.299 (900)	−0.251 (494)	−0.025 (876)	−0.561 (202)
Hidden-state step distance	0.204 (203)	0.265 (927)	0.234 (900)	0.114 (494)	0.061 (876)	0.161 (202)
Hidden/router geometry	−0.054 (203)	0.069 (927)	0.058 (900)	0.001 (494)	−0.118 (876)	0.168 (202)
Top-1 switch rate	−0.359 (203)	−0.421 (927)	−0.312 (900)	−0.341 (494)	−0.401 (876)	−0.515 (202)
Adjacent top-8 overlap	−0.034 (203)	−0.099 (927)	−0.154 (900)	0.064 (494)	0.085 (876)	0.259 (202)
Bin 5						
Router confidence	0.673 (171)	0.452 (859)	0.539 (847)	0.500 (427)	0.528 (767)	0.523 (162)
Top-8 probability mass	0.459 (171)	−0.061 (859)	0.025 (847)	0.088 (427)	0.246 (767)	0.086 (162)
Top-8/top-9 margin	0.414 (171)	−0.154 (859)	−0.086 (847)	−0.023 (427)	0.172 (767)	−0.065 (162)
Top-8/non-top-8 mean gap	0.459 (171)	−0.061 (859)	0.025 (847)	0.088 (427)	0.246 (767)	0.086 (162)
Hidden-state norm	0.027 (171)	−0.313 (859)	−0.337 (847)	−0.278 (427)	0.101 (767)	−0.494 (162)
Hidden-state step distance	0.277 (171)	0.238 (858)	0.277 (847)	0.260 (426)	0.079 (767)	0.343 (162)
Hidden/router geometry	−0.206 (171)	0.062 (858)	0.103 (847)	0.017 (426)	−0.111 (767)	0.230 (162)
Top-1 switch rate	−0.306 (171)	−0.388 (858)	−0.333 (847)	−0.342 (426)	−0.396 (767)	−0.369 (162)
Adjacent top-8 overlap	0.040 (171)	−0.127 (858)	−0.116 (847)	−0.090 (426)	0.062 (767)	0.062 (162)
Bin 6						
Router confidence	0.726 (139)	0.420 (816)	0.473 (803)	0.478 (384)	0.529 (684)	0.502 (133)
Top-8 probability mass	0.473 (139)	−0.031 (816)	−0.026 (803)	0.133 (384)	0.206 (684)	0.209 (133)
Top-8/top-9 margin	0.416 (139)	−0.124 (816)	−0.132 (803)	0.040 (384)	0.122 (684)	0.055 (133)
Top-8/non-top-8 mean gap	0.473 (139)	−0.031 (816)	−0.026 (803)	0.133 (384)	0.206 (684)	0.209 (133)
Hidden-state norm	0.110 (139)	−0.335 (816)	−0.278 (803)	−0.176 (384)	0.081 (684)	−0.410 (133)
Hidden-state step distance	0.566 (139)	0.265 (816)	0.364 (803)	0.248 (383)	0.209 (684)	0.291 (133)
Hidden/router geometry	−0.122 (139)	0.075 (816)	0.081 (803)	−0.001 (383)	−0.100 (684)	−0.118 (133)
Top-1 switch rate	−0.381 (139)	−0.376 (816)	−0.318 (803)	−0.314 (383)	−0.371 (684)	−0.393 (133)
Adjacent top-8 overlap	0.003 (139)	−0.114 (816)	−0.188 (803)	−0.013 (383)	0.062 (684)	0.194 (133)
Bin 7						
Router confidence	0.686 (106)	0.402 (748)	0.510 (752)	0.516 (342)	0.608 (559)	0.533 (102)
Top-8 probability mass	0.467 (106)	−0.043 (748)	0.046 (752)	0.132 (342)	0.286 (559)	0.253 (102)
Top-8/top-9 margin	0.468 (106)	−0.128 (748)	−0.056 (752)	0.054 (342)	0.202 (559)	0.031 (102)
Top-8/non-top-8 mean gap	0.467 (106)	−0.043 (748)	0.046 (752)	0.132 (342)	0.286 (559)	0.253 (102)
Hidden-state norm	0.305 (106)	−0.360 (748)	−0.249 (752)	−0.166 (342)	0.073 (559)	−0.376 (102)
Hidden-state step distance	0.661 (106)	0.306 (748)	0.367 (751)	0.296 (342)	0.239 (559)	0.326 (102)
Hidden/router geometry	−0.385 (106)	0.100 (748)	−0.017 (751)	−0.062 (342)	−0.091 (559)	−0.061 (102)
Top-1 switch rate	−0.576 (106)	−0.386 (748)	−0.346 (751)	−0.256 (342)	−0.367 (559)	−0.552 (102)
Adjacent top-8 overlap	−0.042 (106)	−0.099 (748)	−0.122 (751)	−0.126 (342)	0.060 (559)	0.319 (102)
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	0.832	0.423	0.506	0.500	0.639	0.748
Top-8 probability mass	0.647	0.034	0.055	0.141	0.340	0.427
Top-8/top-9 margin	0.602	−0.062	−0.058	0.058	0.250	0.292
Top-8/non-top-8 mean gap	0.647	0.034	0.055	0.141	0.340	0.427
Hidden-state norm	0.317	−0.340	−0.259	−0.107	0.210	−0.251
Hidden-state step distance	0.671	0.322	0.475	0.312	0.387	0.287

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden/router geometry	−0.393	0.045	−0.052	−0.082	−0.168	−0.080
Top-1 switch rate	−0.494	−0.390	−0.323	−0.293	−0.343	−0.548
Adjacent top-8 overlap	0.085	−0.040	−0.120	−0.044	0.034	0.398
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	0.876	0.446	0.519	0.479	0.667	0.679
Top-8 probability mass	0.629	0.071	0.157	0.170	0.325	0.546
Top-8/top-9 margin	0.603	−0.010	0.042	0.087	0.254	0.435
Top-8/non-top-8 mean gap	0.629	0.071	0.157	0.170	0.325	0.546
Hidden-state norm	0.423	−0.345	−0.216	−0.040	0.274	0.033
Hidden-state step distance	0.820	0.305	0.519	0.354	0.427	0.468
Hidden/router geometry	−0.550	0.022	−0.084	−0.064	−0.225	−0.084
Top-1 switch rate	−0.721	−0.386	−0.321	−0.240	−0.299	−0.533
Adjacent top-8 overlap	0.133	−0.034	−0.094	0.016	−0.002	0.561
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	0.879	0.620	0.675	0.703	0.691	0.871
Top-8 probability mass	0.761	0.362	0.373	0.489	0.400	0.769
Top-8/top-9 margin	0.725	0.263	0.236	0.398	0.335	0.676
Top-8/non-top-8 mean gap	0.761	0.362	0.373	0.489	0.400	0.769
Hidden-state norm	0.562	−0.053	−0.073	0.183	0.445	0.370
Hidden-state step distance	0.896	0.393	0.564	0.513	0.463	0.740
Hidden/router geometry	−0.601	−0.021	−0.045	−0.136	−0.214	−0.465
Top-1 switch rate	−0.837	−0.401	−0.379	−0.286	−0.391	−0.720
Adjacent top-8 overlap	0.510	0.104	−0.011	0.168	0.155	0.729

Table 59. Position windows: metrics versus top-8 probability mass. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	0.488	0.551	0.343	0.254	0.154	0.658
Top-1/top-2 margin	−0.016	−0.146	−0.420	−0.208	−0.015	0.227
Top-8/top-9 margin	0.940	0.970	0.958	0.955	0.913	0.916
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	0.384	0.664	0.727	0.691	0.591	0.485
Hidden-state step distance	−0.166	−0.261	−0.151	0.073	0.061	0.029
Hidden/router geometry	−0.432	−0.432	−0.485	−0.497	−0.448	−0.453
Top-1 switch rate	−0.003	0.237	0.182	0.107	−0.056	−0.295
Adjacent top-8 overlap	0.729	0.709	0.848	0.755	0.697	0.750
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	0.700	0.521	0.305	0.589	0.625	0.755
Top-1/top-2 margin	0.170	−0.092	−0.233	0.040	0.249	0.111
Top-8/top-9 margin	0.964	0.969	0.962	0.957	0.951	0.932
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	0.088	0.758	0.733	0.424	0.306	0.417
Hidden-state step distance	−0.405	−0.194	−0.078	−0.193	−0.364	−0.201
Hidden/router geometry	−0.221	−0.479	−0.462	−0.360	−0.239	−0.322
Top-1 switch rate	0.043	0.170	0.179	0.041	−0.154	−0.020
Adjacent top-8 overlap	0.750	0.694	0.773	0.697	0.756	0.570
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	0.773	0.502	0.473	0.705	0.771	0.735
Top-1/top-2 margin	0.303	−0.114	−0.072	0.132	0.243	0.075
Top-8/top-9 margin	0.965	0.957	0.951	0.964	0.970	0.927

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	−0.076	0.671	0.573	0.177	0.265	0.334
Hidden-state step distance	−0.414	−0.106	−0.283	−0.161	−0.335	−0.299
Hidden/router geometry	−0.301	−0.447	−0.305	−0.292	−0.298	−0.434
Top-1 switch rate	−0.160	0.221	0.016	0.011	−0.151	0.002
Adjacent top-8 overlap	0.773	0.599	0.726	0.682	0.778	0.552
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	0.839	0.494	0.588	0.728	0.797	0.743
Top-1/top-2 margin	0.370	−0.104	−0.022	0.233	0.222	0.086
Top-8/top-9 margin	0.973	0.963	0.958	0.968	0.976	0.934
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	−0.092	0.589	0.525	0.178	0.311	0.367
Hidden-state step distance	−0.358	−0.034	−0.243	−0.108	−0.292	−0.133
Hidden/router geometry	−0.429	−0.404	−0.317	−0.333	−0.328	−0.478
Top-1 switch rate	−0.120	0.196	−0.001	0.005	−0.091	0.165
Adjacent top-8 overlap	0.788	0.589	0.702	0.700	0.798	0.462
Bin 4						
Router confidence	0.761 (204)	0.497 (927)	0.563 (900)	0.713 (494)	0.792 (876)	0.772 (202)
Top-1/top-2 margin	0.265 (204)	−0.024 (927)	−0.035 (900)	0.171 (494)	0.257 (876)	0.098 (202)
Top-8/top-9 margin	0.974 (204)	0.960 (927)	0.963 (900)	0.963 (494)	0.969 (876)	0.932 (202)
Top-8/non-top-8 mean gap	1.000 (204)	1.000 (927)	1.000 (900)	1.000 (494)	1.000 (876)	1.000 (202)
Hidden-state norm	0.199 (204)	0.499 (927)	0.533 (900)	0.292 (494)	0.419 (876)	0.207 (202)
Hidden-state step distance	−0.167 (203)	0.081 (927)	−0.231 (900)	−0.049 (494)	−0.147 (876)	−0.197 (202)
Hidden/router geometry	−0.421 (203)	−0.407 (927)	−0.387 (900)	−0.378 (494)	−0.374 (876)	−0.397 (202)
Top-1 switch rate	−0.016 (203)	0.190 (927)	0.003 (900)	0.047 (494)	−0.071 (876)	0.150 (202)
Adjacent top-8 overlap	0.649 (203)	0.545 (927)	0.703 (900)	0.711 (494)	0.739 (876)	0.404 (202)
Bin 5						
Router confidence	0.814 (171)	0.533 (859)	0.555 (847)	0.719 (427)	0.791 (767)	0.747 (162)
Top-1/top-2 margin	0.459 (171)	−0.061 (859)	0.025 (847)	0.088 (427)	0.246 (767)	0.086 (162)
Top-8/top-9 margin	0.966 (171)	0.961 (859)	0.965 (847)	0.961 (427)	0.972 (767)	0.907 (162)
Top-8/non-top-8 mean gap	1.000 (171)	1.000 (859)	1.000 (847)	1.000 (427)	1.000 (767)	1.000 (162)
Hidden-state norm	0.352 (171)	0.469 (859)	0.490 (847)	0.310 (427)	0.500 (767)	0.245 (162)
Hidden-state step distance	0.109 (171)	−0.011 (858)	−0.104 (847)	−0.036 (426)	−0.070 (767)	−0.022 (162)
Hidden/router geometry	−0.542 (171)	−0.378 (858)	−0.383 (847)	−0.366 (426)	−0.414 (767)	−0.412 (162)
Top-1 switch rate	0.022 (171)	0.173 (858)	0.032 (847)	0.132 (426)	0.003 (767)	0.196 (162)
Adjacent top-8 overlap	0.588 (171)	0.566 (858)	0.662 (847)	0.664 (426)	0.728 (767)	0.367 (162)
Bin 6						
Router confidence	0.778 (139)	0.600 (816)	0.537 (803)	0.747 (384)	0.767 (684)	0.778 (133)
Top-1/top-2 margin	0.473 (139)	−0.031 (816)	−0.026 (803)	0.133 (384)	0.206 (684)	0.209 (133)
Top-8/top-9 margin	0.970 (139)	0.966 (816)	0.964 (803)	0.960 (384)	0.967 (684)	0.943 (133)
Top-8/non-top-8 mean gap	1.000 (139)	1.000 (816)	1.000 (803)	1.000 (384)	1.000 (684)	1.000 (133)
Hidden-state norm	0.442 (139)	0.462 (816)	0.488 (803)	0.340 (384)	0.482 (684)	0.319 (133)
Hidden-state step distance	0.387 (139)	−0.030 (816)	−0.094 (803)	0.105 (383)	0.086 (684)	−0.022 (133)
Hidden/router geometry	−0.439 (139)	−0.379 (816)	−0.350 (803)	−0.363 (383)	−0.376 (684)	−0.522 (133)
Top-1 switch rate	−0.114 (139)	0.213 (816)	0.068 (803)	0.175 (383)	0.028 (684)	0.068 (133)
Adjacent top-8 overlap	0.499 (139)	0.540 (816)	0.613 (803)	0.601 (383)	0.700 (684)	0.535 (133)
Bin 7						
Router confidence	0.795 (106)	0.596 (748)	0.559 (752)	0.722 (342)	0.797 (559)	0.748 (102)
Top-1/top-2 margin	0.467 (106)	−0.043 (748)	0.046 (752)	0.132 (342)	0.286 (559)	0.253 (102)
Top-8/top-9 margin	0.979 (106)	0.958 (748)	0.966 (752)	0.961 (342)	0.968 (559)	0.899 (102)
Top-8/non-top-8 mean gap	1.000 (106)	1.000 (748)	1.000 (752)	1.000 (342)	1.000 (559)	1.000 (102)
Hidden-state norm	0.648 (106)	0.506 (748)	0.459 (752)	0.409 (342)	0.464 (559)	0.364 (102)
Hidden-state step distance	0.467 (106)	−0.010 (748)	−0.037 (751)	0.104 (342)	0.118 (559)	−0.068 (102)
Hidden/router geometry	−0.656 (106)	−0.371 (748)	−0.334 (751)	−0.365 (342)	−0.478 (559)	−0.481 (102)
Top-1 switch rate	−0.317 (106)	0.260 (748)	0.042 (751)	0.156 (342)	−0.021 (559)	−0.034 (102)
Adjacent top-8 overlap	0.530 (106)	0.494 (748)	0.653 (751)	0.582 (342)	0.702 (559)	0.419 (102)
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	0.864	0.662	0.612	0.726	0.792	0.789

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-1/top-2 margin	0.647	0.034	0.055	0.141	0.340	0.427
Top-8/top-9 margin	0.940	0.962	0.966	0.963	0.974	0.923
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	0.597	0.493	0.430	0.505	0.552	0.364
Hidden-state step distance	0.483	0.086	−0.020	0.123	0.248	0.039
Hidden/router geometry	−0.565	−0.369	−0.353	−0.426	−0.495	−0.443
Top-1 switch rate	−0.334	0.252	0.030	0.065	−0.040	−0.090
Adjacent top-8 overlap	0.554	0.460	0.629	0.618	0.657	0.559
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	0.835	0.701	0.670	0.762	0.788	0.891
Top-1/top-2 margin	0.629	0.071	0.157	0.170	0.325	0.546
Top-8/top-9 margin	0.982	0.965	0.963	0.967	0.973	0.948
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	0.687	0.446	0.423	0.564	0.645	0.359
Hidden-state step distance	0.626	0.012	0.056	0.244	0.270	0.232
Hidden/router geometry	−0.732	−0.335	−0.325	−0.514	−0.501	−0.479
Top-1 switch rate	−0.402	0.218	0.056	0.162	−0.085	−0.112
Adjacent top-8 overlap	0.543	0.484	0.586	0.602	0.697	0.548
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	0.940	0.737	0.727	0.799	0.822	0.884
Top-1/top-2 margin	0.761	0.362	0.373	0.489	0.400	0.769
Top-8/top-9 margin	0.979	0.955	0.957	0.969	0.974	0.954
Top-8/non-top-8 mean gap	1.000	1.000	1.000	1.000	1.000	1.000
Hidden-state norm	0.750	0.498	0.272	0.630	0.670	0.602
Hidden-state step distance	0.764	0.361	0.353	0.414	0.423	0.656
Hidden/router geometry	−0.751	−0.408	−0.220	−0.469	−0.523	−0.672
Top-1 switch rate	−0.663	0.073	0.052	0.086	0.070	−0.511
Adjacent top-8 overlap	0.774	0.410	0.464	0.492	0.645	0.719

Table 60. Position windows: metrics versus top-8/top-9 margin. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	0.310	0.456	0.252	0.111	−0.083	0.425
Top-1/top-2 margin	−0.070	−0.226	−0.505	−0.309	−0.091	−0.081
Top-8 probability mass	0.940	0.970	0.958	0.955	0.913	0.916
Top-8/non-top-8 mean gap	0.940	0.970	0.958	0.955	0.913	0.916
Hidden-state norm	0.538	0.712	0.787	0.779	0.733	0.678
Hidden-state step distance	−0.041	−0.209	−0.096	0.152	0.188	0.077
Hidden/router geometry	−0.468	−0.472	−0.516	−0.534	−0.524	−0.443
Top-1 switch rate	0.150	0.280	0.235	0.176	0.087	−0.004
Adjacent top-8 overlap	0.585	0.674	0.805	0.688	0.570	0.541
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	0.603	0.412	0.171	0.453	0.476	0.631
Top-1/top-2 margin	0.114	−0.180	−0.341	−0.080	0.146	−0.119
Top-8 probability mass	0.964	0.969	0.962	0.957	0.951	0.932
Top-8/non-top-8 mean gap	0.964	0.969	0.962	0.957	0.951	0.932
Hidden-state norm	0.189	0.774	0.790	0.534	0.443	0.509
Hidden-state step distance	−0.318	−0.136	0.002	−0.115	−0.267	−0.203
Hidden/router geometry	−0.266	−0.482	−0.479	−0.389	−0.314	−0.357
Top-1 switch rate	0.110	0.270	0.269	0.094	−0.055	0.227
Adjacent top-8 overlap	0.701	0.626	0.711	0.661	0.695	0.372

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	0.684	0.369	0.321	0.609	0.682	0.622
Top-1/top-2 margin	0.235	−0.213	−0.211	0.044	0.173	−0.145
Top-8 probability mass	0.965	0.957	0.951	0.964	0.970	0.927
Top-8/non-top-8 mean gap	0.965	0.957	0.951	0.964	0.970	0.927
Hidden-state norm	0.019	0.701	0.644	0.264	0.339	0.406
Hidden-state step distance	−0.344	−0.051	−0.203	−0.111	−0.273	−0.318
Hidden/router geometry	−0.362	−0.464	−0.323	−0.306	−0.339	−0.415
Top-1 switch rate	−0.081	0.338	0.147	0.070	−0.082	0.248
Adjacent top-8 overlap	0.720	0.502	0.632	0.637	0.736	0.361
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	0.789	0.362	0.467	0.633	0.729	0.663
Top-1/top-2 margin	0.332	−0.204	−0.138	0.140	0.155	−0.100
Top-8 probability mass	0.973	0.963	0.958	0.968	0.976	0.934
Top-8/non-top-8 mean gap	0.973	0.963	0.958	0.968	0.976	0.934
Hidden-state norm	−0.043	0.650	0.595	0.266	0.354	0.412
Hidden-state step distance	−0.304	0.007	−0.168	−0.071	−0.249	−0.148
Hidden/router geometry	−0.431	−0.418	−0.333	−0.366	−0.356	−0.458
Top-1 switch rate	−0.089	0.315	0.110	0.054	−0.024	0.385
Adjacent top-8 overlap	0.756	0.501	0.622	0.672	0.759	0.274
Bin 4						
Router confidence	0.728 (204)	0.373 (927)	0.457 (900)	0.623 (494)	0.714 (876)	0.703 (202)
Top-1/top-2 margin	0.223 (204)	−0.133 (927)	−0.131 (900)	0.071 (494)	0.169 (876)	−0.078 (202)
Top-8 probability mass	0.974 (204)	0.960 (927)	0.963 (900)	0.963 (494)	0.969 (876)	0.932 (202)
Top-8/non-top-8 mean gap	0.974 (204)	0.960 (927)	0.963 (900)	0.963 (494)	0.969 (876)	0.932 (202)
Hidden-state norm	0.216 (204)	0.560 (927)	0.596 (900)	0.376 (494)	0.465 (876)	0.261 (202)
Hidden-state step distance	−0.154 (203)	0.096 (927)	−0.173 (900)	−0.039 (494)	−0.132 (876)	−0.191 (202)
Hidden/router geometry	−0.419 (203)	−0.425 (927)	−0.395 (900)	−0.382 (494)	−0.391 (876)	−0.373 (202)
Top-1 switch rate	0.013 (203)	0.290 (927)	0.106 (900)	0.108 (494)	−0.015 (876)	0.370 (202)
Adjacent top-8 overlap	0.632 (203)	0.482 (927)	0.625 (900)	0.660 (494)	0.708 (876)	0.216 (202)
Bin 5						
Router confidence	0.774 (171)	0.426 (859)	0.444 (847)	0.618 (427)	0.737 (767)	0.643 (162)
Top-1/top-2 margin	0.414 (171)	−0.154 (859)	−0.086 (847)	−0.023 (427)	0.172 (767)	−0.065 (162)
Top-8 probability mass	0.966 (171)	0.961 (859)	0.965 (847)	0.961 (427)	0.972 (767)	0.907 (162)
Top-8/non-top-8 mean gap	0.966 (171)	0.961 (859)	0.965 (847)	0.961 (427)	0.972 (767)	0.907 (162)
Hidden-state norm	0.355 (171)	0.520 (859)	0.541 (847)	0.406 (427)	0.507 (767)	0.291 (162)
Hidden-state step distance	0.111 (171)	−0.002 (858)	−0.085 (847)	−0.034 (426)	−0.057 (767)	−0.057 (162)
Hidden/router geometry	−0.514 (171)	−0.384 (858)	−0.399 (847)	−0.360 (426)	−0.419 (767)	−0.419 (162)
Top-1 switch rate	0.020 (171)	0.253 (858)	0.130 (847)	0.186 (426)	0.057 (767)	0.419 (162)
Adjacent top-8 overlap	0.573 (171)	0.509 (858)	0.589 (847)	0.627 (426)	0.692 (767)	0.173 (162)
Bin 6						
Router confidence	0.734 (139)	0.496 (816)	0.438 (803)	0.665 (384)	0.688 (684)	0.714 (133)
Top-1/top-2 margin	0.416 (139)	−0.124 (816)	−0.132 (803)	0.040 (384)	0.122 (684)	0.055 (133)
Top-8 probability mass	0.970 (139)	0.966 (816)	0.964 (803)	0.960 (384)	0.967 (684)	0.943 (133)
Top-8/non-top-8 mean gap	0.970 (139)	0.966 (816)	0.964 (803)	0.960 (384)	0.967 (684)	0.943 (133)
Hidden-state norm	0.478 (139)	0.523 (816)	0.521 (803)	0.414 (384)	0.512 (684)	0.361 (133)
Hidden-state step distance	0.379 (139)	−0.019 (816)	−0.087 (803)	0.105 (383)	0.097 (684)	−0.056 (133)
Hidden/router geometry	−0.415 (139)	−0.384 (816)	−0.338 (803)	−0.358 (383)	−0.391 (684)	−0.485 (133)
Top-1 switch rate	−0.091 (139)	0.293 (816)	0.173 (803)	0.230 (383)	0.092 (684)	0.244 (133)
Adjacent top-8 overlap	0.475 (139)	0.484 (816)	0.538 (803)	0.551 (383)	0.656 (684)	0.387 (133)
Bin 7						
Router confidence	0.767 (106)	0.481 (748)	0.477 (752)	0.651 (342)	0.729 (559)	0.662 (102)
Top-1/top-2 margin	0.468 (106)	−0.128 (748)	−0.056 (752)	0.054 (342)	0.202 (559)	0.031 (102)
Top-8 probability mass	0.979 (106)	0.958 (748)	0.966 (752)	0.961 (342)	0.968 (559)	0.899 (102)
Top-8/non-top-8 mean gap	0.979 (106)	0.958 (748)	0.966 (752)	0.961 (342)	0.968 (559)	0.899 (102)
Hidden-state norm	0.671 (106)	0.567 (748)	0.486 (752)	0.483 (342)	0.492 (559)	0.485 (102)
Hidden-state step distance	0.493 (106)	−0.000 (748)	−0.020 (751)	0.094 (342)	0.096 (559)	−0.119 (102)
Hidden/router geometry	−0.625 (106)	−0.386 (748)	−0.309 (751)	−0.361 (342)	−0.486 (559)	−0.480 (102)

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-1 switch rate	−0.293 (106)	0.348 (748)	0.168 (751)	0.212 (342)	0.035 (559)	0.216 (102)
Adjacent top-8 overlap	0.508 (106)	0.428 (748)	0.568 (751)	0.538 (342)	0.684 (559)	0.215 (102)
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	0.789	0.557	0.520	0.651	0.717	0.715
Top-1/top-2 margin	0.602	−0.062	−0.058	0.058	0.250	0.292
Top-8 probability mass	0.940	0.962	0.966	0.963	0.974	0.923
Top-8/non-top-8 mean gap	0.940	0.962	0.966	0.963	0.974	0.923
Hidden-state norm	0.659	0.549	0.481	0.555	0.573	0.413
Hidden-state step distance	0.546	0.084	−0.033	0.118	0.239	−0.025
Hidden/router geometry	−0.620	−0.374	−0.343	−0.434	−0.494	−0.448
Top-1 switch rate	−0.376	0.344	0.134	0.145	0.014	0.116
Adjacent top-8 overlap	0.590	0.399	0.564	0.553	0.637	0.375
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	0.791	0.594	0.579	0.675	0.725	0.855
Top-1/top-2 margin	0.603	−0.010	0.042	0.087	0.254	0.435
Top-8 probability mass	0.982	0.965	0.963	0.967	0.973	0.948
Top-8/non-top-8 mean gap	0.982	0.965	0.963	0.967	0.973	0.948
Hidden-state norm	0.709	0.516	0.470	0.612	0.657	0.388
Hidden-state step distance	0.611	0.046	0.042	0.233	0.264	0.216
Hidden/router geometry	−0.730	−0.353	−0.285	−0.518	−0.510	−0.501
Top-1 switch rate	−0.395	0.315	0.181	0.215	−0.048	0.015
Adjacent top-8 overlap	0.554	0.418	0.504	0.547	0.685	0.443
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	0.903	0.602	0.622	0.720	0.763	0.841
Top-1/top-2 margin	0.725	0.263	0.236	0.398	0.335	0.676
Top-8 probability mass	0.979	0.955	0.957	0.969	0.974	0.954
Top-8/non-top-8 mean gap	0.979	0.955	0.957	0.969	0.974	0.954
Hidden-state norm	0.779	0.583	0.334	0.699	0.661	0.637
Hidden-state step distance	0.759	0.410	0.333	0.418	0.421	0.627
Hidden/router geometry	−0.761	−0.483	−0.210	−0.488	−0.520	−0.660
Top-1 switch rate	−0.659	0.207	0.186	0.172	0.112	−0.415
Adjacent top-8 overlap	0.801	0.325	0.392	0.430	0.612	0.618

Table 61. Position windows: metrics versus top-8/non-top-8 mean gap. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	0.488	0.551	0.343	0.254	0.154	0.658
Top-1/top-2 margin	−0.016	−0.146	−0.420	−0.208	−0.015	0.227
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.940	0.970	0.958	0.955	0.913	0.916
Hidden-state norm	0.384	0.664	0.727	0.691	0.591	0.485
Hidden-state step distance	−0.166	−0.261	−0.151	0.073	0.061	0.029
Hidden/router geometry	−0.432	−0.432	−0.485	−0.497	−0.448	−0.453
Top-1 switch rate	−0.003	0.237	0.182	0.107	−0.056	−0.295
Adjacent top-8 overlap	0.729	0.709	0.848	0.755	0.697	0.750
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	0.700	0.521	0.305	0.589	0.625	0.755
Top-1/top-2 margin	0.170	−0.092	−0.233	0.040	0.249	0.111
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.964	0.969	0.962	0.957	0.951	0.932

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden-state norm	0.088	0.758	0.733	0.424	0.306	0.417
Hidden-state step distance	−0.405	−0.194	−0.078	−0.193	−0.364	−0.201
Hidden/router geometry	−0.221	−0.479	−0.462	−0.360	−0.239	−0.322
Top-1 switch rate	0.043	0.170	0.179	0.041	−0.154	−0.020
Adjacent top-8 overlap	0.750	0.694	0.773	0.697	0.756	0.570
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	0.773	0.502	0.473	0.705	0.771	0.735
Top-1/top-2 margin	0.303	−0.114	−0.072	0.132	0.243	0.075
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.965	0.957	0.951	0.964	0.970	0.927
Hidden-state norm	−0.076	0.671	0.573	0.177	0.265	0.334
Hidden-state step distance	−0.414	−0.106	−0.283	−0.161	−0.335	−0.299
Hidden/router geometry	−0.301	−0.447	−0.305	−0.292	−0.298	−0.434
Top-1 switch rate	−0.160	0.221	0.016	0.011	−0.151	0.002
Adjacent top-8 overlap	0.773	0.599	0.726	0.682	0.778	0.552
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	0.839	0.494	0.588	0.728	0.797	0.743
Top-1/top-2 margin	0.370	−0.104	−0.022	0.233	0.222	0.086
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.973	0.963	0.958	0.968	0.976	0.934
Hidden-state norm	−0.092	0.589	0.525	0.178	0.311	0.367
Hidden-state step distance	−0.358	−0.034	−0.243	−0.108	−0.292	−0.133
Hidden/router geometry	−0.429	−0.404	−0.317	−0.333	−0.328	−0.478
Top-1 switch rate	−0.120	0.196	−0.001	0.005	−0.091	0.165
Adjacent top-8 overlap	0.788	0.589	0.702	0.700	0.798	0.462
Bin 4						
Router confidence	0.761 (204)	0.497 (927)	0.563 (900)	0.713 (494)	0.792 (876)	0.772 (202)
Top-1/top-2 margin	0.265 (204)	−0.024 (927)	−0.035 (900)	0.171 (494)	0.257 (876)	0.098 (202)
Top-8 probability mass	1.000 (204)	1.000 (927)	1.000 (900)	1.000 (494)	1.000 (876)	1.000 (202)
Top-8/top-9 margin	0.974 (204)	0.960 (927)	0.963 (900)	0.963 (494)	0.969 (876)	0.932 (202)
Hidden-state norm	0.199 (204)	0.499 (927)	0.533 (900)	0.292 (494)	0.419 (876)	0.207 (202)
Hidden-state step distance	−0.167 (203)	0.081 (927)	−0.231 (900)	−0.049 (494)	−0.147 (876)	−0.197 (202)
Hidden/router geometry	−0.421 (203)	−0.407 (927)	−0.387 (900)	−0.378 (494)	−0.374 (876)	−0.397 (202)
Top-1 switch rate	−0.016 (203)	0.190 (927)	0.003 (900)	0.047 (494)	−0.071 (876)	0.150 (202)
Adjacent top-8 overlap	0.649 (203)	0.545 (927)	0.703 (900)	0.711 (494)	0.739 (876)	0.404 (202)
Bin 5						
Router confidence	0.814 (171)	0.533 (859)	0.555 (847)	0.719 (427)	0.791 (767)	0.747 (162)
Top-1/top-2 margin	0.459 (171)	−0.061 (859)	0.025 (847)	0.088 (427)	0.246 (767)	0.086 (162)
Top-8 probability mass	1.000 (171)	1.000 (859)	1.000 (847)	1.000 (427)	1.000 (767)	1.000 (162)
Top-8/top-9 margin	0.966 (171)	0.961 (859)	0.965 (847)	0.961 (427)	0.972 (767)	0.907 (162)
Hidden-state norm	0.352 (171)	0.469 (859)	0.490 (847)	0.310 (427)	0.500 (767)	0.245 (162)
Hidden-state step distance	0.109 (171)	−0.011 (858)	−0.104 (847)	−0.036 (426)	−0.070 (767)	−0.022 (162)
Hidden/router geometry	−0.542 (171)	−0.378 (858)	−0.383 (847)	−0.366 (426)	−0.414 (767)	−0.412 (162)
Top-1 switch rate	0.022 (171)	0.173 (858)	0.032 (847)	0.132 (426)	0.003 (767)	0.196 (162)
Adjacent top-8 overlap	0.588 (171)	0.566 (858)	0.662 (847)	0.664 (426)	0.728 (767)	0.367 (162)
Bin 6						
Router confidence	0.778 (139)	0.600 (816)	0.537 (803)	0.747 (384)	0.767 (684)	0.778 (133)
Top-1/top-2 margin	0.473 (139)	−0.031 (816)	−0.026 (803)	0.133 (384)	0.206 (684)	0.209 (133)
Top-8 probability mass	1.000 (139)	1.000 (816)	1.000 (803)	1.000 (384)	1.000 (684)	1.000 (133)
Top-8/top-9 margin	0.970 (139)	0.966 (816)	0.964 (803)	0.960 (384)	0.967 (684)	0.943 (133)
Hidden-state norm	0.442 (139)	0.462 (816)	0.488 (803)	0.340 (384)	0.482 (684)	0.319 (133)
Hidden-state step distance	0.387 (139)	−0.030 (816)	−0.094 (803)	0.105 (383)	0.086 (684)	−0.022 (133)
Hidden/router geometry	−0.439 (139)	−0.379 (816)	−0.350 (803)	−0.363 (383)	−0.376 (684)	−0.522 (133)
Top-1 switch rate	−0.114 (139)	0.213 (816)	0.068 (803)	0.175 (383)	0.028 (684)	0.068 (133)
Adjacent top-8 overlap	0.499 (139)	0.540 (816)	0.613 (803)	0.601 (383)	0.700 (684)	0.535 (133)
Bin 7						
Router confidence	0.795 (106)	0.596 (748)	0.559 (752)	0.722 (342)	0.797 (559)	0.748 (102)
Top-1/top-2 margin	0.467 (106)	−0.043 (748)	0.046 (752)	0.132 (342)	0.286 (559)	0.253 (102)

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8 probability mass	1.000 (106)	1.000 (748)	1.000 (752)	1.000 (342)	1.000 (559)	1.000 (102)
Top-8/top-9 margin	0.979 (106)	0.958 (748)	0.966 (752)	0.961 (342)	0.968 (559)	0.899 (102)
Hidden-state norm	0.648 (106)	0.506 (748)	0.459 (752)	0.409 (342)	0.464 (559)	0.364 (102)
Hidden-state step distance	0.467 (106)	-0.010 (748)	-0.037 (751)	0.104 (342)	0.118 (559)	-0.068 (102)
Hidden/router geometry	-0.656 (106)	-0.371 (748)	-0.334 (751)	-0.365 (342)	-0.478 (559)	-0.481 (102)
Top-1 switch rate	-0.317 (106)	0.260 (748)	0.042 (751)	0.156 (342)	-0.021 (559)	-0.034 (102)
Adjacent top-8 overlap	0.530 (106)	0.494 (748)	0.653 (751)	0.582 (342)	0.702 (559)	0.419 (102)
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	0.864	0.662	0.612	0.726	0.792	0.789
Top-1/top-2 margin	0.647	0.034	0.055	0.141	0.340	0.427
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.940	0.962	0.966	0.963	0.974	0.923
Hidden-state norm	0.597	0.493	0.430	0.505	0.552	0.364
Hidden-state step distance	0.483	0.086	-0.020	0.123	0.248	0.039
Hidden/router geometry	-0.565	-0.369	-0.353	-0.426	-0.495	-0.443
Top-1 switch rate	-0.334	0.252	0.030	0.065	-0.040	-0.090
Adjacent top-8 overlap	0.554	0.460	0.629	0.618	0.657	0.559
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	0.835	0.701	0.670	0.762	0.788	0.891
Top-1/top-2 margin	0.629	0.071	0.157	0.170	0.325	0.546
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.982	0.965	0.963	0.967	0.973	0.948
Hidden-state norm	0.687	0.446	0.423	0.564	0.645	0.359
Hidden-state step distance	0.626	0.012	0.056	0.244	0.270	0.232
Hidden/router geometry	-0.732	-0.335	-0.325	-0.514	-0.501	-0.479
Top-1 switch rate	-0.402	0.218	0.056	0.162	-0.085	-0.112
Adjacent top-8 overlap	0.543	0.484	0.586	0.602	0.697	0.548
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	0.940	0.737	0.727	0.799	0.822	0.884
Top-1/top-2 margin	0.761	0.362	0.373	0.489	0.400	0.769
Top-8 probability mass	1.000	1.000	1.000	1.000	1.000	1.000
Top-8/top-9 margin	0.979	0.955	0.957	0.969	0.974	0.954
Hidden-state norm	0.750	0.498	0.272	0.630	0.670	0.602
Hidden-state step distance	0.764	0.361	0.353	0.414	0.423	0.656
Hidden/router geometry	-0.751	-0.408	-0.220	-0.469	-0.523	-0.672
Top-1 switch rate	-0.663	0.073	0.052	0.086	0.070	-0.511
Adjacent top-8 overlap	0.774	0.410	0.464	0.492	0.645	0.719

Table 62. Position windows: metrics versus hidden-state norm. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	-0.315	0.099	0.041	-0.189	-0.371	-0.109
Top-1/top-2 margin	-0.226	-0.303	-0.459	-0.369	-0.099	-0.449
Top-8 probability mass	0.384	0.664	0.727	0.691	0.591	0.485
Top-8/top-9 margin	0.538	0.712	0.787	0.779	0.733	0.678
Top-8/non-top-8 mean gap	0.384	0.664	0.727	0.691	0.591	0.485
Hidden-state step distance	0.215	-0.209	-0.046	0.143	0.146	0.133
Hidden/router geometry	-0.399	-0.415	-0.485	-0.475	-0.444	-0.262
Top-1 switch rate	0.390	0.262	0.224	0.185	0.068	0.342
Adjacent top-8 overlap	0.123	0.536	0.652	0.530	0.396	0.130
Bin 1						

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Finite traces	429	960	960	668	1,269	269
Router confidence	−0.418	0.190	−0.137	−0.212	−0.215	−0.074
Top-1/top-2 margin	−0.324	−0.180	−0.437	−0.384	−0.129	−0.492
Top-8 probability mass	0.088	0.758	0.733	0.424	0.306	0.417
Top-8/top-9 margin	0.189	0.774	0.790	0.534	0.443	0.509
Top-8/non-top-8 mean gap	0.088	0.758	0.733	0.424	0.306	0.417
Hidden-state step distance	0.228	−0.160	0.048	0.145	0.129	0.171
Hidden/router geometry	−0.345	−0.385	−0.486	−0.408	−0.384	−0.295
Top-1 switch rate	0.278	0.205	0.263	0.179	0.153	0.432
Adjacent top-8 overlap	0.049	0.557	0.609	0.398	0.232	0.011
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	−0.516	0.079	−0.113	−0.333	−0.110	−0.182
Top-1/top-2 margin	−0.431	−0.242	−0.355	−0.431	−0.142	−0.554
Top-8 probability mass	−0.076	0.671	0.573	0.177	0.265	0.334
Top-8/top-9 margin	0.019	0.701	0.644	0.264	0.339	0.406
Top-8/non-top-8 mean gap	−0.076	0.671	0.573	0.177	0.265	0.334
Hidden-state step distance	0.288	−0.003	−0.058	0.296	0.154	0.082
Hidden/router geometry	−0.375	−0.405	−0.387	−0.381	−0.308	−0.223
Top-1 switch rate	0.356	0.301	0.182	0.247	0.147	0.398
Adjacent top-8 overlap	−0.072	0.398	0.427	0.193	0.218	0.048
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	−0.393	−0.089	−0.114	−0.294	−0.014	−0.112
Top-1/top-2 margin	−0.399	−0.303	−0.354	−0.383	−0.109	−0.517
Top-8 probability mass	−0.092	0.589	0.525	0.178	0.311	0.367
Top-8/top-9 margin	−0.043	0.650	0.595	0.266	0.354	0.412
Top-8/non-top-8 mean gap	−0.092	0.589	0.525	0.178	0.311	0.367
Hidden-state step distance	0.261	0.095	−0.042	0.222	0.108	0.090
Hidden/router geometry	−0.128	−0.451	−0.415	−0.375	−0.299	−0.305
Top-1 switch rate	0.338	0.398	0.189	0.183	0.134	0.386
Adjacent top-8 overlap	−0.110	0.251	0.416	0.182	0.261	0.055
Bin 4						
Router confidence	−0.185 (204)	−0.163 (927)	−0.113 (900)	−0.153 (494)	0.117 (876)	−0.182 (202)
Top-1/top-2 margin	−0.246 (204)	−0.313 (927)	−0.299 (900)	−0.251 (494)	−0.025 (876)	−0.561 (202)
Top-8 probability mass	0.199 (204)	0.499 (927)	0.533 (900)	0.292 (494)	0.419 (876)	0.207 (202)
Top-8/top-9 margin	0.216 (204)	0.560 (927)	0.596 (900)	0.376 (494)	0.465 (876)	0.261 (202)
Top-8/non-top-8 mean gap	0.199 (204)	0.499 (927)	0.533 (900)	0.292 (494)	0.419 (876)	0.207 (202)
Hidden-state step distance	0.211 (203)	0.108 (927)	−0.059 (900)	0.197 (494)	0.006 (876)	0.023 (202)
Hidden/router geometry	−0.484 (203)	−0.505 (927)	−0.473 (900)	−0.352 (494)	−0.376 (876)	−0.220 (202)
Top-1 switch rate	0.288 (203)	0.339 (927)	0.165 (900)	0.109 (494)	−0.019 (876)	0.425 (202)
Adjacent top-8 overlap	0.040 (203)	0.245 (927)	0.401 (900)	0.253 (494)	0.425 (876)	−0.100 (202)
Bin 5						
Router confidence	0.032 (171)	−0.103 (859)	−0.144 (847)	−0.112 (427)	0.210 (767)	−0.213 (162)
Top-1/top-2 margin	0.027 (171)	−0.313 (859)	−0.337 (847)	−0.278 (427)	0.101 (767)	−0.494 (162)
Top-8 probability mass	0.352 (171)	0.469 (859)	0.490 (847)	0.310 (427)	0.500 (767)	0.245 (162)
Top-8/top-9 margin	0.355 (171)	0.520 (859)	0.541 (847)	0.406 (427)	0.507 (767)	0.291 (162)
Top-8/non-top-8 mean gap	0.352 (171)	0.469 (859)	0.490 (847)	0.310 (427)	0.500 (767)	0.245 (162)
Hidden-state step distance	0.340 (171)	0.060 (858)	−0.104 (847)	0.095 (426)	0.104 (767)	0.017 (162)
Hidden/router geometry	−0.518 (171)	−0.420 (858)	−0.459 (847)	−0.350 (426)	−0.483 (767)	−0.343 (162)
Top-1 switch rate	0.140 (171)	0.302 (858)	0.208 (847)	0.135 (426)	−0.059 (767)	0.333 (162)
Adjacent top-8 overlap	0.111 (171)	0.237 (858)	0.389 (847)	0.312 (426)	0.457 (767)	0.079 (162)
Bin 6						
Router confidence	0.133 (139)	−0.091 (816)	−0.135 (803)	−0.050 (384)	0.190 (684)	−0.092 (133)
Top-1/top-2 margin	0.110 (139)	−0.335 (816)	−0.278 (803)	−0.176 (384)	0.081 (684)	−0.410 (133)
Top-8 probability mass	0.442 (139)	0.462 (816)	0.488 (803)	0.340 (384)	0.482 (684)	0.319 (133)
Top-8/top-9 margin	0.478 (139)	0.523 (816)	0.521 (803)	0.414 (384)	0.512 (684)	0.361 (133)
Top-8/non-top-8 mean gap	0.442 (139)	0.462 (816)	0.488 (803)	0.340 (384)	0.482 (684)	0.319 (133)
Hidden-state step distance	0.359 (139)	0.057 (816)	−0.137 (803)	0.191 (383)	0.105 (684)	0.114 (133)
Hidden/router geometry	−0.656 (139)	−0.435 (816)	−0.439 (803)	−0.449 (383)	−0.381 (684)	−0.344 (133)

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-1 switch rate	−0.016 (139)	0.372 (816)	0.125 (803)	0.122 (383)	−0.036 (684)	0.229 (133)
Adjacent top-8 overlap	0.140 (139)	0.205 (816)	0.379 (803)	0.277 (383)	0.448 (684)	0.161 (133)
Bin 7						
Router confidence	0.365 (106)	−0.078 (748)	−0.118 (752)	0.001 (342)	0.216 (559)	−0.042 (102)
Top-1/top-2 margin	0.305 (106)	−0.360 (748)	−0.249 (752)	−0.166 (342)	0.073 (559)	−0.376 (102)
Top-8 probability mass	0.648 (106)	0.506 (748)	0.459 (752)	0.409 (342)	0.464 (559)	0.364 (102)
Top-8/top-9 margin	0.671 (106)	0.567 (748)	0.486 (752)	0.483 (342)	0.492 (559)	0.485 (102)
Top-8/non-top-8 mean gap	0.648 (106)	0.506 (748)	0.459 (752)	0.409 (342)	0.464 (559)	0.364 (102)
Hidden-state step distance	0.405 (106)	0.021 (748)	−0.142 (751)	0.178 (342)	0.106 (559)	0.063 (102)
Hidden/router geometry	−0.564 (106)	−0.464 (748)	−0.360 (751)	−0.413 (342)	−0.545 (559)	−0.468 (102)
Top-1 switch rate	−0.194 (106)	0.375 (748)	0.127 (751)	0.093 (342)	−0.107 (559)	0.293 (102)
Adjacent top-8 overlap	0.339 (106)	0.250 (748)	0.410 (751)	0.367 (342)	0.510 (559)	0.060 (102)
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	0.367	−0.029	−0.107	0.115	0.318	−0.015
Top-1/top-2 margin	0.317	−0.340	−0.259	−0.107	0.210	−0.251
Top-8 probability mass	0.597	0.493	0.430	0.505	0.552	0.364
Top-8/top-9 margin	0.659	0.549	0.481	0.555	0.573	0.413
Top-8/non-top-8 mean gap	0.597	0.493	0.430	0.505	0.552	0.364
Hidden-state step distance	0.526	0.065	−0.205	0.184	0.231	0.249
Hidden/router geometry	−0.674	−0.485	−0.282	−0.444	−0.503	−0.495
Top-1 switch rate	−0.280	0.388	0.145	0.102	−0.083	0.182
Adjacent top-8 overlap	0.369	0.237	0.356	0.371	0.500	0.230
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	0.466	−0.023	−0.055	0.165	0.424	0.172
Top-1/top-2 margin	0.423	−0.345	−0.216	−0.040	0.274	0.033
Top-8 probability mass	0.687	0.446	0.423	0.564	0.645	0.359
Top-8/top-9 margin	0.709	0.516	0.470	0.612	0.657	0.388
Top-8/non-top-8 mean gap	0.687	0.446	0.423	0.564	0.645	0.359
Hidden-state step distance	0.524	0.057	−0.159	0.283	0.341	0.254
Hidden/router geometry	−0.763	−0.467	−0.305	−0.477	−0.548	−0.505
Top-1 switch rate	−0.333	0.444	0.174	0.143	−0.114	0.017
Adjacent top-8 overlap	0.429	0.153	0.354	0.357	0.539	0.340
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	0.625	0.118	−0.147	0.284	0.507	0.412
Top-1/top-2 margin	0.562	−0.053	−0.073	0.183	0.445	0.370
Top-8 probability mass	0.750	0.498	0.272	0.630	0.670	0.602
Top-8/top-9 margin	0.779	0.583	0.334	0.699	0.661	0.637
Top-8/non-top-8 mean gap	0.750	0.498	0.272	0.630	0.670	0.602
Hidden-state step distance	0.647	0.201	−0.068	0.328	0.390	0.520
Hidden/router geometry	−0.757	−0.492	−0.224	−0.414	−0.473	−0.660
Top-1 switch rate	−0.585	0.281	0.022	0.099	−0.094	−0.228
Adjacent top-8 overlap	0.725	0.199	0.318	0.380	0.561	0.446

Table 63. Position windows: metrics versus hidden-state step distance. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	−0.420	−0.327	−0.335	−0.312	−0.439	−0.162
Top-1/top-2 margin	0.045	0.063	0.128	−0.105	−0.069	−0.019
Top-8 probability mass	−0.166	−0.261	−0.151	0.073	0.061	0.029
Top-8/top-9 margin	−0.041	−0.209	−0.096	0.152	0.188	0.077
Top-8/non-top-8 mean gap	−0.166	−0.261	−0.151	0.073	0.061	0.029

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden-state norm	0.215	-0.209	-0.046	0.143	0.146	0.133
Hidden/router geometry	-0.125	-0.188	-0.117	-0.271	-0.307	0.005
Top-1 switch rate	0.464	0.380	0.201	0.433	0.536	0.219
Adjacent top-8 overlap	-0.530	-0.571	-0.239	-0.228	-0.345	-0.250
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	-0.479	-0.284	-0.285	-0.386	-0.506	-0.242
Top-1/top-2 margin	-0.050	0.098	0.025	-0.156	-0.235	0.000
Top-8 probability mass	-0.405	-0.194	-0.078	-0.193	-0.364	-0.201
Top-8/top-9 margin	-0.318	-0.136	0.002	-0.115	-0.267	-0.203
Top-8/non-top-8 mean gap	-0.405	-0.194	-0.078	-0.193	-0.364	-0.201
Hidden-state norm	0.228	-0.160	0.048	0.145	0.129	0.171
Hidden/router geometry	-0.204	-0.129	-0.129	-0.254	-0.173	0.020
Top-1 switch rate	0.323	0.331	0.284	0.413	0.444	0.205
Adjacent top-8 overlap	-0.662	-0.530	-0.368	-0.468	-0.614	-0.379
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	-0.459	-0.250	-0.313	-0.334	-0.426	-0.233
Top-1/top-2 margin	-0.055	0.178	0.034	-0.119	-0.153	0.096
Top-8 probability mass	-0.414	-0.106	-0.283	-0.161	-0.335	-0.299
Top-8/top-9 margin	-0.344	-0.051	-0.203	-0.111	-0.273	-0.318
Top-8/non-top-8 mean gap	-0.414	-0.106	-0.283	-0.161	-0.335	-0.299
Hidden-state norm	0.288	-0.003	-0.058	0.296	0.154	0.082
Hidden/router geometry	-0.151	-0.207	-0.180	-0.271	-0.076	0.073
Top-1 switch rate	0.374	0.269	0.344	0.400	0.419	0.168
Adjacent top-8 overlap	-0.645	-0.476	-0.554	-0.433	-0.593	-0.457
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	-0.366	-0.142	-0.249	-0.244	-0.362	-0.047
Top-1/top-2 margin	-0.098	0.188	0.077	-0.101	-0.087	0.156
Top-8 probability mass	-0.358	-0.034	-0.243	-0.108	-0.292	-0.133
Top-8/top-9 margin	-0.304	0.007	-0.168	-0.071	-0.249	-0.148
Top-8/non-top-8 mean gap	-0.358	-0.034	-0.243	-0.108	-0.292	-0.133
Hidden-state norm	0.261	0.095	-0.042	0.222	0.108	0.090
Hidden/router geometry	-0.117	-0.315	-0.204	-0.299	-0.093	-0.101
Top-1 switch rate	0.377	0.278	0.324	0.379	0.381	0.176
Adjacent top-8 overlap	-0.582	-0.402	-0.526	-0.421	-0.551	-0.417
Bin 4						
Finite traces	203	927	900	494	876	202
Router confidence	-0.111	-0.044	-0.204	-0.109	-0.184	-0.052
Top-1/top-2 margin	0.204	0.265	0.234	0.114	0.061	0.161
Top-8 probability mass	-0.167	0.081	-0.231	-0.049	-0.147	-0.197
Top-8/top-9 margin	-0.154	0.096	-0.173	-0.039	-0.132	-0.191
Top-8/non-top-8 mean gap	-0.167	0.081	-0.231	-0.049	-0.147	-0.197
Hidden-state norm	0.211	0.108	-0.059	0.197	0.006	0.023
Hidden/router geometry	-0.134	-0.307	-0.176	-0.239	-0.140	0.092
Top-1 switch rate	0.189	0.255	0.323	0.247	0.334	0.125
Adjacent top-8 overlap	-0.591	-0.346	-0.537	-0.338	-0.460	-0.412
Bin 5						
Finite traces	171	858	847	426	767	162
Router confidence	0.079	-0.061	-0.029	-0.026	-0.134	0.125
Top-1/top-2 margin	0.277	0.238	0.277	0.260	0.079	0.343
Top-8 probability mass	0.109	-0.011	-0.104	-0.036	-0.070	-0.022
Top-8/top-9 margin	0.111	-0.002	-0.085	-0.034	-0.057	-0.057
Top-8/non-top-8 mean gap	0.109	-0.011	-0.104	-0.036	-0.070	-0.022
Hidden-state norm	0.340	0.060	-0.104	0.095	0.104	0.017
Hidden/router geometry	-0.248	-0.229	-0.139	-0.278	-0.208	0.047
Top-1 switch rate	0.109	0.258	0.221	0.210	0.267	0.154

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Adjacent top-8 overlap	-0.434	-0.360	-0.433	-0.344	-0.411	-0.375
Bin 6						
Finite traces	139	816	803	383	684	133
Router confidence	0.425	-0.082	0.053	0.029	0.019	0.008
Top-1/top-2 margin	0.566	0.265	0.364	0.248	0.209	0.291
Top-8 probability mass	0.387	-0.030	-0.094	0.105	0.086	-0.022
Top-8/top-9 margin	0.379	-0.019	-0.087	0.105	0.097	-0.056
Top-8/non-top-8 mean gap	0.387	-0.030	-0.094	0.105	0.086	-0.022
Hidden-state norm	0.359	0.057	-0.137	0.191	0.105	0.114
Hidden/router geometry	-0.351	-0.301	-0.094	-0.434	-0.267	-0.020
Top-1 switch rate	-0.094	0.236	0.175	0.206	0.197	0.086
Adjacent top-8 overlap	-0.230	-0.405	-0.443	-0.270	-0.278	-0.260
Bin 7						
Finite traces	106	748	751	342	559	102
Router confidence	0.498	-0.052	0.114	0.086	0.109	-0.000
Top-1/top-2 margin	0.661	0.306	0.367	0.296	0.239	0.326
Top-8 probability mass	0.467	-0.010	-0.037	0.104	0.118	-0.068
Top-8/top-9 margin	0.493	-0.000	-0.020	0.094	0.096	-0.119
Top-8/non-top-8 mean gap	0.467	-0.010	-0.037	0.104	0.118	-0.068
Hidden-state norm	0.405	0.021	-0.142	0.178	0.106	0.063
Hidden/router geometry	-0.437	-0.233	-0.057	-0.352	-0.268	-0.136
Top-1 switch rate	-0.289	0.214	0.159	0.128	0.251	-0.122
Adjacent top-8 overlap	-0.119	-0.334	-0.375	-0.267	-0.245	-0.172
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	0.547	0.041	0.185	0.118	0.258	0.081
Top-1/top-2 margin	0.671	0.322	0.475	0.312	0.387	0.287
Top-8 probability mass	0.483	0.086	-0.020	0.123	0.248	0.039
Top-8/top-9 margin	0.546	0.084	-0.033	0.118	0.239	-0.025
Top-8/non-top-8 mean gap	0.483	0.086	-0.020	0.123	0.248	0.039
Hidden-state norm	0.526	0.065	-0.205	0.184	0.231	0.249
Hidden/router geometry	-0.648	-0.294	-0.155	-0.366	-0.351	-0.157
Top-1 switch rate	-0.359	0.208	0.090	0.094	0.081	-0.170
Adjacent top-8 overlap	-0.006	-0.288	-0.350	-0.201	-0.169	-0.016
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	0.764	-0.003	0.205	0.167	0.313	0.253
Top-1/top-2 margin	0.820	0.305	0.519	0.354	0.427	0.468
Top-8 probability mass	0.626	0.012	0.056	0.244	0.270	0.232
Top-8/top-9 margin	0.611	0.046	0.042	0.233	0.264	0.216
Top-8/non-top-8 mean gap	0.626	0.012	0.056	0.244	0.270	0.232
Hidden-state norm	0.524	0.057	-0.159	0.283	0.341	0.254
Hidden/router geometry	-0.664	-0.310	-0.104	-0.369	-0.414	-0.204
Top-1 switch rate	-0.506	0.223	0.089	0.145	0.046	-0.161
Adjacent top-8 overlap	0.037	-0.357	-0.356	-0.104	-0.104	0.122
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	0.823	0.215	0.394	0.354	0.470	0.596
Top-1/top-2 margin	0.896	0.393	0.564	0.513	0.463	0.740
Top-8 probability mass	0.764	0.361	0.353	0.414	0.423	0.656
Top-8/top-9 margin	0.759	0.410	0.333	0.418	0.421	0.627
Top-8/non-top-8 mean gap	0.764	0.361	0.353	0.414	0.423	0.656
Hidden-state norm	0.647	0.201	-0.068	0.328	0.390	0.520
Hidden/router geometry	-0.603	-0.385	-0.139	-0.407	-0.371	-0.519
Top-1 switch rate	-0.759	0.187	0.042	-0.020	-0.002	-0.489
Adjacent top-8 overlap	0.488	-0.187	-0.235	-0.051	-0.005	0.516

Table 64. Position windows: metrics versus hidden/router geometry. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	0.013	-0.097	0.094	0.085	0.224	-0.201
Top-1/top-2 margin	0.037	0.108	0.225	0.222	0.087	-0.064
Top-8 probability mass	-0.432	-0.432	-0.485	-0.497	-0.448	-0.453
Top-8/top-9 margin	-0.468	-0.472	-0.516	-0.534	-0.524	-0.443
Top-8/non-top-8 mean gap	-0.432	-0.432	-0.485	-0.497	-0.448	-0.453
Hidden-state norm	-0.399	-0.415	-0.485	-0.475	-0.444	-0.262
Hidden-state step distance	-0.125	-0.188	-0.117	-0.271	-0.307	0.005
Top-1 switch rate	-0.254	-0.239	-0.149	-0.221	-0.270	0.100
Adjacent top-8 overlap	-0.194	-0.200	-0.354	-0.303	-0.134	-0.312
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	0.110	-0.175	0.073	0.033	0.076	-0.064
Top-1/top-2 margin	0.074	0.080	0.224	0.130	0.058	0.150
Top-8 probability mass	-0.221	-0.479	-0.462	-0.360	-0.239	-0.322
Top-8/top-9 margin	-0.266	-0.482	-0.479	-0.389	-0.314	-0.357
Top-8/non-top-8 mean gap	-0.221	-0.479	-0.462	-0.360	-0.239	-0.322
Hidden-state norm	-0.345	-0.385	-0.486	-0.408	-0.384	-0.295
Hidden-state step distance	-0.204	-0.129	-0.129	-0.254	-0.173	0.020
Top-1 switch rate	-0.244	-0.161	-0.232	-0.177	-0.083	-0.185
Adjacent top-8 overlap	-0.065	-0.265	-0.311	-0.205	-0.140	-0.124
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	0.015	-0.098	0.049	0.022	-0.101	-0.196
Top-1/top-2 margin	0.056	0.013	0.109	0.101	-0.047	0.033
Top-8 probability mass	-0.301	-0.447	-0.305	-0.292	-0.298	-0.434
Top-8/top-9 margin	-0.362	-0.464	-0.323	-0.306	-0.339	-0.415
Top-8/non-top-8 mean gap	-0.301	-0.447	-0.305	-0.292	-0.298	-0.434
Hidden-state norm	-0.375	-0.405	-0.387	-0.381	-0.308	-0.223
Hidden-state step distance	-0.151	-0.207	-0.180	-0.271	-0.076	0.073
Top-1 switch rate	-0.233	-0.204	-0.102	-0.141	0.025	-0.169
Adjacent top-8 overlap	-0.140	-0.161	-0.147	-0.212	-0.259	-0.146
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	-0.242	-0.042	0.033	-0.011	-0.143	-0.212
Top-1/top-2 margin	-0.070	0.063	0.104	0.075	-0.024	0.069
Top-8 probability mass	-0.429	-0.404	-0.317	-0.333	-0.328	-0.478
Top-8/top-9 margin	-0.431	-0.418	-0.333	-0.366	-0.356	-0.458
Top-8/non-top-8 mean gap	-0.429	-0.404	-0.317	-0.333	-0.328	-0.478
Hidden-state norm	-0.128	-0.451	-0.415	-0.375	-0.299	-0.305
Hidden-state step distance	-0.117	-0.315	-0.204	-0.299	-0.093	-0.101
Top-1 switch rate	-0.090	-0.236	-0.142	-0.179	-0.028	-0.186
Adjacent top-8 overlap	-0.286	-0.092	-0.160	-0.183	-0.243	-0.150
Bin 4						
Finite traces	203	927	900	494	876	202
Router confidence	-0.104	-0.005	-0.019	-0.121	-0.173	-0.085
Top-1/top-2 margin	-0.054	0.069	0.058	0.001	-0.118	0.168
Top-8 probability mass	-0.421	-0.407	-0.387	-0.378	-0.374	-0.397
Top-8/top-9 margin	-0.419	-0.425	-0.395	-0.382	-0.391	-0.373
Top-8/non-top-8 mean gap	-0.421	-0.407	-0.387	-0.378	-0.374	-0.397
Hidden-state norm	-0.484	-0.505	-0.473	-0.352	-0.376	-0.220
Hidden-state step distance	-0.134	-0.307	-0.176	-0.239	-0.140	0.092
Top-1 switch rate	-0.154	-0.272	-0.096	-0.109	0.011	-0.204
Adjacent top-8 overlap	-0.174	-0.117	-0.219	-0.228	-0.273	-0.131
Bin 5						
Finite traces	171	858	847	426	767	162

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Router confidence	-0.285	-0.076	-0.002	-0.135	-0.214	-0.088
Top-1/top-2 margin	-0.206	0.062	0.103	0.017	-0.111	0.230
Top-8 probability mass	-0.542	-0.378	-0.383	-0.366	-0.414	-0.412
Top-8/top-9 margin	-0.514	-0.384	-0.399	-0.360	-0.419	-0.419
Top-8/non-top-8 mean gap	-0.542	-0.378	-0.383	-0.366	-0.414	-0.412
Hidden-state norm	-0.518	-0.420	-0.459	-0.350	-0.483	-0.343
Hidden-state step distance	-0.248	-0.229	-0.139	-0.278	-0.208	0.047
Top-1 switch rate	-0.076	-0.195	-0.108	-0.197	0.020	-0.176
Adjacent top-8 overlap	-0.221	-0.120	-0.232	-0.204	-0.298	-0.173
Bin 6						
Finite traces	139	816	803	383	684	133
Router confidence	-0.217	-0.097	0.012	-0.060	-0.181	-0.311
Top-1/top-2 margin	-0.122	0.075	0.081	-0.001	-0.100	-0.118
Top-8 probability mass	-0.439	-0.379	-0.350	-0.363	-0.376	-0.522
Top-8/top-9 margin	-0.415	-0.384	-0.338	-0.358	-0.391	-0.485
Top-8/non-top-8 mean gap	-0.439	-0.379	-0.350	-0.363	-0.376	-0.522
Hidden-state norm	-0.656	-0.435	-0.439	-0.449	-0.381	-0.344
Hidden-state step distance	-0.351	-0.301	-0.094	-0.434	-0.267	-0.020
Top-1 switch rate	-0.011	-0.214	-0.020	-0.206	-0.000	-0.051
Adjacent top-8 overlap	-0.089	-0.078	-0.272	-0.150	-0.253	-0.281
Bin 7						
Finite traces	106	748	751	342	559	102
Router confidence	-0.502	-0.071	-0.069	-0.181	-0.270	-0.262
Top-1/top-2 margin	-0.385	0.100	-0.017	-0.062	-0.091	-0.061
Top-8 probability mass	-0.656	-0.371	-0.334	-0.365	-0.478	-0.481
Top-8/top-9 margin	-0.625	-0.386	-0.309	-0.361	-0.486	-0.480
Top-8/non-top-8 mean gap	-0.656	-0.371	-0.334	-0.365	-0.478	-0.481
Hidden-state norm	-0.564	-0.464	-0.360	-0.413	-0.545	-0.468
Hidden-state step distance	-0.437	-0.233	-0.057	-0.352	-0.268	-0.136
Top-1 switch rate	0.138	-0.217	0.071	-0.174	0.022	0.062
Adjacent top-8 overlap	-0.272	-0.136	-0.315	-0.142	-0.398	-0.263
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	-0.446	-0.074	-0.163	-0.234	-0.297	-0.200
Top-1/top-2 margin	-0.393	0.045	-0.052	-0.082	-0.168	-0.080
Top-8 probability mass	-0.565	-0.369	-0.353	-0.426	-0.495	-0.443
Top-8/top-9 margin	-0.620	-0.374	-0.343	-0.434	-0.494	-0.448
Top-8/non-top-8 mean gap	-0.565	-0.369	-0.353	-0.426	-0.495	-0.443
Hidden-state norm	-0.674	-0.485	-0.282	-0.444	-0.503	-0.495
Hidden-state step distance	-0.648	-0.294	-0.155	-0.366	-0.351	-0.157
Top-1 switch rate	0.364	-0.226	0.009	-0.057	0.075	-0.028
Adjacent top-8 overlap	-0.354	-0.091	-0.240	-0.262	-0.426	-0.303
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	-0.650	-0.115	-0.178	-0.248	-0.311	-0.333
Top-1/top-2 margin	-0.550	0.022	-0.084	-0.064	-0.225	-0.084
Top-8 probability mass	-0.732	-0.335	-0.325	-0.514	-0.501	-0.479
Top-8/top-9 margin	-0.730	-0.353	-0.285	-0.518	-0.510	-0.501
Top-8/non-top-8 mean gap	-0.732	-0.335	-0.325	-0.514	-0.501	-0.479
Hidden-state norm	-0.763	-0.467	-0.305	-0.477	-0.548	-0.505
Hidden-state step distance	-0.664	-0.310	-0.104	-0.369	-0.414	-0.204
Top-1 switch rate	0.393	-0.276	0.041	-0.187	0.042	-0.074
Adjacent top-8 overlap	-0.337	-0.031	-0.236	-0.286	-0.354	-0.160
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	-0.650	-0.131	-0.047	-0.276	-0.322	-0.565
Top-1/top-2 margin	-0.601	-0.021	-0.045	-0.136	-0.214	-0.465
Top-8 probability mass	-0.751	-0.408	-0.220	-0.469	-0.523	-0.672
Top-8/top-9 margin	-0.761	-0.483	-0.210	-0.488	-0.520	-0.660

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8/non-top-8 mean gap	-0.751	-0.408	-0.220	-0.469	-0.523	-0.672
Hidden-state norm	-0.757	-0.492	-0.224	-0.414	-0.473	-0.660
Hidden-state step distance	-0.603	-0.385	-0.139	-0.407	-0.371	-0.519
Top-1 switch rate	0.667	-0.249	0.117	0.001	-0.027	0.232
Adjacent top-8 overlap	-0.769	-0.073	-0.244	-0.262	-0.361	-0.396

Table 65. Position windows: metrics versus top-1 switch rate. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	-0.397	-0.114	-0.235	-0.230	-0.396	-0.575
Top-1/top-2 margin	-0.337	-0.320	-0.434	-0.350	-0.282	-0.672
Top-8 probability mass	-0.003	0.237	0.182	0.107	-0.056	-0.295
Top-8/top-9 margin	0.150	0.280	0.235	0.176	0.087	-0.004
Top-8/non-top-8 mean gap	-0.003	0.237	0.182	0.107	-0.056	-0.295
Hidden-state norm	0.390	0.262	0.224	0.185	0.068	0.342
Hidden-state step distance	0.464	0.380	0.201	0.433	0.536	0.219
Hidden/router geometry	-0.254	-0.239	-0.149	-0.221	-0.270	0.100
Adjacent top-8 overlap	-0.405	-0.243	-0.088	-0.306	-0.498	-0.722
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	-0.193	-0.244	-0.426	-0.195	-0.449	-0.258
Top-1/top-2 margin	-0.357	-0.510	-0.524	-0.345	-0.477	-0.619
Top-8 probability mass	0.043	0.170	0.179	0.041	-0.154	-0.020
Top-8/top-9 margin	0.110	0.270	0.269	0.094	-0.055	0.227
Top-8/non-top-8 mean gap	0.043	0.170	0.179	0.041	-0.154	-0.020
Hidden-state norm	0.278	0.205	0.263	0.179	0.153	0.432
Hidden-state step distance	0.323	0.331	0.284	0.413	0.444	0.205
Hidden/router geometry	-0.244	-0.161	-0.232	-0.177	-0.083	-0.185
Adjacent top-8 overlap	-0.374	-0.354	-0.167	-0.398	-0.502	-0.688
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	-0.389	-0.207	-0.415	-0.200	-0.351	-0.187
Top-1/top-2 margin	-0.437	-0.512	-0.470	-0.390	-0.462	-0.541
Top-8 probability mass	-0.160	0.221	0.016	0.011	-0.151	0.002
Top-8/top-9 margin	-0.081	0.338	0.147	0.070	-0.082	0.248
Top-8/non-top-8 mean gap	-0.160	0.221	0.016	0.011	-0.151	0.002
Hidden-state norm	0.356	0.301	0.182	0.247	0.147	0.398
Hidden-state step distance	0.374	0.269	0.344	0.400	0.419	0.168
Hidden/router geometry	-0.233	-0.204	-0.102	-0.141	0.025	-0.169
Adjacent top-8 overlap	-0.501	-0.358	-0.399	-0.445	-0.480	-0.675
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	-0.277	-0.258	-0.334	-0.203	-0.285	-0.010
Top-1/top-2 margin	-0.417	-0.493	-0.419	-0.358	-0.416	-0.478
Top-8 probability mass	-0.120	0.196	-0.001	0.005	-0.091	0.165
Top-8/top-9 margin	-0.089	0.315	0.110	0.054	-0.024	0.385
Top-8/non-top-8 mean gap	-0.120	0.196	-0.001	0.005	-0.091	0.165
Hidden-state norm	0.338	0.398	0.189	0.183	0.134	0.386
Hidden-state step distance	0.377	0.278	0.324	0.379	0.381	0.176
Hidden/router geometry	-0.090	-0.236	-0.142	-0.179	-0.028	-0.186
Adjacent top-8 overlap	-0.467	-0.419	-0.446	-0.470	-0.422	-0.640
Bin 4						
Finite traces	203	927	900	494	876	202
Router confidence	-0.217	-0.206	-0.290	-0.150	-0.268	-0.025

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-1/top-2 margin	-0.359	-0.421	-0.312	-0.341	-0.401	-0.515
Top-8 probability mass	-0.016	0.190	0.003	0.047	-0.071	0.150
Top-8/top-9 margin	0.013	0.290	0.106	0.108	-0.015	0.370
Top-8/non-top-8 mean gap	-0.016	0.190	0.003	0.047	-0.071	0.150
Hidden-state norm	0.288	0.339	0.165	0.109	-0.019	0.425
Hidden-state step distance	0.189	0.255	0.323	0.247	0.334	0.125
Hidden/router geometry	-0.154	-0.272	-0.096	-0.109	0.011	-0.204
Adjacent top-8 overlap	-0.345	-0.420	-0.464	-0.431	-0.448	-0.670
Bin 5						
Finite traces	171	858	847	426	767	162
Router confidence	-0.166	-0.197	-0.277	-0.082	-0.184	-0.001
Top-1/top-2 margin	-0.306	-0.388	-0.333	-0.342	-0.396	-0.369
Top-8 probability mass	0.022	0.173	0.032	0.132	0.003	0.196
Top-8/top-9 margin	0.020	0.253	0.130	0.186	0.057	0.419
Top-8/non-top-8 mean gap	0.022	0.173	0.032	0.132	0.003	0.196
Hidden-state norm	0.140	0.302	0.208	0.135	-0.059	0.333
Hidden-state step distance	0.109	0.258	0.221	0.210	0.267	0.154
Hidden/router geometry	-0.076	-0.195	-0.108	-0.197	0.020	-0.176
Adjacent top-8 overlap	-0.280	-0.439	-0.437	-0.347	-0.372	-0.651
Bin 6						
Finite traces	139	816	803	383	684	133
Router confidence	-0.257	-0.156	-0.214	-0.031	-0.163	-0.042
Top-1/top-2 margin	-0.381	-0.376	-0.318	-0.314	-0.371	-0.393
Top-8 probability mass	-0.114	0.213	0.068	0.175	0.028	0.068
Top-8/top-9 margin	-0.091	0.293	0.173	0.230	0.092	0.244
Top-8/non-top-8 mean gap	-0.114	0.213	0.068	0.175	0.028	0.068
Hidden-state norm	-0.016	0.372	0.125	0.122	-0.036	0.229
Hidden-state step distance	-0.094	0.236	0.175	0.206	0.197	0.086
Hidden/router geometry	-0.011	-0.214	-0.020	-0.206	-0.000	-0.051
Adjacent top-8 overlap	-0.352	-0.422	-0.469	-0.417	-0.372	-0.610
Bin 7						
Finite traces	106	748	751	342	559	102
Router confidence	-0.455	-0.129	-0.228	-0.062	-0.169	-0.172
Top-1/top-2 margin	-0.576	-0.386	-0.346	-0.256	-0.367	-0.552
Top-8 probability mass	-0.317	0.260	0.042	0.156	-0.021	-0.034
Top-8/top-9 margin	-0.293	0.348	0.168	0.212	0.035	0.216
Top-8/non-top-8 mean gap	-0.317	0.260	0.042	0.156	-0.021	-0.034
Hidden-state norm	-0.194	0.375	0.127	0.093	-0.107	0.293
Hidden-state step distance	-0.289	0.214	0.159	0.128	0.251	-0.122
Hidden/router geometry	0.138	-0.217	0.071	-0.174	0.022	0.062
Adjacent top-8 overlap	-0.327	-0.428	-0.428	-0.409	-0.379	-0.749
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	-0.393	-0.094	-0.242	-0.083	-0.170	-0.231
Top-1/top-2 margin	-0.494	-0.390	-0.323	-0.293	-0.343	-0.548
Top-8 probability mass	-0.334	0.252	0.030	0.065	-0.040	-0.090
Top-8/top-9 margin	-0.376	0.344	0.134	0.145	0.014	0.116
Top-8/non-top-8 mean gap	-0.334	0.252	0.030	0.065	-0.040	-0.090
Hidden-state norm	-0.280	0.388	0.145	0.102	-0.083	0.182
Hidden-state step distance	-0.359	0.208	0.090	0.094	0.081	-0.170
Hidden/router geometry	0.364	-0.226	0.009	-0.057	0.075	-0.028
Adjacent top-8 overlap	-0.393	-0.444	-0.463	-0.486	-0.414	-0.670
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	-0.589	-0.126	-0.206	-0.059	-0.178	-0.232
Top-1/top-2 margin	-0.721	-0.386	-0.321	-0.240	-0.299	-0.533
Top-8 probability mass	-0.402	0.218	0.056	0.162	-0.085	-0.112
Top-8/top-9 margin	-0.395	0.315	0.181	0.215	-0.048	0.015
Top-8/non-top-8 mean gap	-0.402	0.218	0.056	0.162	-0.085	-0.112

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden-state norm	−0.333	0.444	0.174	0.143	−0.114	0.017
Hidden-state step distance	−0.506	0.223	0.089	0.145	0.046	−0.161
Hidden/router geometry	0.393	−0.276	0.041	−0.187	0.042	−0.074
Adjacent top-8 overlap	−0.335	−0.455	−0.479	−0.444	−0.396	−0.717
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	−0.746	−0.279	−0.174	−0.085	−0.124	−0.632
Top-1/top-2 margin	−0.837	−0.401	−0.379	−0.286	−0.391	−0.720
Top-8 probability mass	−0.663	0.073	0.052	0.086	0.070	−0.511
Top-8/top-9 margin	−0.659	0.207	0.186	0.172	0.112	−0.415
Top-8/non-top-8 mean gap	−0.663	0.073	0.052	0.086	0.070	−0.511
Hidden-state norm	−0.585	0.281	0.022	0.099	−0.094	−0.228
Hidden-state step distance	−0.759	0.187	0.042	−0.020	−0.002	−0.489
Hidden/router geometry	0.667	−0.249	0.117	0.001	−0.027	0.232
Adjacent top-8 overlap	−0.694	−0.603	−0.490	−0.555	−0.330	−0.883

Table 66. Position windows: metrics versus adjacent top-8 overlap. Spearman ρ ; model and protocol as specified above.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Bin 0						
Finite traces	500	960	960	675	1,280	272
Router confidence	0.430	0.380	0.290	0.122	0.138	0.646
Top-1/top-2 margin	−0.163	−0.248	−0.421	−0.276	−0.150	0.397
Top-8 probability mass	0.729	0.709	0.848	0.755	0.697	0.750
Top-8/top-9 margin	0.585	0.674	0.805	0.688	0.570	0.541
Top-8/non-top-8 mean gap	0.729	0.709	0.848	0.755	0.697	0.750
Hidden-state norm	0.123	0.536	0.652	0.530	0.396	0.130
Hidden-state step distance	−0.530	−0.571	−0.239	−0.228	−0.345	−0.250
Hidden/router geometry	−0.194	−0.200	−0.354	−0.303	−0.134	−0.312
Top-1 switch rate	−0.405	−0.243	−0.088	−0.306	−0.498	−0.722
Bin 1						
Finite traces	429	960	960	668	1,269	269
Router confidence	0.500	0.431	0.218	0.315	0.486	0.511
Top-1/top-2 margin	0.035	−0.048	−0.290	−0.110	0.144	0.342
Top-8 probability mass	0.750	0.694	0.773	0.697	0.756	0.570
Top-8/top-9 margin	0.701	0.626	0.711	0.661	0.695	0.372
Top-8/non-top-8 mean gap	0.750	0.694	0.773	0.697	0.756	0.570
Hidden-state norm	0.049	0.557	0.609	0.398	0.232	0.011
Hidden-state step distance	−0.662	−0.530	−0.368	−0.468	−0.614	−0.379
Hidden/router geometry	−0.065	−0.265	−0.311	−0.205	−0.140	−0.124
Top-1 switch rate	−0.374	−0.354	−0.167	−0.398	−0.502	−0.688
Bin 2						
Finite traces	347	960	960	634	1,172	254
Router confidence	0.605	0.340	0.358	0.401	0.598	0.429
Top-1/top-2 margin	0.152	−0.108	−0.125	−0.004	0.145	0.244
Top-8 probability mass	0.773	0.599	0.726	0.682	0.778	0.552
Top-8/top-9 margin	0.720	0.502	0.632	0.637	0.736	0.361
Top-8/non-top-8 mean gap	0.773	0.599	0.726	0.682	0.778	0.552
Hidden-state norm	−0.072	0.398	0.427	0.193	0.218	0.048
Hidden-state step distance	−0.645	−0.476	−0.554	−0.433	−0.593	−0.457
Hidden/router geometry	−0.140	−0.161	−0.147	−0.212	−0.259	−0.146
Top-1 switch rate	−0.501	−0.358	−0.399	−0.445	−0.480	−0.675
Bin 3						
Finite traces	268	956	942	564	1,038	234
Router confidence	0.626	0.338	0.404	0.466	0.620	0.334
Top-1/top-2 margin	0.240	−0.073	−0.082	0.087	0.106	0.219

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Top-8 probability mass	0.788	0.589	0.702	0.700	0.798	0.462
Top-8/top-9 margin	0.756	0.501	0.622	0.672	0.759	0.274
Top-8/non-top-8 mean gap	0.788	0.589	0.702	0.700	0.798	0.462
Hidden-state norm	-0.110	0.251	0.416	0.182	0.261	0.055
Hidden-state step distance	-0.582	-0.402	-0.526	-0.421	-0.551	-0.417
Hidden/router geometry	-0.286	-0.092	-0.160	-0.183	-0.243	-0.150
Top-1 switch rate	-0.467	-0.419	-0.446	-0.470	-0.422	-0.640
Bin 4						
Finite traces	203	927	900	494	876	202
Router confidence	0.466	0.256	0.380	0.469	0.552	0.292
Top-1/top-2 margin	-0.034	-0.099	-0.154	0.064	0.085	0.259
Top-8 probability mass	0.649	0.545	0.703	0.711	0.739	0.404
Top-8/top-9 margin	0.632	0.482	0.625	0.660	0.708	0.216
Top-8/non-top-8 mean gap	0.649	0.545	0.703	0.711	0.739	0.404
Hidden-state norm	0.040	0.245	0.401	0.253	0.425	-0.100
Hidden-state step distance	-0.591	-0.346	-0.537	-0.338	-0.460	-0.412
Hidden/router geometry	-0.174	-0.117	-0.219	-0.228	-0.273	-0.131
Top-1 switch rate	-0.345	-0.420	-0.464	-0.431	-0.448	-0.670
Bin 5						
Finite traces	171	858	847	426	767	162
Router confidence	0.399	0.278	0.319	0.379	0.532	0.207
Top-1/top-2 margin	0.040	-0.127	-0.116	-0.090	0.062	0.062
Top-8 probability mass	0.588	0.566	0.662	0.664	0.728	0.367
Top-8/top-9 margin	0.573	0.509	0.589	0.627	0.692	0.173
Top-8/non-top-8 mean gap	0.588	0.566	0.662	0.664	0.728	0.367
Hidden-state norm	0.111	0.237	0.389	0.312	0.457	0.079
Hidden-state step distance	-0.434	-0.360	-0.433	-0.344	-0.411	-0.375
Hidden/router geometry	-0.221	-0.120	-0.232	-0.204	-0.298	-0.173
Top-1 switch rate	-0.280	-0.439	-0.437	-0.347	-0.372	-0.651
Bin 6						
Finite traces	139	816	803	383	684	133
Router confidence	0.244	0.327	0.257	0.396	0.481	0.379
Top-1/top-2 margin	0.003	-0.114	-0.188	-0.013	0.062	0.194
Top-8 probability mass	0.499	0.540	0.613	0.601	0.700	0.535
Top-8/top-9 margin	0.475	0.484	0.538	0.551	0.656	0.387
Top-8/non-top-8 mean gap	0.499	0.540	0.613	0.601	0.700	0.535
Hidden-state norm	0.140	0.205	0.379	0.277	0.448	0.161
Hidden-state step distance	-0.230	-0.405	-0.443	-0.270	-0.278	-0.260
Hidden/router geometry	-0.089	-0.078	-0.272	-0.150	-0.253	-0.281
Top-1 switch rate	-0.352	-0.422	-0.469	-0.417	-0.372	-0.610
Bin 7						
Finite traces	106	748	751	342	559	102
Router confidence	0.254	0.285	0.271	0.323	0.455	0.283
Top-1/top-2 margin	-0.042	-0.099	-0.122	-0.126	0.060	0.319
Top-8 probability mass	0.530	0.494	0.653	0.582	0.702	0.419
Top-8/top-9 margin	0.508	0.428	0.568	0.538	0.684	0.215
Top-8/non-top-8 mean gap	0.530	0.494	0.653	0.582	0.702	0.419
Hidden-state norm	0.339	0.250	0.410	0.367	0.510	0.060
Hidden-state step distance	-0.119	-0.334	-0.375	-0.267	-0.245	-0.172
Hidden/router geometry	-0.272	-0.136	-0.315	-0.142	-0.398	-0.263
Top-1 switch rate	-0.327	-0.428	-0.428	-0.409	-0.379	-0.749
Bin 8						
Finite traces	83	679	724	314	436	83
Router confidence	0.314	0.293	0.313	0.384	0.403	0.398
Top-1/top-2 margin	0.085	-0.040	-0.120	-0.044	0.034	0.398
Top-8 probability mass	0.554	0.460	0.629	0.618	0.657	0.559
Top-8/top-9 margin	0.590	0.399	0.564	0.553	0.637	0.375
Top-8/non-top-8 mean gap	0.554	0.460	0.629	0.618	0.657	0.559
Hidden-state norm	0.369	0.237	0.356	0.371	0.500	0.230

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden-state step distance	−0.006	−0.288	−0.350	−0.201	−0.169	−0.016
Hidden/router geometry	−0.354	−0.091	−0.240	−0.262	−0.426	−0.303
Top-1 switch rate	−0.393	−0.444	−0.463	−0.486	−0.414	−0.670
Bin 9						
Finite traces	65	605	701	291	317	70
Router confidence	0.287	0.344	0.335	0.455	0.399	0.538
Top-1/top-2 margin	0.133	−0.034	−0.094	0.016	−0.002	0.561
Top-8 probability mass	0.543	0.484	0.586	0.602	0.697	0.548
Top-8/top-9 margin	0.554	0.418	0.504	0.547	0.685	0.443
Top-8/non-top-8 mean gap	0.543	0.484	0.586	0.602	0.697	0.548
Hidden-state norm	0.429	0.153	0.354	0.357	0.539	0.340
Hidden-state step distance	0.037	−0.357	−0.356	−0.104	−0.104	0.122
Hidden/router geometry	−0.337	−0.031	−0.236	−0.286	−0.354	−0.160
Top-1 switch rate	−0.335	−0.455	−0.479	−0.444	−0.396	−0.717
Overflow						
Finite traces	54	540	664	270	238	59
Router confidence	0.674	0.364	0.237	0.348	0.424	0.707
Top-1/top-2 margin	0.510	0.104	−0.011	0.168	0.155	0.729
Top-8 probability mass	0.774	0.410	0.464	0.492	0.645	0.719
Top-8/top-9 margin	0.801	0.325	0.392	0.430	0.612	0.618
Top-8/non-top-8 mean gap	0.774	0.410	0.464	0.492	0.645	0.719
Hidden-state norm	0.725	0.199	0.318	0.380	0.561	0.446
Hidden-state step distance	0.488	−0.187	−0.235	−0.051	−0.005	0.516
Hidden/router geometry	−0.769	−0.073	−0.244	−0.262	−0.361	−0.396
Top-1 switch rate	−0.694	−0.603	−0.490	−0.555	−0.330	−0.883

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